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Chapter 1

Preliminaries

Theorem 1.0.1 (Borel-Cantelli Lemma)**.** *Let* $A_1, A_2, \cdots \in \mathscr{F}$.

1. *if* $\sum_n P(A_n) < \infty$, $P(\limsup_n A_n) = 0$
2. *if* $\sum_n P(A_n) = \infty$, *and* $\{A_n\}$ *are independent, then* $P(\limsup_n A_n) = 1$

pf.

- 1
  $\forall m \in \mathbb{N}$, $P(\limsup A_n) \leq P(\cup_{k=m}^{\infty} A_k) \leq \sum_{k=m}^{\infty} P(A_k) \to 0$ as $m \to \infty$ if the sum is convergent.
- 2
  since $(\limsup A_n)^c = \cup_{n=1}^{\infty} \cap_{k=n}^{\infty} A_k^c$, we can reduce the case to the one where $P(\cap_{k=n}^{\infty} A_k^c) = 0$ for each $n \in \mathbb{N}$. For $n, m \in \mathbb{N}$, $P(\cap_{k=n}^{\infty} A_k^c) \leq P(\cap_{k=n}^{n+m} A_k^c) = \Pi_{k=n}^{n+m}(1 - P(A_k)) \leq \Pi_{k=n}^{m+n} e^{-P(A_k)} = e^{-\sum_{k=n}^{n+m} P(A_k)} \to 0$ as $m \to \infty$ if the sum is divergent.

Theorem 1.0.2 (Monotone Convergence Theorem)**.** *Suppose* $X_1, X_2, \ldots$ *are random variables with* $E[X_1] > -\infty$ *and* $\{X_n\} \uparrow X$. *Then* X *is a random variable and* $\lim_{n\to\infty} E[X_n] = E[X]$.

pf. $\{X \leq x\} = \cap_n \{X_n \leq x\} \in \mathscr{F}$, $\forall x \in \mathbb{R}$. Furthermore, by monotonicity, $E[X_1] \leq E[X_2] \leq \cdots \leq E[X]$, so that $\lim_n E[X_n]$ exists.

- CLAIM : $E[X_n] \geq E[X]$
  if $E[X_1] = \infty$, it is trivial.
  if $E[X_1] < \infty$, by replacing X_n by $X_n - X_1$ and X by $X - X_1$, we may assume X_n, X are nonnegative. Then for any random variable $Y \leq X$, write it as $Y = \sum_i v_i \mathbb{1}_{A_i}$ s.t. A_i is any finite partition of Ω wirh $v_i \leq X(\omega)$, $\forall \omega \in A_i$. Choose $\epsilon > 0$ and set $A_{in} = \{\omega \in A_i : X_n(\omega) \geq v_i - \epsilon\}$. Then $\{A_{in}\} \uparrow A_i$ as $n \to$

∞. Furthermore, $E[X_n] \geq \sum_i (v_i - \epsilon)P(A_{in})$. By the continuity of probability, as $n \to \infty$, $\sum(v_i - \epsilon)P(A_{in}) = \sum v_i P(A_i) - \epsilon$. Hence $\lim E[X_n] \geq \sum v_i P(A_i) - \epsilon$. Since this is true for all $\epsilon > 0$, $\lim E[X_n] \geq \sum v_i P(A_i)$ and the claim follows.

Theorem 1.0.3 (Fatou's Lemma)**.** *Given a measure space* $(\Omega, \mathscr{F}, P)$ *define* $X, X_1, X_2, \ldots$ *are random variables on it. Suppose* $\lim X_n(\omega) = X(\omega)$, $\forall \omega \in \Omega$, *i.e.* $\{X_n\}$ *converges toX a.s. If* $X_n \geq C$, $\forall n$ *and some constant* $C > -\infty$,

$$E[\liminf X_n] \leq \liminf E[X_n]$$

pf. Set $Y_n = \inf_{k \geq n} X_k$ and let $Y = \lim Y_n = \liminf X_n$. Then $Y_n \geq C$ and $\{Y_n\} \uparrow Y$ and furthermore, $Y_n \leq X_n$. Hence by the order-preserving property and the monotone convergence theorem, the lemma follows.

Theorem 1.0.4 (Dominated Convergence Theorem)**.** *If* $X, X_1, X_2, \ldots$ *are random variables and if* $\{X_n\} \to X$ *with probability 1 and if there is a random variable* Y *with* $|X_n| \leq Y$, $\forall n$ *and with* $EY < \infty$, *then* $\lim_{n\to\infty} E[X_n] = E[X]$.

pf. $Y + X_n \geq 0$ so by the Fatou's lemma, we get $E[Y] + E[X] = E[X + Y] \leq \liminf E[Y + X_n] = E[Y] + \liminf E[X_n]$. Hence, by canceling the $E[Y]$ terms we see $E[X] \leq \liminf E[X_n]$.
$Y - X_n \geq 0$ so by Fatou's lemma, we get $E[Y] - E[X] \leq E[Y] + \liminf E[-X_n] = E[Y] - \limsup E[X_n]$. Hence by cancelling the $E[Y]$ terms, we see $E[X] \geq \limsup E[X_n]$.

Definition 1.0.1. *A collection* $\{X_n\}$ *of random variables is uniformly integrable if*

$$\lim_{\alpha\to\infty} \sup_n E[|X|_n \mathbb{1}_{|X_n| \geq \alpha}] = 0$$

Theorem 1.0.5 (Uniform Integrability Convergence Theorem)**.** *If* $X, X_1, X_2, \ldots$ *are random variables and if* $\{X_n\} \to X$ *with probability 1, and if* $\{X_n\}$ *are uniformly integrable, then* $\lim_{n\to\infty} E[X_n] = E[X]$.

pf. Let $Y_n = |X_n - X|$ so that $Y_n \to 0$.

- CLAIM : $E[Y_n] \to 0$
  $Y_n = Y_n \mathbb{1}_{Y_n < \alpha} + Y_n \mathbb{1}_{Y_n \geq \alpha}$. Let $Y_n^{(\alpha)} = Y_n \mathbb{1}_{Y_n < \alpha}$. Then for any fixed $\alpha > 0$, $|Y_n^{(\alpha)}| \leq \alpha$ and also $Y_n^{(\alpha)} \to 0$ as $n \to \infty$. So by the bounded convergence theorem, $\lim E[Y_n^{(\alpha)}] = 0$, $\alpha > 0$. By the triangle inequality, $Y_n \leq |X_n| + |X| \leq 2M_n$ where $M_n = \max(|X_n|, |X|)$. Hence if $Y_n \geq \alpha$, $M_n \geq \alpha/2$ and thus either $|X_n| \geq \alpha/2$ or $|X| \geq \alpha/2$. Thus $Y_n \mathbb{1}_{Y_n \geq \alpha} \leq 2M_n \mathbb{1}_{M_n \geq \alpha/2} \leq 2|X_n| \mathbb{1}_{|X_n| \geq \alpha/2} + 2|X| \mathbb{1}_{|X| \geq \alpha/2}$. By taking expectations and supremums, $\sup_n E[Y_n \mathbb{1}_{Y_n \geq \alpha} \leq 2 \sup_n E[|X_n| \mathbb{1}_{|X_n| \geq \alpha/2} + 2E[|X| \mathbb{1}_{|X| \geq \alpha/2}]$. As $\alpha \to \infty$, the first term of the RHS goes to 0 by the uniform integrability assumption. The second term of the RHS goes to 0 by the dominated convergence theorem. Thus $\lim_{\alpha\to\infty} \sup_n E[Y_n \mathbb{1}_{Y_n \geq \alpha}] = 0$. Let $\epsilon > 0$.

Then $\exists \alpha_0 > 0$ s.t. $\sup_n E[Y_n \mathbb{1}_{Y_n \geq \alpha_0}] < \epsilon/2$. Then since $\lim EY_n^{(\alpha)} = 0$, $\exists n_0 \in \mathbb{N}$ s.t. $E[Y_n^{(\alpha_0)}] < \epsilon/2$, $\forall n \geq n_0$. Then $\forall n \geq n_0$,

$$E[Y_n] = E[Y_n^{(\alpha_0)}] + E[Y_n \mathbb{1}_{Y_n \geq \alpha_0}] < \epsilon/2 + \epsilon/2 = \epsilon$$

So $E[Y_n] \to 0$.

Then by the triangle inequality, $|E[X_n] - E[X]| \leq E[Y_n] \to 0$ so the theorem follows.

Definition 1.0.2. *Given any Borel-measurable function f s.t. $f \geq 0$, $\int_{-\infty}^{\infty} f(t)\lambda(dt) = 1$, we can define a law μ by $\mu(B) = \int_{-\infty}^{\infty} f(t)\mathbb{1}_B(t)\lambda(dt)$ and say μ is absolutely continuous w.r.t. λ and f is the density for μ w.r.t. λ and written as $f = \frac{d\mu}{d\lambda}$.*

Lemma 1.0.1 (Hahn Decomposition)**.** *Let ϕ be a finite signed measure on $(\Omega, \mathscr{F})$, i.e. $\phi = \mu - v$ for some finite measures μ, v. Then there is a partition $\Omega = A^+ \cup A^-$ with $A^+, A^- \in \mathscr{F}$ s.t.* $\begin{cases} \phi(E) \geq 0 & \forall E \subseteq A^+ \\ \phi(E) \leq 0 & \forall E \subseteq A^- \end{cases}$

pf. Set $\alpha = \sup\{\phi(A) : A \in \mathscr{F}\}$. Construct a subset A^+ s.t. $\phi(A^+) = \alpha$. Then we can take $A^- = \Omega \setminus A^+$. Then if $E \subseteq A^+$ but $\phi(E) < 0$, $\phi(A^+ \setminus E) = \phi(A^+) - \phi(E) > \phi(A^+) = \alpha$, a contradiction to the definition of α Similarly, if $E \subseteq A^-$ but $\phi(E) > 0$, $\phi(A^+ \cup E) \geq \phi(A^+) + \phi(E) > \alpha$, a contradiction to the definition of α. Thus the lemma follows.

- Construction of A^+

  Choose subsets $A_1, A_2, \ldots \mathscr{F}$ s.t. $\phi(A_n) \to \alpha$. Let $\cup A_i$ and let $\mathscr{G}_n = \{\cap_{k=1}^n A'_k$, $A_{k'} = A_k$ or $A \setminus A_k\}$. Then let $C_n = \cup_{S \in \mathscr{G}_n \phi(S) \geq 0} S$ i.e. C_n is the union of those elements of $\mathscr{G}_n$ with nonnegative measure under ϕ. Finally set $A^+ = \limsup C_n$. Since A_n is a union of certain particular elements of $\mathscr{G}_n$ and C_n is a union of all the ϕ-postive elements of $\mathscr{G}_n$, $\phi(C_n) \geq \phi(A_n)$. Also $C_m \cup \cdots \cup C_n$ is formed from $C_m \cup \cdots \cup C_{n-1}$ by including some additional ϕ-positive elements of $\mathscr{G}_n$ so $\phi(C_m \cup \cdots \cup C_n) \geq \phi(C_m \cup \cdots \cup C_{n-1})$. By induction, $\phi(C_m \cup \cdots \cup C_n) \geq \phi(C_m) \geq \phi(A_m)$. Since this holds for all n, by the continuity of probability on μ, v, $\phi(C_m \cup \ldots) \geq \phi(A_m)$ for all m. Then $\phi(A^+) = \phi(\limsup C_n) = \lim \phi(C_m \cup C_{m+1} \cup \ldots) \geq \lim \phi(A_m) = \alpha$. Hence $\phi(A^+) = A$, and we get the desired set.

Theorem 1.0.6 (Lebesgue Decomposition)**.** *Any probability measure μ on $\mathbb{R}$ can uniquely be decomposed as $\mu = \mu_{disc} + \mu_{ac} + \mu_{sign}$ where they satisfy*

1. *μ_{disc} is discrete, i.e. $\sum_{x \in \mathbb{R}} \mu_{disc}(x) = \mu_{disc}(\mathbb{R})$*
2. *μ_{ac} is absolutely continuous, i.e. $\mu_{ac}(A) = \int_A f\, d\lambda$ for all Borel sets A, for some nonnegative Borel-measurable function f, where λ is Lebesgue measure on $\mathbb{R}$*

3. *μ_{sing} is singular continuous, i.e. $\mu_{sing}(x) = 0$, $\forall x \in \mathbb{R}$, but there is $S \subseteq \mathbb{R}$ with $\lambda(S) = 0$ and $\mu_{sing}(S^c) = 0$.*

pf.

- Existence

  Define $\mu_{disc}(A) = \sum_{x \in A} \mu(x)$ and then $\mu - \mu_{disc}$ has no discrete component. Hence assume μ has no discrete component. By the countable additivity, we may assume μ is supported on $[0,1]$ so that we may take λ to be Lebesgue measure on $[0,1]$. Call a function g a candidate density if $g \geq 0$, and $\int_E g d\lambda \leq \mu(E)$ for all Borel sets E. If g_1, g_2 are candidate densities, so is $\max(g_1, g_2)$, since $\int_E \max(g_1, g_2) d\lambda = \int_{E \cap \{g_1 \geq g_2\}} g_1 d\lambda + \int_{E \cap \{g_1 < g_2\}} g_e d\lambda \leq \mu(E \cap \{g_1 \geq g_2\}) + \mu(E \cap \{g_1 < g_2\}) = \mu(E)$. By the monotone convergence theorem, if $h_1, h_2, \ldots$ are candidate densities and $h_n \uparrow h$, h is also a candidate density. Thus if $g_1, g_2, \ldots$ are candidate densities, so is $\sup_n g_n = \lim \max(g_1, \ldots, g_n)$. Let $\beta = \sup\{\int_{[0,1]} g d\lambda : g \text{ is a candidate density}\}$. Choose candidate densities g_n with $\int_{[0,1]} f d\lambda = \beta$, i.e. f is the largest possible candidate density. Define $\mu_{ac}(A) = \int_A f d\lambda$. Then define $\mu_{sing}(A) = \mu(A) - \mu_{ac}(A)$. Since f was a candidate density, $\mu_{sing}(A) \geq 0$. For each $n \in \mathbb{N}$, let $[0,1] = A_n^+ \cup A_n^-$ be a Hahn decomposition for the signed measure $\phi_n = \mu_{sing} - \frac{1}{n}\lambda$. Set $M = \cup_n A_n^+$, then $M^C = \cup_n A_n^-$, so that $M^c \subseteq A_n^-$ for all n. Then $(\mu_{sing} - \frac{1}{n}\lambda)(M^c) \leq 0$, $\forall n$ so that $\mu_{sing}(M^c) \leq \frac{1}{n}\lambda(M^c)$ for all n. Hence $\mu_{sing}(M^c) = 0$. If $\lambda(M) > 0$, $\exists n \in \mathbb{N}$ s.t. $\lambda(A_n^+) > 0$. For this n, $(\mu_{sing} - \frac{1}{n}\lambda)(E) \geq 0$, i.e. $\mu_{sing}(E) \geq \frac{1}{n}\lambda(E)$, $\forall E \subseteq A_n^+$. But $g = f + \frac{1}{n}\mathbb{1}_{A_n^+}$ is a candidate density and $\int_{0,1] } g d\lambda = \int_{[0,1]} f d\lambda + \frac{1}{n}\lambda_{[0,1]}\mathbb{1}_{A_n^+} d\lambda = \beta + \frac{1}{n}\lambda(A_n^+) > \beta$, contradiction to the maximality of f. Thus $\lambda(M) = 0$. So μ_{sing} must be singular, so the existence follows.

  - $g = f + \frac{1}{n}\mathbb{1}_{A_n^+}$ is a candidate density.
    For any Borel set E, $\int_E g d\lambda = \int_E f d\lambda + \frac{1}{n}\int_E \mathbb{1}_{A_n^+} d\lambda = \mu_{ac}(E) + \frac{1}{n}\lambda(A_n^+ \cap E) \leq \mu_{ac}(E) + \mu_{sing}(A_n^+ \cap E) \leq \mu_{ac} + \mu_{sing}(A) = \mu(E)$.

- Uniqueness

  Suppose $\mu = \mu_{ac} + \mu_{sing} = v_{ac} + v_{sing}$ with $\begin{cases} \mu_{ac}(A) = \int_A f d\lambda \\ v_{ac}(A) = \int_A g d\lambda \end{cases}$ Since μ_{sing}, v_{sing} are singular, $\exists S_1, S_2$ with $\lambda(S_1) = \lambda(S_2) = 0$ and $\mu_{sing}(S_1^c) = v_{sing}(S_2^c) = 0$. Let $S = S_1 \cup S_2$ and let $B = \{\omega \in S^c : f(\omega) < g(\omega)\}$. Then $g - f > 0$ on B but $\int_B (g - f) d\lambda = \mu_{ac}(B) - v_{ac}(B) = \mu(B) - v(B) = 0$. Similarly $\lambda(f > g) = 0$ Thus $\lambda(f = g) = 1$ and the uniqueness follows.

Corollary 1.0.1 (Radon-Nikodym Theorem)**.** *If $v << \mu$, then there exists a $\mathscr{F}$-measurable function $f : X \to [0.\infty)$ s.t. for any measurable set $A \in \mathscr{F}$, $v(A) = \inf_A f d\mu$.*

pf. If μ is absolutely continuous, $\mu << \lambda$.
Suppose $\mu << \lambda$ and let $\mu = \mu_{disc} + \mu_{ac} + \mu_{sing}$ as in the previous theorem. Since $\lambda(x) = 0,\ \forall x,\ \mu(x) = 0$ as well, so that $\mu_{disc} = 0$. Similarly, if S is s.t. $\lambda(S) = 0$ and $\mu_{sing}(S^c) = 0,\ \mu(S) = 0$ so that $\mu_{sing}(S) = 0$. Thus $\mu_{sing} = 0$. Hence $\mu = \mu_{ac}$, i.e. it is absolutely continuous.
$\to$ if $\mu(A) = \int_A f d\lambda$ for all Borel sets A, we write $\frac{d\mu}{d\lambda} = f$ and call f the density, or Radon-Nikodym derivative of μ w.r.t. λ.

Let E be an arbitrary set, and let $\mathscr{P}(E)$ denote the set of all subsets of E.

Definition 1.0.3. *A subset $\mathscr{M}$ of $\mathscr{P}(E)$ is called a monotone class if the following properties hold:*

1. *$E \in \mathscr{M}$*
2. *if $A, B \in \mathscr{M}$ and $A \subset B$, then $B \setminus A \in \mathscr{M}$*
3. *if $(A_n)_{n\geq 0}$ is an increasing sequence of subsets of E s.t. $A_n \in \mathscr{M}$ for every $n \geq 0$, then $\cup_{n\geq 0} A_n \in \mathscr{M}$*

A σ-field is a monotone class. The intersection of an arbitrary collection of monotone classes is again a monotone class. If $\mathscr{C}$ is an arbitrary subset of $\mathscr{P}(E)$, the monotone class generated by $\mathscr{C}$ which is denoted by $\mathscr{M}(\mathscr{C})$ is defined by

$$\mathscr{M}(\mathscr{C}) = \cap_{\mathscr{M}\ monotone\ class,\ \mathscr{C}\subset\mathscr{M}} \mathscr{M}$$

Lemma 1.0.2 (Monotone Class Argument)**.** *If $\mathscr{C} \subset \mathscr{P}(E)$ is a stable under finite intersections, then $\mathscr{M}(\mathscr{C}) = \sigma(\mathscr{C})$.*

pf. Since any σ-field is a monotone class, $\mathscr{M}(\mathscr{C}) \subset \sigma(\mathscr{C})$.
A monotone class is a σ-field iff it is stable under finite intersections. $\forall A \in \mathscr{P}(E)$, set $\mathscr{M}_A = \{B \in \mathscr{M}(\mathscr{C}) : A \cap B \in \mathscr{M}(\mathscr{C})\}$. Fix $A \in \mathscr{C}$. Since $\mathscr{C}$ is stable under finite intersections, $\mathscr{C} \subset \mathscr{M}_A$. Then $E \in \mathscr{M}_A$ trivially. If $B, B' \in \mathscr{M}_A$ and $B \subset B'$, $A \cap (B \setminus B') = (A \cap B') \setminus (A \cap B) \in \mathscr{M}(\mathscr{C})$ and therefore $B' \setminus B \in \mathscr{M}_A$. If $B_n \in \mathscr{M}_A$, $\forall n \geq 0$ and the sequence (B_n) is increasing $A \cap (\cup B_n) = \cup(A \cap B_n) \in \mathscr{M}(\mathscr{C})$ and therefore $\cup B_n \in \mathscr{M}_A$. Therefore $\mathscr{M}_A$ is a monotone class that contains $\mathscr{C}$. Thus $\mathscr{M}(\mathscr{C}) \subset \mathscr{M}_A$. So $\forall A \in \mathscr{C}$, $\forall B \in \mathscr{M}(\mathscr{C})$, $A \cap B \in \mathscr{M}(\mathscr{C})$. Fix $A \in \mathscr{M}(\mathscr{C})$. Then since $\mathscr{C} \subset \mathscr{M}_A$, $\mathscr{M}_A$ is a monotone class s.t. $\mathscr{M}(\mathscr{C}) \subset \mathscr{M}_A$. Thus $\mathscr{M}(\mathscr{C})$ is stable under finite intersections.

Chapter 2

Brownian Motion

2.1 Gaussian Process

Define random variables on a probability space $(\Omega, \mathscr{F}, P)$. For every real $p \geq 1$, $L^p(\Omega, \mathscr{F}, P)$ denotes the space of all real random variables X s.t. $|X|^p$ is integrable. The space L^p is equipped with the usual norm.

Definition 2.1.1. *A real random variable X is said to be a standard Gaussian variable if its law of density*

$$p_X(x) = \frac{1}{\sqrt{2\pi}} \exp(-\frac{x^2}{2})$$

w.r.t. Lebesgue measure on $\mathbb{R}$.

$\rightarrow$ The complex Laplace transform is given by
$E[e^{\lambda X}] = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{\lambda x} e^{-x^2/2} dx = e^{\lambda^2/2} \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-(x-\lambda)^2/2} dx = e^{\lambda^2/2} \rightarrow E[e^{zX}] = e^{z^2/2}$, $\forall z \in \mathbb{C}$. So $E[e^{zX}]$ is well-defined for all $z \in \mathbb{C}$ and defines a holomorphic function on $\mathbb{C}$. By analytic continuity, the continuity of $E[e^{zX}] = e^{z^2/2}$ for all $z \in \mathbb{R}$ implies for every $z \in \mathbb{C}$ the above identity holds. Let $z = i\xi$, $\xi \in \mathbb{R}$, $E[e^{i\xi X}] = e^{-\xi^2/2} = 1 + i\xi E[X] + \cdots + \frac{(i\xi)^n}{n!} E[X^n] + O(|\xi|^{n+1})$. Then as $\xi \rightarrow 0$, $E[X] = 0$
,$E[X^2] = 1$ and $\forall n \in \mathbb{Z}_{>0}$, $E[X^{2n}] = \frac{(2n)!}{2^n n!}$, $E[X^{2n+1}] = 0$.

Definition 2.1.2. *If $\sigma > 0$, $m \in \mathbb{R}$ we say that a real random variable Y is Gaussian with $\mathcal{N}(m, \sigma^2)$-distribution if Y satisfies any of the three equivalent properties:*

1. *$Y = \sigma X + m$, where X is a standard Gaussian variable*
2. *the law of Y has density $p_Y(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp(-\frac{(y-m)^2}{2\sigma^2})$*
3. *the characteristic function of Y is $E[e^{i\xi Y}] = \exp(im\xi - \frac{\sigma^2}{2}\xi^2)$*

Proposition 2.1.1. *Let $(X_n)_{n\geq 1}$ be a sequence of real random variables s.t. $\forall n \geq 1$, X_n follows the $\mathcal{N}(m_n, \sigma_n^2)$-distribution. Suppose X_n converges in L^2 to X. Then*

1. *the random variable X follows the $\mathcal{N}(m, \sigma^2)$-distribution where $m = \lim m_n$, $\sigma = \lim \sigma_n$.*

2. *the convergence also holds in all L^p spaces, $1 \leq p < \infty$*

pf.

- the convergence in L^2 implies $\begin{cases} m_n = E[X_n] \text{ converges to } E[X] \\ \sigma_n^2 = var(X_n) \text{ converges to } var(X) \end{cases}$ as $n \to \infty$. Then setting $m = E[X]$, $\sigma^2 = var(X)$, $\forall \xi \in \mathbb{R}$,

$$E[e^{i\xi X}] = \lim E[e^{i\xi X_n}] = \lim \exp(im_n\xi - \frac{\sigma_n^2}{2}\xi^2) = \exp(im\xi - \frac{\sigma^2}{2}\xi^2)$$

  so X follows $\mathcal{N}(m, \sigma^2)$-distribution.

- Since X_n has the same distribution as $\sigma_n N + m_n$, where N is a standard Gaussian variable and since the sequence $(m_n), (\sigma_n)$ are bounded, $\sup_n E[|X_n|^q] < \infty$, $\forall q \geq 1$. Thus
  $\sup_n E[|X_n - X|^q] < \infty$, $\forall q \geq 1$. Let $p \geq 1$. The sequence $Y_n = |X_n - X|^p$ converges in probability to 0 and is uniformly integrable because it is bounded in L^2. Thus this sequence converges 0 in L^1.

Definition 2.1.3. *A centered Gaussian space is a closed linear subspace of $L^2(\Omega, \mathcal{F}, P)$ which contains only centered Gaussian variables.*

Definition 2.1.4. *Let $(E, \mathcal{E})$ be a measurable space and let T be an arbitrary index set. A random process indexed by T with values in E is a collection $(X_t)_{t\in T}$ of random variables with values in E. IF the measurable space $(E, \mathcal{E})$ is not specified, $E = \mathbb{R}, \mathcal{E} = \mathcal{B}(\mathbb{R})$ the Borel σ-field on $\mathbb{R}$.*

Definition 2.1.5. *A real-valued random process $(X_t)_{t\in T}$ is called a centered Gaussian process if any finite linear combination of the variables X_t, $t \in T$ is centered Gaussian.*

Proposition 2.1.2. *If $(X_t)_{t\in T}$ is a Gaussian process, the closed linear subspace of L^2 spanned by the variables X_t, $t \in T$ is a Gaussian space, which is called the Gaussian space generated by the process X.*

pf. An L^2-limit of centered Gaussian variables is still centered Gaussian by the proposition1.1.1, so the proposition follows.

Definition 2.1.6. *Let H be a collection of random variables defined on $(\Omega, \mathscr{F}, P)$. The σ-field generated by H, denoted by $\sigma(H)$ is the smallest σ-field on Ω s.t. all variables $\xi \in H$ are measurable for this σ-field. If $\mathscr{C}$ is a collection of subsets of Ω, write $\sigma(\mathscr{C})$ for the smallest σ-field on Ω that contains all elements of $\mathscr{C}$.*

Theorem 2.1.1. *Let H be a centered Gaussian space and let $(H_i)_{i\in I}$ be a collection of linear subspaces of H. Then the subspaces H_i, $i \in I$ are pairwise orthogonal in L^2 iff the σ-fields $\sigma(H_i)$, $i \in I$ are independent.*

pf.

- suppose σ-fields $\sigma(H_i)$ are independent. Then if $i \neq j$, $X \in H_i$, $Y \in H_j$, $E[XY] = E[X]E[Y] = 0$ so that the linear spaces H_i are pairwise orthogonal.

- suppose the linear spaces H_i are pairwise orthogonal.
  CLAIM : If $i_1, \ldots, i_p \in I$ are distinct, the σ-fields $\sigma(H_{i_1}), \ldots, \sigma(H_{i_p})$ are independent
  pf. For every $j \in \{1, \ldots, p\}$, we can find an orthonormal basis $(\eta_1^j, \ldots, \eta_{m_j}^j)$ of linear subspace of L^2 spanned by $\{\xi_1^j, \ldots, \xi_{n_j}^j\}$. The covariance matrix of the vector $(\eta_1^1, \ldots, \eta_{m_1}^1, \eta_1^2, \ldots, \eta_{m_2}^2, \ldots, \eta_1^p, \ldots, \eta_{m_p}^p)$ is then the identity matrix since for $i \neq j, E[\eta_l^i \eta_r^j] = 0$ given $H_i \perp H_j$. Moreover this vector is Gaussian because its components belong to H. By the proposition 1.1.2, the components of the latter vector are independent random variables. Equivalently the vectors $(\xi_1^1, \ldots, \xi_{n_1}^1), \ldots, (\xi_1^p, \ldots, \xi_{n_p}^p)$ are independent.
  By the definition of the independence of an infinite collection of σ-fields, the theorem follows.

Corollary 2.1.1. *Let H be a centered Gaussian space and let K be a closed linear subspace of H. Let p_K denote the orthogonal projection onto K in the Hilbert space L^2 and let $X \in H$.*

- $E[X|\sigma(K)] = p_K(X)$

- *Let $\sigma^2 = E[(X - p_K(X))^2]$. Then for every Boreal subset A of $\mathbb{R}$, the random variable $P(X \in A|\sigma(L)]$ is given by $P(X \in A|\sigma(K))(\omega) = \mathscr{Q}(\omega, A)$, where $\mathscr{Q}(\omega, \cdot)$ denotes the $\mathscr{N}(p_K(X)(\omega), \sigma^2)$-distribution.*

pf.

- let $Y = X - p_K(X)$. Then Y is orthogonal to K and by the previous theorem Y is independent of $\sigma(K)$. Then $E[X|\sigma(K)] = E[p_K(X)|\sigma(K)] + E[Y|\sigma(K)] = p_K(X) + E[Y] = p_K(X)$

- for every nonnegative measurable function f on $\mathbb{R}_{\geq 0}$, $E[f(x)|\sigma(K)] = E[f(p_K(X)+Y)|\sigma(K)] = \int p_Y(dy) f(p_K(X)+y)$ where P_Y is the law of Y which is $\mathcal{N}(0,\sigma^2)$ since Y is centered Gaussian with variance σ^2. In the second equality, we use thee fact : if Z is $\mathcal{G}$-measurable r.v. ad if Y is independent of $\mathcal{G}$, the for every nonnegative measurable function g, $E[g(Y,Z)|\mathcal{G}] = \int g(y,Z) P_Y(dy)$.

$\rightarrow$ for general random variable X in L^2, $E[X|\sigma(K)] = p_{L^2(\Omega,\sigma(K),P)}(X)$ which coincides with the orthogonal projection onto the space K which is much smaller than $L^2(\Omega,\sigma(K),P)$ by the previous corollary i.

Definition 2.1.7. *Let $(X_t)_{t\in T}$ be a centered Gaussian process. The covariance function of X is the function* $\begin{array}{rcl} \Gamma : T\times T & \rightarrow & \mathbb{R} \\ (s,t) & \mapsto & cov(X_s,X_t) = E[X_s X_t] \end{array}$. *This function characterises the collection of finite-dimensional marginal distributions of the process X, that is the collection consisting for every choice of the distinct indices $t_1,\dots,t_p \in T$ of the law of the vector $(X_{t_1},\dots,X_{t_p})$.*

$\rightarrow$ this vector is a centered Gaussian vector in $\mathbb{R}^p$ with covariance matrix $(\Gamma(t_i,t_j))_{1\leq i,j\leq p}$.

Theorem 2.1.2. *Let Γ be a symmetric function of positive type on $T\times T$. There exists on an appropriate probability space $(\Omega,\mathcal{F},P)$, a centered Gaussian process whose covariance function is Γ.*

e.g. $T=\mathbb{R}$, μ a finite measure on $\mathbb{R}$ which is symmetric. Set for every $s,t\in\mathbb{R}\Gamma(s,t) = \int e^{i\xi(t-s)}\mu(d\xi)$. Then Γ has the desirable properties ($\sum_{\mathbb{R}\times\mathbb{R}} c(s)c(t)\Gamma(s,t) = \int |\sum_{\mathbb{R}} c(s)e^{i\xi s}|^2 \mu(d\xi) \geq 0$).
$\rightarrow \Gamma(s,t)$ only depends on $t-s$, and therefore any centered Gaussian process $(X_t)_{t\in\mathbb{R}}$ with covariance function Γ is stationary($(X_{t_1+t},\dots,X_{t_n+t}) = (x_{t_1},X_{t_2},\dots,X_{t_n})$ for any choice of $t_1,\dots,t_n,t\in\mathbb{R}$).
$\Leftrightarrow$any Gaussian process X indexed by $\mathbb{R}$ has a covariance of the preceding type and the measure μ is called the spectral measure of X.

Definition 2.1.8. *Let $(E,\mathcal{E})$ be a measurable space, and let μ be a σ-finite measure on $(E,\mathcal{E})$. A Gaussian white noise with intensity μ is an isometry G from $L^2(E,\mathcal{E},\mu)$ into a centered Gaussian space.*

Let $A_1,\dots,A_n\in\mathcal{E}$ be disjoint and s.t. $\mu(A_j)<\infty$, $\forall j$. Then the vector $(G(A_1),\dots,G(A_n))$ is a Gaussian vector in $\mathbb{R}^n$ and its covariance matrix is diagonal since if $i\neq j$, $E[G(A_i)G(A_j)] =< \mathbb{1}_{A_i},\mathbb{1}_{A_j} >_{L^2(E,\mathcal{E},\mu)} = 0$. By the previous proposition, the variables $G(A_1),\dots,G(A_n)$ are independent.

Suppose $A\in\mathcal{E}$ is s.t. $\mu(A)<\infty$ and A is the disjoint union of a countable collection $A_1,A_2,\dots$ of measurable subsets of E. Then $\mathbb{1}_A = \sum_{j=1}^{\infty}\mathbb{1}_{A_j}$ where the series converges in $L^2(E,\mathcal{E},\mu)$ and by the isometry property it implies $G(A) = \sum_{j=1}^{\infty} G(A_j)$, where the series converges in $L^2(\Omega,\mathcal{F},P)$.

Proposition 2.1.3. *Let $(E,\mathscr{E})$ be a measurable space and let μ be a σ-finite measure on $(E,\mathscr{E})$. There exists on an appropriate probability space $(\Omega,\mathscr{F},P)$, a Gaussian white noise with intensity μ.*

pf. Let $(f_i, i \in I)$ be a total orthonormal system in the Hilbert space $L^2(E,\mathscr{E},\mu)$. For every $f \in L^2(E,\mathscr{E},\mu)$, $f = \sum_{i\in I} \alpha_i f_i$, where the coefficients $\alpha_i =< f, f_i >$ are s.t. $\sum_{i\in I} \alpha_i^2 = ||f||^2 < \infty$. On an appropriate probability space $(\Omega,\mathscr{F},P)$ we can construct a collection $(X_i)_{i\in I}$, indexed by the same index set I, of independent $\mathscr{N}(0,1)$ r.v.s. Set $G(f) = \sum_{i\in I} \alpha_i X_i$. It converges in L^2 since X_i, $i \in I$ form an orthonormal system in L^2. Then clearly G takes values in the Gaussian space generated by X_i, $i \in I$. Also G is an isometry since it maps the orthonormal basis $(f_i,\ i \in I)$ to an orthonormal system.
$\rightarrow T = L^2(E,\mathscr{E},\mu)$, $\Gamma(f,g) =< f,g >_{L^2(E,\mathscr{E},\mu)}$ we get a Gaussian process $(X_f, f \in L^2(E,\mathscr{E},\mu))$ and we just have to take $G(f) = X_f$.

Proposition 2.1.4. *Let G be a Gaussian white noise on $(E,\mathscr{E})$ with intensity μ. Let $A \in \mathscr{E}$ be s.t. $\mu(A) < \infty$. Assume there is a sequence of partitions of A,*
$A = A_1^n \cap \cdots \cap A_{k_n}^n$ whose mesh tends to 0 in the sense that $\lim(\sum_j \mu(A_j^n)) = 0$. Then

$$\lim_{n\to\infty} \sum_{j=1}^{k_n} G(A_j^n)^2 = \mu(A)$$

in L^2.

pf. For every fixed n, the variables $G(A_1^n),\ldots,G(A_{k_n}^n)$ are independent. Furthermore $E[G(A_j^n)^2] = \mu(A_j^n)$. Then $E[(\sum_{j=1}^{k_n} G(A_j^n)^2 - \mu(A))^2] = \sum var(G(A_j^n)^2) = 2\sum_j \mu(A_j^n)^2$, given by $var(X^2) = E[X^4] - \sigma^4 = 3\sigma^4 - \sigma^4 = 2\sigma^4$. Then $\sum_{j=1}^{k_n} \mu(A_j^n)^2 \leq (\sup_j \mu(A_j^n))\mu(A) \to 0$ as $n \to \infty$.

e.g. **KALMAN FILTERING**

Let $(\epsilon_n)_{n\geq 0}, (\eta_n)_{n\geq 0}$ be two independent sequences of independent Gaussian random variables s.t. $\forall n, \epsilon_n$ is distributed according to $\mathscr{N}(0,\sigma^2)$ and η_n is distributed according to $\mathscr{N}(0,\delta^2)$, where $\sigma,\delta > 0$. Consider two other sequences $(X_n)_{n\geq 0}, (Y_n)_{n\geq 0}$ defined by the properties $X_0 = 0$ and $\forall n \geq 0$, $\begin{cases} X_{n+1} = a_n X_n + \epsilon_{n+1} \\ Y_n = cX_n + \eta_n \end{cases}$
where c, a_n are postive constants. Set $\begin{array}{c} \hat{X}_{n/n} = E[X_n|Y_0,\ldots,Y_n] \\ \hat{X}_{n+1/n} = E[X_{n+1}|Y_0,\ldots,Y_n] \end{array}$.

- $\hat{X}_{n+1/n} = E[X_{n+1}|Y_0,\ldots,Y_n] = E[a_n X_n + \epsilon_{n+1}|Y_0,\ldots,Y_n] = a_n E[X_n|Y_0,\ldots,Y_n] + E[\epsilon_{n+1}] = a_n \hat{X}_{n/n}$

- Denote $Z_n \equiv Y_n - c\hat{X}_{n/n-1}$. $\forall n \geq 1$, $\hat{X}_{n/n} = \hat{X}_{n/n-1} + \frac{E[X_n Z_n]}{E[Z_n^2]} Z_n$. Define $P_n \equiv E[(X_n - \hat{X}_{n/n-1})^2]$
  $\rightarrow E[Z_n^2] = E[(Y_n - c\hat{X}_{n/n-1})^2]$
  $= E[Y_n^2 - 2c\hat{X}_{n/n-1}Y_n + \hat{X}_{n/n-1}^2] = E[c^2X_n^2 + 2X_n\eta_n + \eta_n^2 - c^2\hat{X}_{n/n-1}^2 - 2c(cX_n - \eta_n)\hat{X}_{n/n-1}]$
  $= c^2P_n + \delta^2 - 2E[c\eta_n\hat{X}_{n/n-1}] = c^2P_n^2 + \delta^2$
  $E[X_nZ_n] = E[X_n(Y_n - c\hat{X}_{n/n-1})] = E[X_n(cX_n + \eta_n - c\hat{X}_{n/n-1})]$
  $= E[c(X_n - \hat{X}_{n/n-1})^2 + c\hat{X}_{n/n-1}X_n - c\hat{X}_{n/n-1}^2] = cP_n + E[(X_n - \hat{X}_{n/n-1})\hat{X}_{n/n-1}] = cP_n$
  $\rightarrow \forall n \geq 1, \hat{X}_{n+1/n} = E[X_{n+1}|Y_0, \ldots, Y_n] = a_nE[X_n|Y_0, \ldots, Y_n] = a_nX_{n/n}$
  $= a_n(\hat{X}_{n/n-1} + \frac{E[X_nZ_n]}{E[Z_n^2]}Z_n) = a_n(\hat{X}_{n/n-1} + \frac{cP_n}{c^2P_n+\delta^2}Z_n)$

- P_n
  when $n = 1$,
  $P_1 = E[(X_1 - \hat{X}_{1/0})^2] = E[(a_0X_0 + \epsilon_0 - \hat{X}_{n/n-1})^2] = E[a_0^2X_0^2 + \epsilon_0^2 + \hat{X}_{1/0}^2 + 2(a_0X_0\epsilon_0 - a_0X_0\hat{X}_{1/0} - \epsilon_0\hat{X}_{1/0})] = \sigma^2 + E[a_0X_0(a_0X_0 - \hat{X}_{1/0})] + E[\hat{X}_{1/0}(\hat{X}_{1/0} - a_0X_0)] + E[2a_0X_0\epsilon_0 - \epsilon_0\hat{X}_{1/0}] = \sigma^2$
  when $n > 1$,
  $P_{n+1} = E[(X_{n+1} - \hat{X}_{n+1/n})^2] = E[(X_{n+1} - a_n(\hat{X}_{n/n-1} + \frac{cP_n}{c^2P_n+\delta^2}Z_n))^2] = E[(a_nX_n + \epsilon_{n+1} - a_n(\hat{X}_{n/n-1} + \frac{cP_n}{c^2P_n+\delta^2}Z_n))^2] = E[(a_n(X_n - \hat{X}_{n/n-1}) + \epsilon_{n+1} - a_nZ_n\frac{cP_n}{c^2P_n+\delta^2})^2] = a_n^2P_n + \sigma^2 + E[(a_nZ_n\frac{cP_n}{c^2P_n+\delta^2})^2] + 2E[a_n(X_n - \hat{X}_{n/n-1})(\epsilon_{n+1} - a_n\frac{cP_n}{c^2P_n+\delta^2}) - \epsilon_{n+1}a_n\frac{cP_n}{c^2P_n+\delta^2}]$
  $= \sigma^2 + a_n^2P_n + a_n^2\frac{c^2P_n^2}{(c^2P_n+\delta^2)^2}EZ_n^2 - 2E[a_n(X_n - \hat{X}_{n/n-1})a_n\frac{cP_n}{c^2P_n+\delta^2}] = \sigma^2 + a_n^2(P_n + \frac{c^2P_n^2}{c^2P_n+\delta^2} - 2\frac{c^2P_n^2}{c^2P_n+\delta^2}) = \sigma^2 + a_n^2\frac{\delta^2P_n}{cP_n^2+\delta^2}$

2.2 pre-Brownian Motion

Definition 2.2.1. *Let G be a Gaussian white noise on $\mathbb{R}_+$ whose intensity is Lebesgue measure. The random process $(B_t)_{t\in\mathbb{R}_{>0}}$ defined by*

$$B_t = G(\mathbb{1}_{[0,t]})$$

is called pre-Brownian motion.

Proposition 2.2.1. *Pre-Brownian motion is a centered Gaussian process with covariance*

$$K(s,t) = \min\{s,t\} \equiv s \wedge t$$

pf. By the definition of a Gaussian white noise, the variables B_t belong to a common Gaussian space, and therefore $(B_t)_{t\geq 0}$ is a Gaussian process. For $s,t \geq 0$, $E[B_sB_t] = E[G([0,s])G([0,t])] = \int_0^\infty dr\mathbb{1}_{[0,s]}(r)\mathbb{1}_{[0,t]}(r) = s \wedge t$.

Proposition 2.2.2. *Let $(X_t)_{t\geq 0}$ be a real-valued random process. TFAE*

1. *$(X_t)_{t\geq 0}$ is a pre-Brownian motion*
2. *$(X_t)_{t\geq 0}$ is a centered Gaussian process with covariance $K(s,t) = s \wedge t$*
3. *$X_0 = 0$ a.s. and for every $0 \leq s < t$, the random variable $X_t - X_s$ is independent of $\sigma(X_t, r \leq s)$ and distributed according to $\mathcal{N}(0, t-s)$*
4. *$X_0 = 0$ and for every choice of $0 = t_0 < t_1 < \cdots < t_p$, the variables $X_{t_i} - X_{t_{i-1}}$, $1 \leq i \leq p$ are independent and for every $1 \leq i \leq p$, the variable $X_{t_i} - X_{t_{t-1}}$ are distributed according to $\mathcal{N}(0, t_i - t_{i-1})$.*

pf.

- $1 \to 2$ follows by the previous proposition
- $2 \to 3$
  Assume $(X_t)_{t\geq 0}$ to be a centered Gaussian process with covariance $K(s,t) = s \wedge t$ and let H be the Gaussian space generated by $(X_t)_{t\geq 0}$. Then X_0 is distributed according to $\mathcal{N}(0,0)$ and therefore $X_0 = 0$. Fix $s > 0$, and write H_s for the vector space spanned by $(X_r, 0 \leq r \leq x)$ and $\tilde{H}_s$ for the vector space spanned by $(X_{s+u} - X_s, u \geq 0)$. Then H_s and $\tilde{H}_s$ are orthogonal since $\forall r \in [0,s], \forall u \geq 0$, $E[X_r(X_{s+u} - X_s)] = r \wedge (s+u) - r \wedge s = r - r = 0$. Since $H_s, \tilde{H}_s$ are subspaces of H, by the theorem 1.1. , $\sigma(H_s), \sigma(\tilde{H}_s)$ are independent. In particular if we fix $t > s$, $X_t - X_s$ is independent of $\sigma(H_s) = \sigma(X_r, r \leq s)$. By the form of the covariance function, $X_t - X_s \sim \mathcal{N}(0, t-s)$.
- $3 \to 4$ By taking $s = t_{p-1}, t = t_p$ we obtain the independency of $X_{t_p} - X_{t_{p-1}}$ from $(X_{t_1}, \ldots, X_{t_{p-1}})$. By similar argument, $X_{t_i} - X_{t_{i-1}}$ are independent for all $1 \leq i \leq p$.
- $4 \to 1$
  X is a centered Gaussian process by the hypothesis. Then if f is a step function on $\mathbb{R}_{>0}$ of the form $f = \sum_{i=1}^n \lambda_i \mathbb{1}_{(t_{i-1}, t_i]}$ where $0 = t_0 < t_1 < \ldots, t_p$, we set $G(f) = \sum \lambda_i (X_{t_i} - X_{t_{i-1}})$. Suppose f, g are two step functions. Write $\begin{cases} f = \sum \lambda_i \mathbb{1}_{(t_{i-1}, t_i]} \\ g = \sum \mu_i \mathbb{1}_{(t_{i-1}, t_i]} \end{cases}$ with the same subdivision $0 = t_0 < \cdots < t_p$ for f, g. Then $E[G(f)G(g)] = \int_{\mathbb{R}_{\geq 0}} f(t)g(t)dt$ so that G is an isometry from the vector space of step functions on $\mathbb{R}_{>0}$ into the Gaussian space H generated by H. Since step functions are dense in $L^2(\mathbb{R}_{\geq 0}, \mathscr{B}(\mathbb{R}_{\geq 0}), dt)$, we can extend the mapping $f \mapsto G(f)$ to an isometry from $L^2(\mathbb{R}_{\geq 0}, \mathscr{B}(\mathbb{R}_{\geq 0}), dt)$ into the Gaussian space H. Finally, $G([0,t]) = X_t - X_0 = X_t$ by construction.

Corollary 2.2.1. *Let $(B_t)_{t\geq 0}$ be a pre-Brownian motion. Then for every choice of $0 = t_0 < \cdots < t_n$, the law of the vector $(B_{t_1},\ldots,B_{t_n})$ as density*

$$p(x_1,\ldots,x_n) = \frac{1}{(2\pi)^{n/2}\sqrt{t_1(t_2-t_1)\ldots(t_n-t_{n-1})}}\exp(-\sum\frac{(x_i-x_{i-1})^2}{2(t_i-t_{t-1})})$$

where $x_0 = 0$.

pf. The r.v.s $B_{t_1}, B_{t_2}-B_{t_1},\ldots,B_{t_n}-B_{t_{n-1}}$ are independent with respective distribution $\mathcal{N}(0,t_1),\mathcal{N}(0,t_2-t_1),\ldots,\mathcal{N}(0,t_n-t_{n-1})$. Hence $(B_{t_1},B_{t_2}-B_{t_1},\ldots,B_{t_n}-B_{t_{n-1}})$ has density
$q(y_1,\ldots,q_n) = \frac{1}{(2\pi)^{n/2}\sqrt{t_1(t_2-t_1)\ldots(t_n-t_{n-1})}}\exp(-\sum\frac{y_i^2}{2(t_i-t_{i-1})})$ and the change of variables $x_i = y_1+\cdots+y_i$, $\forall i\in\{1,\ldots,n\}$ completes the argument.

Proposition 2.2.3. *Let B be a pre-Brownian motion. Then*

1. *Symmetry : $-B$ is also a pre-Brownian motion*
2. *Invariance under Scaling : $\forall\lambda>0$, $B_t^\lambda = \frac{1}{\lambda}B_{\lambda^2 t}$ is also a pre-Brownian motion*
3. *Markov Property : $\forall s\geq 0$, the process $B_t^{(s)} = B_{s+t}-B_s$ is a pre-Brownian motion and is independent of $\sigma(B_r, r\leq s)$*

pf. The σ-field generated by $B^{(s)}$ is $\sigma(\tilde{H}_s)$ which is independent of $\sigma(H_s) = \sigma(B_r, r\leq s)$. By the previous proposition the independency of $B_{t_i}^{(s)}-B_{t_{i-1}}^{(s)} = B_{s+t_i}-B_{s+t_{i-1}}$ implies $B_t^{(s)}$ is a pre-Brownian motion.

Let B be a pre-Brownian motion and let G be the associated Gaussian white noise. If f is a step function, there is an explicit formula for $G(f)$ in terms of B; $\begin{array}{c} G(f) = \int_0^\infty f(s)dB_s \\ G(f\mathbb{1}_{[s,t]}) = \int_s^t f(r)dB_r \end{array}$. Then if $u<v$, $\int_u^v dB_s = G((u,v]) = G([0,v])-G([0,u]) = B_v - B_u$.
$\rightarrow f\mapsto\int_0^\infty f(s)dB_s$ is called the **Wiener integral w.r.t.**B and it is distributed according to
$\mathcal{N}(0,\int_0^\infty f(s)^2 ds)$.

Definition 2.2.2. *Let $(X_t)_{t\in T}$ be a random process with values in E. The sample paths of X are the mappings $t\mapsto X_t(\omega)$ obtained when fixing $\omega\in\Omega$. The sample paths of X thus form a collection of mappings from T to E, indexed by $\omega\in\Omega$.*

Definition 2.2.3. *Let $(X_t)_{t\in T},(\tilde{X}_t)_{t\in T}$ be two random processes indexed by the same index set T and with values in the same metric space E. We say $\tilde{X}$ is a modification of X if $\forall t\in T$, $P(\tilde{X}_t = X_t) = 1$.*

$\rightarrow \tilde{X}$ has the same finite-dimensional marginals as X. So if X is a pre-Brownian motion, so is $\tilde{X}$.

Definition 2.2.4. *The process $\tilde{X}$ is said to be indistinguishable from X if there exists a negligible subset N of Ω s.t. $\forall \omega \in \Omega \setminus N$, $\forall t \in T$, $\tilde{X}_t(\omega) = X_t(\omega)$, i.e. $\tilde{X}$ is indistinguishable from X if $P(\forall t \in T, X_t = \tilde{X}_t) = 1$.*

Indistinguishability implies modification, but not vice versa. But in the special setting, they are equivalent : suppose $T = I$ is an interval of $\mathbb{R}$. If the sample paths of $X, \tilde{X}$ are continuous, then $\tilde{X}$ is a modification of X iff $\tilde{X}$ is indistinguishable form X.

Lemma 2.2.1. *Let f be a mapping defined on D and with values in the metric space (E,d). Assume $\exists \alpha \in \mathbb{R}_{>0}$ and a constant $K < \infty$ s.t. for every integer $n \geq 1$, $\forall i \in \{1,2,\ldots,2^n - 1\}$, $d(f((i-1)2^{-n}), f(i2^{-n})) \leq K2^{-n\alpha}$. Then $\forall s,t \in D$*

$$d(f(s), f(t)) \leq \frac{2K}{1-2^{-\alpha}}|t-s|^{\alpha}$$

Theorem 2.2.1 (Kolmogorov's Lemma)**.** *Let $X = (X_t)_{t\in I}$ be a random process indexed by a bounded interval I of $\mathbb{R}$ and taking values in a complete metric space (E,d). Assume there exist three reals $q, \epsilon, C > 0$ s.t. $\forall s,t \in I$, $E[d(X_s,X_t)^q] \leq C|t-s|^{1+\epsilon}$. Then there is a modification $\tilde{X}$ of X whose sample paths are Holder continuous w.r.t. α for all $\alpha \in (0, \frac{\epsilon}{q})$, i.e. $\forall \omega \in \Omega$, $\forall \alpha \in (0, \frac{\epsilon}{q})$, $\exists C_\alpha(\omega) < \infty$ s.t. $\forall s,t \in I$, $d(\tilde{X}_s(\omega), \tilde{X}_t(\omega)) \leq C_\alpha(\omega)|t-s|^{\alpha}$. In particular $\tilde{X}$ is a modification of X with continuous sample paths.*

pf. To simplify the presentation, take $I = [0,1]$. Fix $\alpha \in (0, \frac{\epsilon}{q})$.

- continuity of sample paths
  By the hypothesis, $\forall \alpha > 0$, $\forall s,t \in I$, $P(d(X_s,X_t) \geq a) \leq a^{-q}E[d(X_s,X_t)^q] \leq Ca^{-q}|t-s|^{1+\epsilon}$. With $s = (i-1)2^{-n}$, $t = i2^{-n}$, $a = 2^{-n\alpha}$,
  $P(d(X_{(i-1)2^{-n}}, X_{i2^{-n}}) \geq 2^{-n\alpha}) \leq C2^{nq\alpha}2^{-(1+\epsilon)n}$. By summing over i,
  $P(\cup_{i=1}^{2n}\{d(X_{(i-1)2^{-n}}, X_{i2^{-n}}) \geq 2^{-n\alpha}\}) \leq 2^n C2^{nq\alpha-(1+\epsilon)n} = C2^{-n(\epsilon-q\alpha)}$. By the assumption, $\epsilon - q\alpha > 0$. By summing over n

  $$\sum_{n=1}^{\infty} P(\cup\{d(X_{(i-1)2^{-n}}, X_{i2^{-n}}) \geq 2^{-n\alpha}\}) < \infty$$

  By the Borel-Cantelli lemma, with probability one, we can find a finite integer $n_0(\omega)$ s.t.
  $\forall n \geq n_0(\omega)$, $\forall i \in \{1,\ldots,2^n\}$, $d(X_{(i-1)2^{-n}}, X_{i2^{-n}}) \leq 2^{-n\alpha}$. Then the constant

  $$K_\alpha(\omega) \equiv \sup_{n\geq 1}(\sup_{1\leq i\leq 2^n} \frac{d(X_{(i-1)2^{-n}}, X_{i2^{-n}})}{2^{-n\alpha}}) < \infty$$

a.s. By the previous lemma, on the event $\{K_\alpha(\omega) < \infty\}$, we have $\forall s,t \in D$, $d(X_s, X_t) \le C_\alpha(\omega)|t-s|^\alpha$ where $C_\alpha(\omega) = 2(1-2^{-\alpha})^{-1}K_\alpha(\omega)$. Thus on the event $\{K_\alpha(\omega) < \infty\}$, the mapping $t \mapsto X_t(\omega)$ is Holder continuous on D, hence uniformly continuous on D. Since (E,d) is complete, this mapping has a unique continuous extension to $I = [0,1]$ which is also Holder with exponent α. Thus $\forall t \in [0,1]$, $\tilde{X}_t(\omega) = \begin{cases} \lim_{s\to t, s\in D} X_s(\omega) & K_\alpha(\omega) < \infty \\ x_0 & K_\alpha(\omega) = \infty \end{cases}$ where x_0 is a point of E which can be fixed arbitrarily. Then it is a random variable. Consequently the sample paths of the process $\tilde{X}$ are Holder with exponent $\alpha \in [0,1]$.

- modification
  Fix $t \in [0,1]$. By the hypothesis $\lim_{s\to t} X_s = X_t$ in probability. By the definition, $\tilde{X}_t$ is the almost sure limit of X_s when $s \to t$, $s \in D$ and therefore $X_t = \tilde{X}_t$ a.s.

Corollary 2.2.2. *Let $B = (B_t)_{t\ge 0}$ be a pre-Brownian motion. The process B has a modification whose sample paths are continuous and even locally Holder continuous with exponent $\frac{1}{2} - \delta$ for all $\delta \in (0, \frac{1}{2})$.*

pf. If $s < t$, the random variable $B_t - B_s$ is distributed according to $\mathcal{N}(0, t-s)$ and thus $B_t - B_s$ has the same law as $\sqrt{t-s}U$ where $U \sim \mathcal{N}(0,1)$. Then $\forall q > 0$, $E[|B_t - B_s|^q] = (t-s)^{q/2}E[|U|^q] = C_q(t-s)^{q/2}$ where $C_q = E[|U|^q] < \infty$. Taking $q > 2$, then by the previous theorem with $\epsilon = \frac{q}{2} - 1$, B has a modification whose sample paths are locally Holder continuous with exponent α for every $\alpha < (q-2)/(2q)$.

2.3 Brownian Motion

Definition 2.3.1. *A process $(B_t)_{t\ge 0}$ is a Brownian motion if*

1. *$(B_t)_{t\ge 0}$ is a pre-Brownian motion*
2. *all sample paths of B are continuous*

e.g. **the Wiener measure**

Let $C(\mathbb{R}_{\ge 0}, \mathbb{R})$ be the space of all continuous functions from $\mathbb{R}_{\ge 0}$ to $\mathbb{R}$. Then get the σ-field $\mathscr{C}$ defined as the smallest σ-field on $C(\mathbb{R}_{\ge 0}, \mathbb{R})$ for which the coordinate mapping $w \mapsto w(t)$ are measurable for every $t \ge 0$. Given a Brownian motion B, we can consider the mapping $\begin{array}{ccc} \Omega & \to & C(\mathbb{R}_{\ge 0}, \mathbb{R}) \\ w & \mapsto & (t \mapsto B_t(w)) \end{array}$, which is measurable. The Wiener measure is the image of the probability measure $P(d\omega)$ under this mapping, denoted by $W(dw)$. Thus it is a probability measure on $C(\mathbb{R}_{\ge 0}, \mathbb{R})$ and for every measurable subset $A \subseteq C(\mathbb{R}_{\ge 0}, \mathbb{R})$, $W(A) = P(B_\cdot \in A)$ where $B_\cdot$ stands for the random continuous function $t \mapsto B_t(\omega)$.

$\to A = \{w \in C(\mathbb{R}_{\geq 0}, \mathbb{R}) : w(t_0) \in A_0, \ldots, w(t_n) \in A_n\}$ where $0 = t_0 < t_1 < \cdots < t_n$ and $A_0, \ldots, A_n \in \mathscr{B}(\mathbb{R})$. Then
$W(A) = P(B_{t_0} \in A_0, \ldots, B_{t_n} \in A_n) = \mathbb{1}_{A_0}(0) \int_{A_1 \times \cdots \times A_n} \frac{dx_1 \ldots dx_n}{(2\pi)^{n/2}\sqrt{t_1(t_2-t_1)\ldots(t_n-t_{n-1})}} \exp(-\sum \frac{(x_i - x_{i-1})^2}{2(t_i - t_{i-1})})$.
$\to$ the Wiener measure does not depend on the choice of the Brownian motion B.

$\Omega = C(\mathbb{R}_{\geq 0}, \mathbb{R})$, $\mathscr{F} = \mathscr{C}$, $P(dw) = W(dw)$. The canonical process $X_t(w) = w(t)$ is a Brownian motion defined on the probability space above, and this is the canonical construction of Brownian motion.

$$\mathscr{F}_t \equiv \sigma(B_s, s \leq t)$$

for every $t \geq 0$. Then $\mathscr{F}_s \subseteq \mathscr{F}_t$ if $s \leq t$. Set $\mathscr{F}_{0+} = \cap_{s>0} \mathscr{F}_s$.

Theorem 2.3.1 (Blumenthal's Zero-One Law)**.** *The σ-field $\mathscr{F}_{0+}$ is trivial in the sense that $P(A) = 0$ or 1 for every $A \in \mathscr{F}_{0+}$.*

pf. Let $0 < t_1 < \cdots < t_k$ and let $g : \mathbb{R}^j \to \mathbb{R}$ be a bounded continuous function. Fix $A \in \mathscr{F}_{0+}$. Then by the continuity $E[\mathbb{1}_A g(B_{t_1}, \ldots, B_{t_k})] = \lim_{\epsilon \to 0} E[\mathbb{1}_A g(B_{t_1} - B_\epsilon, \ldots, B_{t_k} - B_{t_\epsilon})]$. If $0 < \epsilon < t_1$, the variables $B_{t_i} - B_\epsilon$ are independent of $\mathscr{F}_\epsilon$ and therefore also of $\mathscr{F}_{0+}$. Then $E[\mathbb{1}_A g(B_{t_1}, \ldots, B_{t_k})] = \lim_{\epsilon \to 0} P(A) E[g(B_{t_1} - B_\epsilon, \ldots, B_{t_k} - B_\epsilon)] = P(A) E[g(B_{t_1}, \ldots, B_{t_k})]$. Thus $\mathscr{F}_{0+} \perp \sigma(B_{t_1}, \ldots, B_{t_k})$. This holds for any finite collection $\{t_1, \ldots, t_k\}$ of strictly positive reals, $\mathscr{F}_{0+}$ is independent of B_t when $t \to 0$. But $\sigma(B_t, t > 0) = \sigma(B_t, t \geq 0)$ since B_0 is the pointwise limit of B_t when $t \to 0$. Since $\mathscr{F}_{0+} \subseteq \sigma(B_t, t \geq 0)$, $\mathscr{F}_{0+}$ is independent of itself. Thus the theorem follows.

Proposition 2.3.1. *1. a.s. for every $\epsilon > 0$, $\sup_{0 \leq s \leq \epsilon} B_s > 0$, $\inf_{0 \leq s \leq \epsilon} B_s < 0$*

2. $\forall a \in \mathbb{R}$, let $T_a \equiv \inf\{t \geq 0, B_t = a\}$. Then a.s. $\forall a \in \mathbb{R}$, $T_a < \infty$. Consequently a.s. $\lim \sum B_t = +\infty$, $\liminf B_t = -\infty$.

pf.

- Let (ϵ_p) be a sequence of positive reals strictly decreasing to 0 and let $A = \cap_p \{\sup_{0 \leq s \leq \epsilon_p} B_s > 0\}$. Since it is a monotone decreasing intersection, A is $\mathscr{F}_{0+}$ measurable. On the other hand, $\begin{cases} P(A) = \lim \downarrow P(\sup_{0 \leq s \leq \epsilon_p} B_s > 0) \\ P(\sup_{0 \leq s \leq \epsilon_p} B_s > 0) \geq P(B_{\epsilon_p} > 0) = \frac{1}{2} \end{cases}$, i.e. $P(A) \geq 1/2$. By the Blumenthal's zero-one law, $P(A) = 1$ and therefore a.s. $\forall \epsilon > 0$, $\sup_{0 \leq s \leq \epsilon} B_s > 0$. The result for $\inf_{0 \leq s \leq \epsilon} B_s$ is obtained by replacing B by $-B$.

- $1 = P(\sup_{0 \leq s \leq 1} B_s > 0) = \lim_{\delta \downarrow 0} \uparrow P(\sup_{0 \leq s \leq 1} B_x > \delta)$. By the scale invariance property with $\lambda = \delta$, $\forall \delta > 0$, $P(\sup_{0 \leq s \leq 1} B_s > \delta) = P(\sup_{0 \leq s \leq \frac{1}{\delta^2}} B_s^\delta > 1)$

$= P(\sup_{0\le s\le \frac{1}{\delta^2}} B_s > 1)$ given the Wiener measure is the same for any Brownian motion. If we let $\delta \to 0$, $P(\sup_{s\ge 0} B_s > 1) = \lim_{\delta\downarrow 0} \uparrow P(\sup_{0\le s\le \frac{1}{\delta^2}} B_s > 1) = 1$.
Then again by scaling invariance property, for every $M > 0$,
$P(\sup_{s\le 0} B_s > M) = 1$. Replacing B by $-B$ gives $P(\int_{s\ge 0} B_s < -M) = 1$.
A continuous function $f : \mathbb{R}_{\ge 0} \to \mathbb{R}$ can visit all reals only if $\limsup_{t\to+\infty} f(t) = +\infty$ and $\liminf_{t\to+\infty} = -\infty$, so the last statement follows.

Corollary 2.3.1. *a.s. the function $t \mapsto B_t$ is not monotone on any nontrivial interval.*

pf. By the previous proposition 1 and Markov property, a.s. for every rational $q \in \mathbb{Q}_{\ge 0}$, $\epsilon > 0$, $\sup_{q\le t\le q+\epsilon} B_t > B_q$, $\inf_{q\le t\le q+\epsilon} B_t < B_q$. So the corollary follows.

Proposition 2.3.2. *Let $0 = t_0^n < t_1^n < \cdots < t_{p_n}^n$ be a sequence of subdivisions of $[0,t]$ whose mesh tends to 0. Then*

$$\lim_{n\to\infty} \sum_{i=1}^{p_n} (B_{t_i^n} - B_{t_{i--1}^n})^2 = t$$

in L^2.

pf. Writing $B_{t_i^n} - B_{t_{i-1}^n} = G((t_{i-1}^n, t_i^n])$, where G is the Gaussian white noise associated with B, the proposition follows.

Corollary 2.3.2. *a.s. the function $t \mapsto B_r$ has infinite variation on any nontrivial interval.*

pf. By the Markov property, reduce the case to the interval $[0,t]$ for some fixed $t > 0$. By the previous proposition, $\sum_{i=1}^{p_n} (B_{t_i^n} - B_{t_{i-1}^n})^2 \le (\sup_{1\le i\le p_n} |B_{t_i^n} - B_{t_{i-1}^n}|_\times \sum_{i=1}^{p_n} |B_{t_i^n} - B_{t_{i-1}^n}|$. The supremum inside tends to 0 by the continuity of sample paths. But the LHS tends to t a.s. Thus $\sum |B_{t_i^n} - B_{t_{i-1}^n}| \to \infty$ a.s.

2.3.1 The Strong Markov Property of Brownian Motion

Definition 2.3.2. *A random variable T with values in $[0,\infty]$ is a stopping time if for every $t \ge 0$, $\{T \le t\} \in \mathscr{F}_t$.*

$\to$ if T is a stopping time, $\forall t > 0$, $\{T < t\} = \cup_{q\in[0,t))\cap\mathbb{Q}} \{T \le q\} \in \mathscr{F}_t$

Definition 2.3.3. *Let T be a stopping time. The σ-field of the past before T is*

$$\mathscr{F}_T \equiv \{A \in \mathscr{F}_\infty : \forall t \ge 0,\ A \cap \{T \le t\} \in \mathscr{F}_t\}$$

$\rightarrow \mathscr{F}_T$ is a σ-field and the random variable T is $\mathscr{F}_T$-measurable.

Set $\mathbb{1}_{T<\infty}B_T(\omega) = \begin{cases} B_{T(\omega)}(\omega) & T(\omega) < \infty \\ 0 & T(\omega) = \infty \end{cases}$ Then $\mathbb{1}_{T<\infty}B_T = \lim_{n\to\infty}\sum_{i=0}^{\infty}\mathbb{1}_{\{i2^{-n}\le T<(i+1)2^n\}}B_{i2^{-n}}$ $= \lim\sum\mathbb{1}_{\{T<(i+1)2^{-n}\}}\mathbb{1}_{\{i2^{-n}\le T\}}B_{i2^{-n}}$ is $\mathscr{F}_T$-measurable. Also $\forall s \ge 0$, $B_s\mathbb{1}_{\{s\le T\}}$ is $\mathscr{F}_T$-measurable because if A is a Borel subset of $\mathbb{R}$ not containing 0, $\{B_s\mathbb{1}_{\{s\le T\}} \in A\} \cap \{T \le t\} = \begin{cases} \emptyset & t < s \\ \{B_s \in A\} \cap \{s \le T \le t\} & t \ge s \end{cases}$ which is $\mathscr{F}_t$-measurable in both cases.

Theorem 2.3.2 (Strong Markov Property)**.** *Let T be a stopping time. Assume $P(T<\infty) > 0$ and set $\forall t \ge 0$, $B_t^{(T)} = \mathbb{1}_{\{T<\infty\}}(B_{T+t} - B_T)$. Then under the probability measure $P(\cdot|T<\infty)$, the process $(B_t^{(T)})_{t\ge 0}$ is a Brownian motion independent of $\mathscr{F}_T$.*

pf.

- $T < \infty$ a.s.
  Fix $A \in \mathscr{F}_T$, $0 \le t_1 < \cdots < t_p$ and let F be a bounded continuous function from $\mathbb{R}^p$ to $\mathbb{R}_{\ge 0}$.

  - $E[\mathbb{1}_A F(B_{t_1}^{(T)}, \ldots, B_{t_p}^{(T)})] = P(A)E[F(B_{t_1}, \ldots, B_{t_p})]$
    pf. for every integer $n \ge 1$, and for every real $t \ge 0$, $[t]_n$ is the smallest real of the form $k2^{-n}$ with $k \in \mathbb{Z}_{\ge 0}$, belonging to the interval $[t, \infty)$. Set $[\infty]_n = \infty$ by convention. Since a.s. $F(B_{t_1}^{(T)}, \ldots, B_{t_p}^{(T)}) = \lim_{n\to\infty}F(B_{t_1}^{([T]_n)}, \ldots, B_{t_p}^{([T]_n)})$, by the dominated convergence
    $E[\mathbb{1}_A F(B_{t_1}^{(T)}, \ldots, B_{t_p}^{(T)})] = \lim E[\mathbb{1}_A F(B_{t_1}^{([T]_n)}, \ldots, B_{t_p}^{([T]_n)})]$
    $= \lim\sum_{k=0}^{\infty}E[\mathbb{1}_A\mathbb{1}_{\{(k-1)2^{-n}<T\le k2^{-n}\}}F(B_{k2^{-n}+t_1} - B_{k2^{-n}}, \ldots, B_{k2^{-n}+t_p} - B_{k2^{-n}})]$
    by the definition of $[T]_n$. Since $A \in \mathscr{F}_T$, $A \cap \{(k-1)2^{-n} < T \le k2^{-n}\} = (A \cap \{T \le k2^{-n}\}) \cap \{T \le (k-1)2^{-n}\}^c$ is $\mathscr{F}_{k2^{-n}}$-measurable. By the Markov property, $E[\mathbb{1}_{A\cap\{(k-1)2^{-n}<T\le k2^{-n}\}}F(B_{k2^{-n}+t_1}, \ldots, B_{k2^{-n}+t_p} - B_{k2^{-n}})] = P(A \cap \{(k-1)2^{-n} < T \le k2^{-n}\})E[F(B_{t_1}, \ldots, B_{t_p})]$. By summing them over k gives the above claim.

  Let$A = \Omega$. The process $(B_t^{(T)})_{t\ge 0}$ has the same finite-dimensional marginal distribution as B and is thus a Brownian motion. Then for every choice of $0 \le t < \cdots < t_p$, $(B_{t_1}^{(T)}, \ldots, B_{t_p}^{(T)})$ is independent of $\mathscr{F}_T$ and therefore $B^{(T)}$ is independent of $\mathscr{F}_T$.

- $P(T = \infty) > 0$
  $E[\mathbb{1}_{A\cap\{T<\infty\}}F(B_{t_1}^{(T)}, \ldots, B_{t_p}^{(T)})] = P(A \cap \{T < \infty\})E[F(B_{t_1}, \ldots, B_{t_p})]$ By the similar argement the theorem follows.

Theorem 2.3.3 (Reflection Principle)**.** *For every $t > 0$, set $S_t = \sup_{s \le t} B_s$. Then if $a \ge 0$, $b \in (-\infty, a]$, we have*

$$P(S_t \ge a, B_t \le b) = P(B_t \ge 2a - b)$$

In particular, S_t has the same distribution as $|B_t|$.

pf. The stopping time $T_a = \inf\{t \ge 0 : B_t = a\} < \infty$ a.s. by the previous proposition. Then $P(S_t \ge a, B_t \le b) = P(T_a \ge t, B_t \le b) = P(T_a \le t, B^{(T_a)}_{t-T_a} \le b - a)$. since $B^{(T_a)}_{t-T_a} = B_t - B_{T_a} = B_t - a$ on the event $\{T_a \le t\}$. Write $B' = B^{(T_a)}$ so that by the previous theorem B' is a Brownian motion independent of $\mathscr{F}_{T_a}$ hence in particular of T_a. Since B' has the same distribution as $-B'$, the pair (T_a, B') also has the same distribution as $(T_a, -B')$. Let $H = \{(s,w) \in \mathbb{R}_{\ge 0} \times C(\mathbb{R}_{\ge 0}, \mathbb{R}) : s \le t, w(t-s) \le b - a\}$. Then $P((T_a, B') \in H) = P((T_a, -B') \in H) = P(T_a \le t, -B^{(T_a)}_{t-T_a} \le b - a) = P(T_a \le t, B_t \ge 2a - b) = P(B_t \ge 2a - b)$ because $\{B_t \ge 2a - b\}$ is a.s. contained in $\{T_a \le t\}$.
$P(S_t \ge a) = P(S_t \ge a, B_t \ge a) + P(S_t \ge a, B_t \le a) = 2P(B_t \ge a) = P(|B_t| \ge a)$ so the last statement follows.
$\rightarrow (S_t, B_t)$ has the density

$$g(a,b) = \frac{2(2a-b)}{\sqrt{2\pi t^3}} \exp\left(-\frac{(2a-b)^2}{2t}\right) \mathbb{1}_{\{a>0, b<a\}}$$

Corollary 2.3.3. *For every $a > 0$, T_a has the same distribution as $\frac{a^2}{B_1^2}$ and has the density $f(t) = \frac{a}{\sqrt{2\pi t^3}} \exp(-\frac{a^2}{2t}) \mathbb{1}_{\{t>0\}}$.*

pf. By the previous theorem, for every $t \ge 0$, $P(T_a \le t) = P(S_t \ge a) = P(|B_t| \ge a) = P(B_t^2 \ge a^2) = P(tB_1^2 \ge a^2) = P(\frac{a^2}{B_1^2} \le t)$. Since $B_1 \sim \mathcal{N}(0,1)$, the corollary follows.

Definition 2.3.4. *If Z is a real r.v., a process $(X_t)_{t \ge 0}$ is a real Brownian motion started from Z if we can write $X_t = Z + B_t$ where B_t is a real Brownian motion started from 0 and is independent of Z.*

Definition 2.3.5. *A random process $B_t = (B_t^1, \ldots, B_t^d)$ with values in $\mathbb{R}^d$ is a d-dimensional Brownian motion started from 0 if its components $B^1, \ldots, B^d$ are independent real Brownian motions started from 0. If Z is a random variable with values in $\mathbb{R}^d$, $X_t = (X_t^1, \ldots, X_t^d)$ is a process with values in $\mathbb{R}^d$, we say X is a d-dimensional Brownian motion started from Z if we can write $X_t = Z + B_t$ where B is a d-dimensional Brownian motion started from 0 and is independent of Z.*

For every real $a \ge 0$, $T_a \equiv \inf\{t \ge 0 : B_t = a\}$. Then the process $(T_a)_{a \ge 0}$ has stationary independent increments.
pf. Let $0 < a < b$. $T_b - T_a = \inf\{t \ge s : B_t = b : B_s = a\}$ since $b > a$. Then $P(T_b - T_a \le$

$t) = P(S_t^s \geq b-a) = P(|B_t^{(s)}| \geq b-a) = P((B_t^s)^2 \geq (b-a)^2) = P(t(B_1^s)^2 \geq (b-a)^2) = P(\frac{a^2}{(B_1^s)^2} \leq t)$ where B_1^s is a Brownian motion by the Strong Markov property. This implies it is independent of $\mathscr{F}_s$ and it has the same distribution as $\frac{(b-a)^2}{(B_1^s)^2}$.

e.g.**Brownian Bridge**
Let $W_t = B_t - tB_1$, $\forall t \in [0,1]$. Then $E[W_tW_t] = E[B_tB_t - 2tB_1B_t + t^2B_1^2] = t - 2t^2 + t^2 = t - t^2$ its covariance is $t(1-t)$. It is a centered Gaussian process since $\forall t \in [0,1]$, it is spanned by B_t, B_1 which are centered Gaussian processes. Then by the proposition 1.2.2, it is a pre-Brownian motion and for $0 < t_1 < \cdots < t_p < 1$, its law $(W_{t_1}, W_{t_2} - W_{t_1}, \ldots, W_{1-t_p})$ has density

$$\frac{1}{\sqrt{2\pi t_1(1-t_1)}}\exp(-\frac{x_1^2}{2t_1(1-t_1)})\frac{1}{\sqrt{2\pi(t_2-t_1)(1-t_2+t_1)}}\exp(-\frac{(x_2-x_1)^2}{2(t_2-t_1)(1-t_2+t_1)})\ldots\frac{1}{\sqrt{2\pi t_p(1-t_p)}}\exp(-\frac{(-x_{t_p})^2}{2t_p(1-t_p)})$$
$$= p_{t_1}(x_1)p_{t_2-t_1}(x_2-x_1)\ldots p_{1-t_p}(-x_{t_p})\frac{1}{\sqrt{1-t_1}}\exp(-\frac{(x_1)^2}{2(1-t_1)})\frac{1}{\sqrt{1-t_2+t_1}}\exp(-\frac{(x_2-x_1)^2}{1-t_2+t_1})\ldots\frac{1}{\sqrt{t_p}}\exp(-\frac{(x_p)^2}{2t_p})$$
$$= p_{t_1}(x_1)p_{t_2-t_1}(x_2-x_1)\ldots p_{1-t_p}(-x_{t_p})(\sqrt{2\pi})^{p+1}p_{1-t_1}(x_1)p_{1-t_2+t_1}(x_2-x_1)\ldots p_{t_p}(-x_{t_p}).$$

This density is equivalent for $(W_{1-t_p}, \ldots, W_{1-t_1})$ by the symmetry of the density function. Thus we have the same distribution for W_t and W_{1-t}, $\forall t \in [0,1]$.
$\rightarrow$ the law of $(W_{t_1}, \ldots, W_{t_p})$ can be interpreted as the conditional law of $(B_{t_1}, \ldots, B_{t_p})$ with $B_1 = 0$ since each of W_{t_i} is a Brownian motion and therefore the distribution laws are equivalent with $W_1 = 0$.

Chapter 3

Martingales

3.1 Filtrations

Definition 3.1.1. *A filtration on* $(\Omega, \mathscr{F}, P)$ *is a collection* $(\mathscr{F}_t)_{0 \le t \le \infty}$ *indexed by* $[0, \infty]$ *of sub-*σ*-fields of* $\mathscr{F}$ *s.t.* $\mathscr{F}_s \subset \mathscr{F}_t, \ \forall s \le t \le \infty$.

e.g. If B is a Brownian motion, the filtration $\mathscr{F}_t = \sigma(B_s, 0 \le s \le t)$, $\mathscr{F}_\infty = \sigma(B_s, s \ge 0)$.

If $X = (X_t, \ t \ge 0)$ is any random process indexed by $\mathbb{R}_{\ge 0}$, the canonical filtration of X is defined by $\mathscr{F}_t^X = \sigma(X_s, 0 \le s \le t)$, $\mathscr{F}_\infty^X = \sigma(X_s, s \ge 0)$.

Let $(\mathscr{F}_t)_{0 \le t \le \infty}$ be a filtration on $(\Omega, \mathscr{F}, P)$. Set for every $t \ge 0$, $\begin{cases} \mathscr{F}_{t+} = \cap_{s > t} \mathscr{F}_s \\ \mathscr{F}_{\infty +} = \mathscr{F}_\infty \end{cases}$
Then $\forall t \in [0, \infty]$, $\mathscr{F}_t \subset \mathscr{F}_{t+}$ and the collection $\mathscr{F}_{t+}$ is also a filtration.

Definition 3.1.2. *A filtration* $(\mathscr{F}_t)$ *is right-continuous if* $\mathscr{F}_{t+} = \mathscr{F}_t, \ \forall t \ge 0$.

$\rightarrow \mathscr{F}_{t+}$ is right-continuous by construction.

Let $(\mathscr{F}_t)$ be filtration and let $\mathscr{N}$ be the class of all $(\mathscr{F}_\infty, P)$-negligible sets, i.e. $A \in \mathscr{N}$ if there exists $A' \in \mathscr{F}_\infty$ s.t. $A \subset A'$ and $P(A') = 0$.

Definition 3.1.3. *A filtration is said to be complete if* $\mathscr{N} \subset \mathscr{F}_0$.

$\rightarrow$ if $(\mathscr{F}_t)$ is not complete, it can be completed by setting $\mathscr{F}' = \mathscr{F}_t \wedge \sigma(\mathscr{N})$ for all $t \in [0, \infty]$ using the notation $\mathscr{F}_t \wedge \sigma(\mathscr{N})$ for the smallest σ-field that contains both of them. The resulting filtration is called the completed canonical fitration of X.

Definition 3.1.4. *A process* $X = (X_t)_{t \ge 0}$ *with values in a measurable space* $(E, \mathscr{E})$ *is said to be measurable if the mapping* $(\omega, t) \mapsto X_t(\omega)$ *defined on* $\Omega \times \mathbb{R}_+$ *equipped with the product* σ*-field* $\mathscr{F} \otimes \mathscr{B}(\mathbb{R}_+)$ *is measurable.*

→This is stronger than saying $\forall t \geq 0$, X_t is $\mathscr{F}$-measurable. But if the sample paths of X are right-continuous, the fact that X_t is $\mathscr{F}$-measurable for every t implies the process is measurable in the above definition-sense.

Fix a filtration $(\mathscr{F}_t)$ on $(\Omega, \mathscr{F}, P)$.

Definition 3.1.5. *A random process $(X_t)_{t\geq 0}$ with values in a measurable space $(E, \mathscr{E})$ is called adapted if $\forall t \geq 0$, X_t is $\mathscr{F}_t$-measurable. This process is said to be progressive if $\forall t \geq 0$, the mapping $(\omega, s) \mapsto X_s(\omega)$ defined on $\Omega \times [0,t]$ is measurable for the σ-field $\mathscr{F}_t \otimes \mathscr{B}([0,t])$.*

→ a progressive process is both adapted and measurable.

Proposition 3.1.1. *Let $(X_t)_{t\geq 0}$ be a random process with values in a metric space (E,d) equipped with its Borel σ-field. Suppose X is adapted and the sample paths of X are right-continuous. Then X is progressive. The same conclusion holds if one replaces right-continuous by left-continuous.*

pf. Fix $t > 0$. $\forall n \geq 1, \forall s \in [0,t]$, define a r.v. X_s^n by setting $X_s^n = X_{kt/n}$ if $s \in [(k-1)t/n, kt/n)$, $k \in \{1,2,\ldots,n\}$ and $X_t^n = X_t$. The right-continuity of sample paths ensures $\forall s \in [0,t]$, $\forall \omega \in \Omega$, $X_s(\omega) = \lim_{n\to\infty} X_s^n(\omega)$. On the other hand, for every Borel subset A of E

$$\begin{aligned} &\{(\omega, s) \in \Omega \times [0,t] : X_s^n(\omega) \in A\} \\ &= (\{X_t \in A\} \times \{t\}) \cup (\cup_{k=1}^n (\{X_{kt/n} \in A\} \times [\frac{(k-1)t}{n}, \frac{kt}{n})) \end{aligned}$$

which belongs to the σ-field $\mathscr{F}_t \otimes \mathscr{B}([0,t])$. Hence $\forall n \geq 1$, $(\omega, s) \mapsto X_s^n(\omega)$ defined on $\Omega \times [0,t]$ is measurable for $\mathscr{F}_t \otimes \mathscr{B}([0,t])$. Since a pointwise limit of measurable functions is also measurable, $(\omega, s) \mapsto X_t(\omega)$ defined on $\Omega \times [0,t]$ is also measurable. Thus X is progressive.

e.g. the collection $\mathscr{P}$ of all sets $A \in \mathscr{F} \otimes \mathscr{B}(\mathbb{R}_+)$ s.t. the process $X_t(\omega) = \mathbb{1}_A(\omega, t)$ is progressive forms a σ-field on $\Omega \times \mathbb{R}_+$, called the progressive σ-field. Then $A \subset \Omega \times \mathbb{R}_\times$ belongs to $\mathscr{P}$ iff $\forall t \geq 0$, $A \cap (\Omega \times [0,t])$ belongs to $\mathscr{F}_t \otimes \mathscr{B}([0,t])$.

Definition 3.1.6. *A random variable $T : \Omega \to [0,\infty]$ is a stopping time of the filtration $(\mathscr{F}_t)$ if $\{T \leq t\} \in \mathscr{F}_t$ for every $t \geq 0$. The σ-field of the past before T is then defined by*

$$\mathscr{F}_T = \{A \in \mathscr{F}_\infty : \forall t \geq 0,\ A \cap \{T \leq t\} \in \mathscr{F}_t\}$$

→ $\mathscr{F}_T$ is indeed a σ-field.

If T is a stopping time, $\{T < t\} \in \mathscr{F}_t$, $\forall t > 0$ by the same argument as in the Brownian case, and $\{T = \infty\} = (\cup_{n=1}^\infty \{T \leq n\})^c \in \mathscr{F}_\infty$.

Proposition 3.1.2. *Write* $\mathscr{G}_t = \mathscr{F}_{t+}$, $\forall t \in [0,\infty]$

1. *A r.t.* $T : \Omega \to [0,\infty]$ *is a stopping time of the filtration* $(\mathscr{G}_t)$ *iff* $\{T < t\} \in \mathscr{F}_t$, $\forall t > 0$. *This is also equivalent to saying that* $T \vee t$ *is* $\mathscr{F}_t$*-measurable* $\forall t > 0$.

2. *Let* T *be a stopping time of the filtration* $(\mathscr{G}_t)$. *Then* $\mathscr{G}_T = \{A \in \mathscr{F}_\infty : \forall t > 0,\ A \cap \{T < t\} \in \mathscr{F}_t\}$

pf.

- Suppose T is a stopping time of the filtration $(\mathscr{G}_t)$. Then $\forall t > 0$, $\{T < t\} = \cap_{q \in \mathbb{Q}_{\geq 0}, q<t}\{T \leq q\} \in \mathscr{F}_t$ because $\{T \leq q\} \in \mathscr{G}_q \subset \mathscr{F}_t$ if $q < t$. Conversely assume $\{T < t\} \in \mathscr{F}_t$, $\forall t > 0$. Then $\forall t \geq 0$, $\forall s > t$, $\{T < t\} = \cap_{q \in \mathbb{Q}_{\geq 0}, t<q<s}\{T < q\} \in \mathscr{F}_s$ and therefore $\{T \leq t\} \in \mathscr{F}_{t+} = \mathscr{F}_t$.
  Then, $T \vee t$ is $\mathscr{F}_t$-measurable for every $t > 0$ is equivalent to saying $\forall s < t$, $\{T \leq s\} \in \mathscr{F}_t$. Taking a sequence of values of s that increases to t, the latter property implies $\{T < t\} \in \mathscr{F}_t$ and therefore T is a stopping time of the filtration $(\mathscr{G}_t)$. Conversely if T is a stopping time of the filtration $(\mathscr{G}_t)$, $\{T \leq s\} \in \mathscr{G}_s \subset \mathscr{F}_t$ whenever $s < t$ and therefore $T \vee t$ is $\mathscr{F}_t$-measurable.

- if $A \in \mathscr{F}_T$, $A \cap \{T \leq t\} \in \mathscr{G}_t$, $\forall t \geq 0$. So $\forall t > 0$, $A \cap \{T < t\} = \cup_{q \in \mathbb{Q}_{\geq 0}, q<t}(A \cap \{T \leq q\}) \in \mathscr{F}_t$ since $A \cap \{T \leq q\} \in \mathscr{G}_q \subset \mathscr{F}_t$, $\forall q < t$. Conversely, assume $A \cap \{T < t\} \in \mathscr{F}_t$, $\forall t > 0$. Then $\forall t \geq 0$, $s > t$, $A \cap \{T \leq t\} = \cap_{q \in \mathbb{Q}_{\geq 0}, t<q<s}(A \cap \{T < q\}) \in \mathscr{F}_s \to A \cap \{T \leq t\} \in \mathscr{F}_{t+} = \mathscr{G}_t$.

- **Properties of Stopping Times and of the Associated σ-fields.**

  1. **for every stopping time T, $\mathscr{F}_T \subset \mathscr{F}_{T+}$. If the filtration $(\mathscr{F}_t)$ is right-continuous, $\mathscr{F}_{T+} = \mathscr{F}_T$.**
  2. **if $T = t$ is a constant stopping time, $\mathscr{F}_T = \mathscr{F}_t$, $\mathscr{F}_{T+} = \mathscr{F}_{t+}$.**
  3. **let T be a stopping time. Then T is $\mathscr{F}_T$-measurable.**
  4. **let T be a stopping time, and $A \in \mathscr{F}_\infty$. Set $T^A(\omega) = \begin{cases} T(\omega) & \omega \in A \\ +\infty & w \notin A \end{cases}$.**
     Then $A \in \mathscr{F}_T$ iff T^A is a stopping time.
     pf. $\forall t \geq 0$, $\{T^A \leq t\} = A \cap \{T \leq t\}$ and the result follows from the definition of $\mathscr{F}_T$.
  5. **let S, T be two stopping times s.t. $S \leq T$. Then $\mathscr{F}_S \subset \mathscr{F}_T$, $\mathscr{F}_{S+} \subset \mathscr{F}_{T+}$.**
     pf. $\forall A \in \mathscr{F}_S$, $A \cap \{T \leq t\} = (A \cap \{S \leq t\}) \cap \{T \leq t\} \in \mathscr{F}_t \to A \in \mathscr{F}_T$. So $\mathscr{F}_S \subset \mathscr{F}_T$ and the latter follows.

6. **let S,T be two stopping times. Then $S \vee T$, $S \wedge T$ are also stopping times and $\mathscr{F}_{S\vee T} = \mathscr{F}_S \cap \mathscr{F}_T$. Furthermore $\{S \le T\}, \{S = T\} \in \mathscr{F}_{S\vee T}$.**
pf. $\begin{array}{l} \{S \vee T \le t\} = \{S \le t\} \cup \{T \le t\} \in \mathscr{F}_t \\ \{S \wedge T \le t\} = \{S \le t\} \cap \{T \le t\} \in \mathscr{F}_t \end{array}$ so that they are stopping times.
By the previous statement, $\mathscr{F}_{S\vee T} \subset (\mathscr{F}_S \cap \mathscr{F}_T)$. If $A \in \mathscr{F}_S \cap \mathscr{F}_T$, $A \cap \{S \vee T \le t\} = (A \cap \{S \le t\}) \cup (A \cap \{T \le t\}) \in \mathscr{F}_t \to A \in \mathscr{F}_{S\vee T} \to \mathscr{F}_{S\vee T} = \mathscr{F}_S \cup \mathscr{F}_T$.
$\forall t \ge 0$, $\begin{array}{r} \{S \le T\} \cap \{T \le t\} = \{S \le t\} \cap \{T \le t\} \cap \{S \vee t \le T \vee t\} \in \mathscr{F}_t \\ \{S \le T\} \cap \{S \le t\} = \{S \vee t \le T \vee t\} \cap \{S \le t\} \in \mathscr{F}_t \end{array}$. Then $\{S \le T\} \in \mathscr{F}_S \cap \mathscr{F}_T = \mathscr{F}_{S\vee T}$. Additionally $\{S = T\} = \{S \le T\} \cap \{T \le S\}$ and the statement follows.

7. **if (S_n) is a monotone increasing sequence of stopping times, then $S = \lim \uparrow S_n$ is also a stopping time.**
pf. $\forall t \ge 0$, $\{S \le t\} = \cap_n \{S_n \le\} \in \mathscr{F}_t$

8. **if (S_n) is a monotone decreasing sequence of stoping times, then $S = \lim \downarrow S_n$ is a stopping time of the filtration $(\mathscr{F}_{t+})$ and $\mathscr{F}_{S+} = \cap_n \mathscr{F}_{S_n+}$.**
pf. $\forall t \ge 0$, $\{S < t\} = \cup_n \{S_n < t\} \in \mathscr{F}_t$. Then by the previous argument 5, $\mathscr{F}_{S+} \subset \mathscr{F}_{S_n+}$, $\forall n$. Conversely, if $a \in \cap_n \mathscr{F}_{S+}$, $A \cap \{S < t\} = \cup_n (A \cap \{S_n < t\}) \in \mathscr{F}_t \to A \in \mathscr{F}_{S+}$.

9. **if (S_n) is a monotone decreasing sequence of stopping times, which is also stationary($\forall \omega$, $\exists N(\omega)$ s.t. $S_n(\omega) = S(\omega)$, $\forall n \ge N(\omega)$), then $S = \lim \downarrow S_n$ is also a stopping time and $\mathscr{F}_S = \cap_n \mathscr{F}_{S_n}$.**
pf. $\forall t \ge 0$, $\{S \le t\} = \cup_n \{S_n \le t\} \in \mathscr{F}_t$. If $A \in \cap_n \mathscr{F}_{S_n}$, $A \cap \{S \le t\} = \cup_n (A \cap \{S_n \le t\}) \in \mathscr{F}_t \to A \in \mathscr{F}_S$

10. **let T be a stopping time. A function $\omega \mapsto Y(\omega)$ defined on the set $\{T < \infty\}$ and taking values in the measurable set $(E, \mathscr{E})$ is $\mathscr{F}_T$-measurable iff $\forall t \ge 0$, the restriction of Y to the set $\{T \le t\}$ is $\mathscr{F}_t$-measurable.**
pf. Assume $\forall t \ge 0$, the restriction of Y to $\{T \le t\}$ is $\mathscr{F}_t$-measurable. Then for every measurable subset A of E, $\{Y \in A\} \cap \{T \le t\} \in \mathscr{F}_t$. Letting $t \to \infty$ $\{Y \in A\} \in \mathscr{F}_\infty \to \{Y \in A\} \in \mathscr{F}_T$. Conversely, if Y is $\mathscr{F}_T$-measurable, $\{Y \in A\} \in \mathscr{F}_T \to \{Y \in A\} \cap \{T \le t\} \in \mathscr{F}_t$.

Theorem 3.1.1. *Let $(X_t)_{t\ge 0}$ be a progressive process with values in a measurable space $(E, \mathscr{E})$ and let T be a stopping time. Then the function $\omega \mapsto X_T(\omega) \equiv X_{T(\omega)}(\omega)$, which is defined on the event $\{T < \infty\}$ is $\mathscr{F}_T$-measurable.*

pf. Let $t \ge 0$. The restriction to $\{T \le t\}$ of the function $\omega \mapsto X_T(\omega)$ is the composition of two mappings $\begin{array}{ccc} \mathscr{F}_t & \to & \mathscr{F}_t \otimes \mathscr{B}([0,t]) \\ \omega & \mapsto & (\omega, T(\omega) \wedge t) \end{array}$ $\begin{array}{ccc} \mathscr{F}_t \otimes \mathscr{B}([0,t]) & \to & \mathscr{E} \\ (\omega, s) & \mapsto & X_s(\omega) \end{array}$ which are

both measurable. Then its restriction to $\{T \le t\}$ is $\mathscr{F}_t$-measurable by the property 10 above.

Proposition 3.1.3. *Let T be a stopping time and let S be an $\mathscr{F}_T$-measurable random variable with values in $[0,\infty]$ s.t. $S \ge T$. Then S is also a stopping time. In particular if T is a stopping time,*

$$T_n = \sum_{k=0}^{\infty} \frac{k+1}{2^n} \mathbb{1}_{\{k2^{-n} < T \le (k+1)2^{-n}\}} + \infty \cdot \mathbb{1}_{T=\infty}$$

defines a sequence of stopping times that decreases to T.

pf. $\forall t \ge 0,\ \{S \le t\} = \{S \le t\} \cap \{T \le t\} \in \mathscr{F}_t$ since $\{S \le t\}$ is $\mathscr{F}_T$-measurable. Since $T_n \ge T$ and T_n is a function of T, $\mathscr{F}_T$-measurable and $T_n \downarrow T$ as $n \uparrow \infty$ by construction.

Proposition 3.1.4. *Let $(X_t)_{t\ge 0}$ be an adapted process with values in a metric space (E,d).*

1. *assume that the sample paths of X are right-continuous and let O be an open subset of E. Then $T_O = \inf\{t \ge 0 : X_t \in O\}$ is a stopping time of the filtration $(\mathscr{F}_{t+})$.*

2. *Assume that the sample paths of X are continuous, and let F be a closed subset of E. Then $T_F = \inf\{t \ge 0 : X_t \in F\}$ is a stopping time of the filtration $(\mathscr{F}_t)$.*

pf.

- $\forall t > 0,\ \{T_O < t\} = \cup_{s\in[0,g)\cap\mathbb{Q}}\{X_s \in O\} \in \mathscr{F}_t$ so the statement follows.
- $\forall t \ge 0,\ \{T_F \le t\} = \{\inf_{0\le s\le t} d(X_s,F) = 0\} = \{\inf_{s\in[0,t]\cap\mathbb{Q}} d(X_s,F) = 0\} \in \mathscr{F}_t$

3.2 Continuous Time Martingales

Fix a filtered probability space $(\Omega,\mathscr{F},(\mathscr{F}_t),P)$.

Definition 3.2.1. *An adapted real-valued process $(X_t)_{t\ge 0}$ s.t. $X_t \in L^1,\ \forall t \ge 0$ is called*

- *a martingale if $\forall 0 \le s < t,\ E[X_t|\mathscr{F}_s] = X_s$*
- *a supermartingale if $\forall 0 \le s < t,\ E[X_t|\mathscr{F}_s] \le X_s$*
- *a submartingale if $\forall 0 \le s < t,\ E[X_t|\mathscr{F}_s] \ge X_s$*

$\rightarrow (X_t)_{t\geq 0}$ is a submartingale$\Leftrightarrow (-X_t)_{T\geq 0}$ is a supermartingale.

e.g. For a random variable $Z \in L^1$, set $X_t = E[Z|\mathscr{F}_t]$, $\forall t \geq 0$. Then we get a martingales.

Definition 3.2.2. *A real-valued process $B = (B_t)_{t\geq 0}$ is an $(\mathscr{F}_t)$-Brownian motion if B is a Brownian motion and if B is adapted and has independent increments w.r.t, $(\mathscr{F}_t)$. Similarly, a process $B = (B_t)_{t\geq 0}$ with values in $\mathbb{R}^d$ is a d-dimensional $(\mathscr{F}_t)$-Brownian motion if B is a d-dimensional Brownian motion and if B is adapted and has independent increments w.r.t. $(\mathscr{F}_t)$.*

$\rightarrow$ if B is a Brownian motion, $(\mathscr{F}_t^B)$ is the completed canonical filtration of B, then B is a $(\mathscr{F}_t^B)$-Brownian motion.

e.g.

1. $\forall f \in L^2(\mathbb{R}_{\geq 0}, \mathscr{B}(\mathbb{R}_{\geq 0}), dt)$, $Z_t = \int_0^t f(x)dB_s$. Then Z has independent increments w.r.t. the canonical filtration of B and thus $\int_0^t f(s)dB_s$ is a martingale.

2. $Z = N$ is a Poisson process with parameter λ and $(\mathscr{F}_t)$ is the canonical filtration of B, then Z has independent increments, and $N_t - \lambda t$ is a Martingale.

Proposition 3.2.1. *Let $(X_t)_{t\geq 0}$ be an adapted process and let $f : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ be a convex function s.t. $E[f(X_t)] < \infty$, $\forall t \geq 0$.*

1. *if $(X_t)_{t\geq 0}$ is a martingale, $(f(X_t))_{t\geq 0}$ is a submartingale.*

2. *if $(X_t)_{t\geq 0}$ is a submartingale and if in addition f is nondecreasing, then $(f(X_t))_{t\geq 0}$ is a submartingale.*

pf. By Jensen's inequality, $\forall s < t$, $E[f(X_t)|\mathscr{F}_s] \geq f(E[X_t|\mathscr{F}_s) \geq f(X_s)$ given f is nondecreasing.
$\rightarrow$ if $(X_t)_{t\geq 0}$ is a martingale, $\forall p \geq 1$, $|X_t|$ is a submartingale, provided that $E[|X|^p] < \infty$, $\forall t \geq 0$.

Proposition 3.2.2. *Let $(X_t)_{t\geq 0}$ be a submartingale or a supermartingale. Then $\forall t > 0$, $\sup_{0\leq s\leq t} E[|X_s|] < \infty$.*

pf. Consider $(X_t)_{t\geq 0}$ a submartingale. Since $(X_t)^+$ is also a submartingale, $\forall s \in [0,t]$, $E[(X_s)^+] \leq E[(X_t)^+]$. On the other hand, since it is a submartingale, we also have $\forall s \in [0,t]$, $E[X_s] \geq E[X_0]$. By combining above two bounds $\sup_{s\in[0,t]} E|X_s| \leq 2E[(X_t)^+] - E[X_0] < \infty$ given $|x| = 2x^+ - x$. So the proposition follows. By inverting the sign, we get a supermartingale case.

Proposition 3.2.3. *Let $(M_t)_{t\geq 0}$ be a square integrable martingale. Let $0 \leq s < t$ and let $s = t_0 < t_1 < \ldots, < t_p = t$ be a subdivision of the interval $[s,t]$. Then*

$$E[\sum_{i=1}^{p}(M_{t_i} - M_{t_{i-1}})^2|\mathscr{F}_s] = E[M_t^2 - M_s^2|\mathscr{F}_s] = E[(M_t - M_s)^2|\mathscr{F}_s]$$

In particular, $E[\sum_{i=1}^{p}(M_{t_i} - M_{t_{i-1}})^2] = E[M_t^2 - M_s^2] = E[(M_t - M_s)^2]$.

pf. $\forall i$, $E[(M_{t_i} - M_{t_{i-1}})^2|\mathscr{F}_s] = E[E[(M_{t_i} - M_{t_{i-1}})^2|\mathscr{F}_{t_{i-1}}]|\mathscr{F}_S] = E[E[M_{t_i}^2|\mathscr{F}_{t_{i-1}}] - 2M_{t_{i-1}}E[M_{t_i}|\mathscr{F}_{t_{i-1}}] + M_{t_{i-1}}^2|\mathscr{F}_s] = E[E[M_{t_i}^2|\mathscr{F}_{t_{i-1}}] - M_{t_{i-1}}^2|\mathscr{F}_s] = E[M_{t_i}^2 - M_{t_{i-1}}^2|\mathscr{F}_s$ by the tower property. Summing over i gives the proposition.

Maximal Inequality for Discrete Supermartingales : if $(Y_n)_{n\in\mathbb{N}}$ is a supermartingales, $\forall\lambda > 0$, $\forall k \in \mathbb{N}$

$$\lambda P(\sup_{n\leq k}|Y_n| > \lambda) \leq E[|Y_0|] + 2E[|Y_k|]$$

Doob's inequality in L^p for Discrete Supermartingales : if $(Y_n)_{n\in\mathbb{N}}$ is a supermartingales, $\forall k \in \mathbb{N}$, $\forall p > 1$,

$$E[\sup_{0\leq n\leq k}|Y_n|^p] \leq (\frac{p}{p-1})^p E[|Y_k|^p]$$

Proposition 3.2.4. 1. ***Maximal Inequality***

let $(X_t)_{t\geq 0}$ be a supermartingale with right-continuous sample paths. Then $\forall t > 0, \forall\lambda > 0$,

$$\lambda P(\sup_{0\leq s\leq t}|X_s| > \lambda) \leq E[|X_0|] + 2E[|X_t|]$$

2. ***Doob's Inequality in L^p***

let $(X_t)_{t\geq 0}$ be a martingale with right-continuous sample paths. Then $\forall t > 0, \forall p > 1$,

$$E[\sup_{0\leq s\leq t}|X_s|^p] \leq (\frac{p}{p-1})^p E[|X_t|^p]$$

.

pf.

- fix $t > 0$ and consider a countable dense subset D of $\mathbb{R}_{\geq 0}$ s.t. $0, t \in D$. Then $D \cap [0,t]$ is the increasing union of a sequence $(D_m)_{m\geq 1}$ of finite subsets $[0,t]$ of the form $D_m = \{t_p^m, \ldots, t_m^m\}$ where $0 = t_0^m < \cdots < t_m^m = t$. For every fixed m, by the discrete time maximal inequality to the sequence $Y_n = X_{t_{n\wedge m}}$, which is a discrete supermartingales w.r.t. the filtration $\mathscr{G}_n = \mathscr{F}_{t_{n\wedge m}}$. Then $\lambda P(\sup_{s\in D_m}|X_s| > \lambda) \leq E[|X_0|] + 2E[|X_t|]$. Since $P(\sup_{s\in D_m}|X_s| > \lambda) \uparrow P(\sup_{s\in D\cap[0,t]}|X_s| > \lambda)$ as $m \uparrow \infty$, we have $\lambda P(\sup_{s\in D\cap[0,t]}|X_s| > \lambda) \leq E[|X_0|] + 2E[|X_t|]$. By the right-continuity of sample paths, $\sup_{s\in D\cap[0,t]}|X_s| = \sup_{s\in[0,t]}|X_s|$ and the proposition follows.

- in the similar setting with above, $\forall m \geq 1$, $E[\sup_{s\in D_m}|X_s|^p] \leq (\frac{p}{p-1})^p E[|X_t|^p]$ by the Doob's inequality in L^p for discrete martingales. Letting m tend to infinity and using monotone convergence theorem gives the statement.

$\rightarrow$ even when the sample paths of the supermaringales are not right-continuous, for every countable dense subset D of $\mathbb{R}_{\geq 0}$, $\forall t > 0$, $P(\sup_{s\in D\cap[0,t]}|X_s| > \lambda) \leq \frac{1}{\lambda}(E[|X_0|]+ 2E[|X_t|]$ and letting $\lambda \rightarrow \infty$, $\sup_{s\in D\cap[0,t]}|X_s| < \infty$ a.s.

Definition 3.2.3. *Let $f : I \rightarrow \mathbb{R}$ be a function defined on a subset I of $\mathbb{R}_{\geq 0}$. If $a < b$, the upcrossing number of f along $[a,b]$, denoted by $M^f_{ab}(I)$ is the maximal integer $k \geq 1$ s.t. $\exists$ a finite increasing sequences $s_1 < t_1 < \cdots < s_l < t_k$ of elements of I s.t.* $\begin{cases} f(s_i) \leq a \\ f(t_i) \geq b \end{cases}$, $\forall i \in \{1,\ldots,k\}$.

$\rightarrow$ $g : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ is said to be cadlag if g is right-continuous and has left-limits at every $t > 0$.

Lemma 3.2.1. *Let E be a countable dense subset of $\mathbb{R}_{\geq 0}$ and let f be a real function defined on D. We assume $\forall T \in D$,*

- *the function f is bounded on $D \cap [0,T]$*
- *for all rationals a,b s.t. $a < b$, $M^f_{ab}(D \cap [0,T]) < \infty$*

Then the right-limit

$$f(t+) \equiv \lim_{s\downarrow\downarrow t, s\in D} f(s)$$

exists for every real $t \geq 0$. Similarly the left-limit $f(t-) \equiv \lim_{s\uparrow\uparrow t, s\in D} f(x)$ exists for every real $t > 0$. Furthermore, the function $g : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ defined by $g(t) = f(t+)$ is cadlag.

Doob's Upcrossing Inequality for Discrete Supermartingales : if $(Y_n)_{n\in\mathbb{N}}$ is a supermartingale, then for every $n \in \mathbb{N}$, $\forall a < b$,

$$E[M^Y_{ab}(n)] \leq \frac{1}{b-a} E[(Y_n - a)^-]$$

Theorem 3.2.1. *Let $(X_t)_{t\geq 0}$ be a supermartingale, and let D be a countable dense subset of $\mathbb{R}_{\geq 0}$.*

1. *for almost every $\omega \in \Omega$, the restriction of the functions $s \mapsto X_s(\omega)$ to the set D has a right-limit $X_{t+}(\omega) \equiv \lim_{s\downarrow\downarrow t, s\in D} X_s(\omega)$ at every $t \in [0,\infty)$ and a left-limit $X_{t-}(\omega) \equiv \lim_{s\uparrow\uparrow t, s\in D} X_s(\omega)$ at every $t \in (0,\infty)$.*

2. *$\forall t \in \mathbb{R}_{\geq 0}$, $X_{t+} \in L^1$, $X_t \geq E[X_{t+}|\mathscr{F}_t]$ with equality if the function $t \mapsto E[X_t]$ is right-continuous. The process $(X_{t+})_{t\geq 0}$ is a supermartingale w.r.t. the filtration $\mathscr{F}_{t+}$). It is a martingale if X is a martingale.*

pf.

- fix $T \in D$. We have $\sup_{s\in D\cap[0,T]}|X_s| < \infty$ a.s. by the previous proposition. We can choose a sequence $(D_m)_{m\geq 1}$ of finite subset of D that increase to $D\cap[0,T]$ and are s.t. $0, T \in D_m$. The Doob's upcrossing inequality for discrete supermartingales give $\forall a < b, \forall m \geq 1$, $E[M^X_{ab}(D_m)] \leq \frac{1}{b-a}E[(X_T - a)^-]$. If we let $m \to \infty$, by the monotone convergence, $E[M^X_{ab}(D \cap [0,T])] \leq \frac{1}{b-a}E[(X_T - a)^-] < \infty \to M^X_{ab}([0,T]\cap D) < \infty$ a.s. Set $N = \cup_{T\in D}(\{\sup_{t\in D\cap[0,T]}|X_t| = \infty\} \cup (\cup_{a,b\in\mathbb{Q},a<b}\{M^X_{ab}(D\cap[0,T]) = \infty\}))$. Then $P(N) = 0$. On the other hand, if $\omega \notin N$, $t \mapsto X_t(\omega)$ satisfies all the assumptions of the previous lemma. So the statement follows.
- Set $X_{t+}(\omega) = \begin{cases} \lim_{s\downarrow\downarrow t, s\in D} X_s(\omega) & \text{the limit exists} \\ 0 & \text{otherwise} \end{cases}$. With this definition, X_{t+} is $\mathscr{F}_{t+}$-measurable. Fix $t \geq 0$ and choose a sequence $(t_n)_{n\geq 0}$ in D s.t. t_n decreases strictly to t as $n \to \infty$. Then by construction, a.s. $X_{t+} = \lim_{n\to} X_{t_n}$. Set $Y_k = X_{t_{-k}}$, $\forall k \leq 0$. Then Y is a backward supermartingale w.r.t. the discrete filtration $\mathscr{H}_k = \mathscr{F}_{t_{-k}}$. Then by the proposition 2.2.2, $\sup_{k\leq 0}E[|Y_k|] < \infty$. The convergence theorem for backward supermartingales implies that the sequence X_{t_n} converges to X_{t+} in L^1. In particular, $X_{t+} \in L^1$. Thanks to the L^1-convergence, we can pass to the limit $n \to \infty$ in the inequality $X_t \geq E[X_{t_n}|\mathscr{F}_t] \to X_t \geq E[X_{t+}|\mathscr{F}_t]$. Furthermore by the L^1-convergence, $E[X_{t+}] = \lim E[X_{t_n}]$. Thus if the function $s \mapsto E[X_s]$ is right-continuous, $E[X_t] = E[X_{t+}] = E[E[X_{t+}|\mathscr{F}_t]]$ and the inequality $X_t \geq E[X_{t+}|\mathscr{F}_t]$ forces $X_t = E[X_{t+}|\mathscr{F}_t[$. Let $s < t$, and let $(s_n)_{n\geq 0}$ be a sequence in D that decreases strictly to s. We may assume $s_n \leq t_n$ for every n. Then as previously $X_{s_n} \to X_{s+}$ in L^1 and thus if $A \in \mathscr{F}_{s+}$, $A \in \mathscr{F}_{s_n}$, $\forall n$. Then we have $E[X_{s+}\mathbb{1}_A] = \lim_{n\to\infty}E[X_{s_n}\mathbb{1}_A] \geq \lim E[X_{t_n}\mathbb{1}_A] = E[X_{t+}\mathbb{1}_A] = E[E[X_{t+}|\mathscr{F}_{s+}]\mathbb{1}_A]$. Since this holds $\forall A \in \mathscr{F}_{s+}$ and since $X_{s+}, E[X_{t+}|\mathscr{F}_{s+}]$ are $\mathscr{F}_{s+}$-measurable, $X_{s+} \geq E[X_{t+}|\mathscr{F}_{s+}]$.

Theorem 3.2.2. *Assume that the filtration $(\mathscr{F}_t)$ is right-continuous and complete. Let $X = (X_t)_{t\geq 0}$ be a supermartingale, s.t. the function $t \mapsto E[X_t]$ is right-continuous. Then X has a modification with cadlag sample paths, which is also an $(\mathscr{F}_t)$-supermartingale.*

pf. Let D be a countable dense subset of $\mathbb{R}_{\geq 0}$. Let N be the negligible set defined before. Set $\forall t \geq 0$, $Y_t(\omega) = \begin{cases} X_{t+}(\omega) & \omega \notin N \\ 0 & \omega \in N \end{cases}$. Then by the previous lemma, the sample paths of Y are cadlag.

The r.v. X_{t+} is $\mathscr{F}_{t+}$-measurable and thus $\mathscr{F}_t$-measurable, since the filtration is right-continuous. As the negligible set N belong to $\mathscr{F}_\infty$, the completeness of the filtration ensures Y_t is $\mathscr{F}_t$-measurable. By the previous theorem 2, $\forall t \geq 0$, $X_t = E[X_{t+}|\mathscr{F}_t] = X_{t+} = Y_t$ a.s. because X_{t+} is $\mathscr{F}_t$-measurable. Consequently, Y is a modification of X. The process Y is adapted to the filtration $(\mathscr{F}_t)$. Since Y is a modification of X, $E[X_t|\mathscr{F}_s] \leq X_s$, $\forall 0 \leq s < t$ implies the same inequality holds for Y.
$\rightarrow$ right-continuity of the filtration is necessary($\Omega = \{\pm 1\}$ with the probability measure P defined by $P(\{-1\}) = P(\{1\}) = 1/2$. Let ϵ be the r.v. $\epsilon(\omega) = \omega$ and let the process $(X_t)_{t\geq 0}$ be defined by $X_t = \begin{cases} 0 & 0 \leq t \leq 1 \\ \epsilon & t > 1 \end{cases}$. Then X is a martingale w.r.t. the canonical filtration $(\mathscr{F}_t^X)$. But no modification of X can be right-continuous at $t = 1$).

Theorem 3.2.3. *Let X be a supermartingale with right-continuous sample paths. Assume that the collection $(X_t)_{t\geq 0}$ is bounded in L^1. Then there exists a random variable $X_\infty \in L^1$ s.t.*

$$\lim_{t\to\infty} X_t = X_\infty$$

a.s.

pf. Let D be a countable dense subset of $\mathbb{R}_{\geq 0}$. Then $\forall T \in D$, $a < b$, $E[M_a^X(D \cap [0,T])] \leq \frac{1}{b-a}E[(X_T - a)^-]$. By the monotone convergence, $\forall a < b$, $E[M_{ab}^X(D)] \leq \frac{1}{b-a}\sup_{t\geq 0} E[(X_t - a)^-] < \infty$ since the collection $(X_t))_{t\geq 0}$ is bounded in L^1. Hence a.s. for all rationals $a < b$, $M_{ab}^X(D) < \infty$. This implies the limit $X_\infty \equiv \lim_{t\to\infty, t\in D} X_t$ exists a.s. in $[-\infty, \infty]$. By the Fatou's lemma, $E[|X_\infty|] \leq \lim_{t\to\infty, t\in D} \inf E[|X_t|] < \infty$, $X_\infty \in L^1$. The right-continuity of sample paths allows us to remove the restriction $t \in D$ in the limit.

Definition 3.2.4. *A martingale $(X_t)_{t\geq 0}$ is said to be closed if there exists a r.v. $Z \in L^1$ s.t. $\forall t \geq 0$, $X_t = E[Z|\mathscr{F}_t]$.*

Theorem 3.2.4. *Let X be a martingale with right-continuous sample paths. Then TFAE*

1. *X is closed*
2. *the collection $(X_t)_{t\geq 0}$ is uniformly integrable*
3. *X_t converges a.s. and in L^1 as $t \to \infty$*

Moreover, if there properties hold, $X_t = E[X_\infty|\mathscr{F}_t]$, $\forall t \geq 0$, where $X_\infty \in L^1$ is the a.s. limit of X_t as $t \to \infty$.

pf.

- $1 \to 2$
  if $Z \in L^1$, the collection of all random variables $E[Z|\mathscr{G}]$ when $\mathscr{G}$ varies over sub-σ-fields of $\mathscr{F}$, is uniformly integrable.

- $2 \to 3$ by the previous theorem, X_t converges to X_∞. By uniform integrability, the latter convergence also holds in L^1.

- $3 \to 1$
  $\forall s \geq 0$, we can pass to the limit $t \to \infty$ in the equality $X_s = E[X_t|\mathscr{F}_s]$ and we get $X_s = E[X_\infty|\mathscr{F}_s]$.

Let $(X_t)_{t\geq 0}$ be a martingale or a supermartingale with right-continuous sample paths and $X_t \to X_\infty$ as $t \to \infty$. Then for every stopping time T, $X_T(\omega) = \mathbb{1}_{\{T(\omega)<\infty\}}X_{T(\omega)}(\omega) + \mathbb{1}_{\{T(\omega)=\infty\}}X_\infty(\omega)$. With this definition X_T, is $\mathscr{F}_T$-measurable.

Theorem 3.2.5 (Optimal Stopping Theorem for Martingales)**.** *Let $(X_t)_{t\geq 0}$ be a uniformly integrable martingale with right-continuous sample paths. Let S, T be two stopping times with $S \leq T$. Then $X_S, X_T \in L^1$ and $X_S = E[X_T|\mathscr{F}_S]$. In particular, for every stopping time S,* $\begin{cases} X_S = E[X_\infty|\mathscr{F}_S] \\ E[X_S] = E[X_\infty] = E[X_0] \end{cases}$.

pf. Set $\forall n \geq 0$ integer, $\begin{cases} T_n = \sum_{k=0}^{\infty} \frac{k+1}{2^n} \mathbb{1}_{\{k2^{-n}<T\leq(k+1)2^{-n}\}} + \infty \cdot \mathbb{1}_{\{T=\infty\}} \\ S_n = \sum_{k=0}^{\infty} \frac{k+1}{2} \mathbb{1}_{\{k2^{-n}<S\leq(k+1)2^{-n}\}} + \infty \cdot \mathbb{1}_{\{T=\infty\}} \end{cases}$. Then by the proposition 2.1.3, they are two sequences of stopping times that decreases to T, S respectively. Moreover $S_n \leq T_n$, $\forall n \geq 0$. For every fixed n, $2^n S_n$, $2^n T_n$ are stopping times of the discrete filtration $\mathscr{H}_k^{(n)} \equiv \mathscr{F}_{k/2^n}$, and $Y_k^{(n)} \equiv X_{k/2^n}$ is a discrete martingale w.r.t. this filtration. From the optional stopping theorem for uniformly integrable discrete martingales, $Y^{(n)}_{2^n S_n}, Y^{(n)}_{2^n T_n}$ are in L^1 and $X_{S_n} = Y^{(n)}_{2^n S_n} = E[Y^{(n)}_{2^n T_n}|\mathscr{H}^{(n)}_{2^n S_n}] = E[X_{T_n}|\mathscr{F}_{S_n}]$. Let $A \in \mathscr{F}_S$. Since $\mathscr{F}_S \subset \mathscr{F}_{S_n}$, $A \in \mathscr{F}_{S_n}$ $\to E[\mathbb{1}_A X_{S_n}] = E[\mathbb{1}_A X_{T_n}]$. By the right-continuity of sample paths, a.s. $\begin{cases} X_S = \lim X_{S_n} \\ X_T = \lim X_{T_n} \end{cases}$. These limits also hold in L^1. By the optional stopping theorem for uniformly integrable discrete marttingales, $X_{S_n} = E[X_\infty|\mathscr{F}_{S_n}]$ for every n and thus (X_{S_n}) is uniformly integrable. The L^1-convergence implies X_S, X_T belong to L^1. Additionally $E[\mathbb{1}_A X_S] = E[\mathbb{1}_A X_T]$ by passing $E[\mathbb{1}_A X_{S_n}] = E[\mathbb{1}_A X_{T_n}]$ to the limit $n \to \infty$. Since this holds for all $A \in \mathscr{F}_S$, and since X_S is $\mathscr{F}_S$-mesurable, $X_S = E[X_T|\mathscr{F}_S]$.

Corollary 3.2.1. *Let $(X_t)_{t\geq 0}$ be a martingale with right-continuous sample paths and let $S \leq T$ be two bounded stopping times. Then X_S, X_T are in L^1 and $X_S = E[X_T|\mathscr{F}_S]$.*

pf. Let $a \geq 0$ s.t. $S \leq T \leq a$. Then apply the previous theorem to $(X_{t\wedge a})_{t\geq 0}$ which is closed by X_a and the corollary follows.

Corollary 3.2.2. *Let $(X_t)_{t\geq 0}$ be a martingle with right-continuous sample paths and let T be a stopping time.*

1. *the process $(X_{t\wedge T})_{t\geq 0}$ is still a martingale*
2. *suppose in addition that the martingale $(X_t)_{t\geq 0}$ is uniformly integrable. Then the process $(X_{t\wedge T})_{t\geq 0}$ is also uniformly integrable martingale and more precisely $\forall t \geq 0,\ X_{t\wedge T} = E[X_T|\mathscr{F}_t]$.*

pf.

- 2
  $t\wedge T$ is a stopping time by the property 6. By the previous theorem $X_{t\wedge T}$ and X_T are in L^1 and $X_{t\wedge T}$ is $\mathscr{F}_{t\wedge T}$-measurable, hence $\mathscr{F}_t$-measurable since $\mathscr{F}_{t\wedge T} \subset \mathscr{F}_t$. Fix $A \in \mathscr{F}_t$. Then $\begin{cases} E[\mathbb{1}_{A\cap\{T\leq t\}}X_T] = E[\mathbb{1}_{A\cap\{T\leq t\}}X_{t\wedge T}] \\ X_{t\wedge T} = E[X_T|\mathscr{F}_{t\wedge T}] \end{cases}$ Since $A\cap\{T>t\} \in \mathscr{F}_t \cap \mathscr{F}_T,\ A\cap\{T>t\} \in \mathscr{F}_{T\wedge t} \to E[\mathbb{1}_{A\cap\{T>t\}}X_T] = E[\mathbb{1}_{A\cap\{T>t\}}X_{t\wedge T}]$. Then with previous equality, the corollary follows.

- 1
  apply 2 to the uniformly integrable martingale $(X_{t\wedge a})_{t\geq 0}$ for any choice of $a \geq 0$.

Theorem 3.2.6. *Let $(Z_t)_{t\geq 0}$ be a nonnegative supermartingale with right-continuous sample paths. Let U,V be two stopping times s.t. $U \leq V$. Then $Z_U, Z_V \in L^1$ and $Z_U \geq E[Z_V|\mathscr{F}_U]$.*

pf.

- Assume U,V are bounded and let $p \geq 1$ be an integer s.t. $U \leq p, V \leq p$. Then for every integer $n \geq 0$, $\begin{matrix} U_n = \sum_{k=0}^{p^{2^n-1}} \frac{k+1}{2^n}\mathbb{1}_{\{k2^{-n}<U\leq(k+1)2^{-n}\}} \\ V_n = \sum_{k=0}^{p^{2^n-1}} \frac{k+1}{2^n}\mathbb{1}_{\{k2^{-n}<V\leq(k+1)2^{-n}\}} \end{matrix}$ in such a way that $(U_n),(V_n)$ are two sequences of bounded stopping times that decreases respectively to U and V, and additionally $U_n \leq V_n$, $\forall n \geq 0$. The right-continuity of the sample paths ensures $\begin{matrix} Z_{U_n} \to Z_U \\ Z_{V_n} \to Z_V \end{matrix}$ a.s. as $n \to \infty$. Then by the optional stopping theorem for discrete supermartingales in the case of bounded stopping times, w.r.t the filtration $(\mathscr{F}_{k/2^{n+1}})_{k\geq 0}$. $\forall n \geq 0$, $Z_{U_{n+1}} \geq E[Z_{U_n}|\mathscr{F}_{U_{n+1}}]$. Setting $Y_n = Z_{U_{-n}}$ and $\mathscr{H}_n = \mathscr{F}_{U_{-n}}$, $\forall n \leq 0$, we get the sequence $(Y_n)_{n\geq 0}$ is a backward supermartingale w.r.t. the filtration $(\mathscr{H}_n)_{n\leq 0}$. Since $\forall n \geq 0$, $E[Z_{U_n}] \leq E[Z_0]$, $(Y_n)_{n\leq 0}$ is bounded in $L^1$and by the convergence theorem for backward supermatingales it converges in L^1. Thus the convergence of Z_{U_n} to Z_U holds in L^1 and similarly for Z_{V_n} to

Z_V. Since $U_v \le V_n$, by the another application of the discrete optional stopping theorem, $E[Z_{U_n}] \ge E[Z_{V_n}]$. By the L^1-convergence of Z_{U_n}, Z_{V_n}, $E[Z_U] \ge E[Z_V]$.

- general case

  $E[Z_{U\wedge p}] \le E[Z_0]$ by the above case for every $p \ge 1$. By Fatou's lemma $E[Z_U] \le E[Z_0] < \infty$ and similarly $E[Z_V] < \infty$. Fix $A \in \mathscr{F}_U \subset \mathscr{F}_V$ and use U^A for the stopping time defined by $U^A(\omega) = \begin{cases} U(\omega) & \omega \in A \\ \infty & \omega \notin A \end{cases}$. Then for every $p \ge 1$, $E[Z_{U^A\wedge p}] \ge E[Z_{V^A\wedge p}]$ by the previous case. By writing each of these two as a sum of expectations over the sets A^c, $A \cap \{U \le p[\}$, $A \cap \{Up\}$, $E[Z_U \mathbb{1}_{A\cap\{U\le p\}}] \ge E[Z_V \mathbb{1}_{A\cap\{U\le p\}}]$. Letting $p \to \infty$, $E[Z_U \mathbb{1}_{A\cap\{U<\infty\}}] \ge E[Z_V \mathbb{1}_{A\cap\{U<\infty\}}]$. On the other hand, $E[Z_U \mathbb{1}_{A\cap\{U=\infty\}}] = E[Z_V \mathbb{1}_{A\cap\{V=\infty\}}]$ is trivial. By adding two inequalities, $E[Z_U \mathbb{1}_A] \ge E[Z_V \mathbb{1}_A] = E[E[Z_V|\mathscr{F}_U]\mathbb{1}_A]$. Since this holds for all $A \in \mathscr{F}_U$, and Z_U is $\mathscr{F}_U$-measurable, the theorem follows.

3.3 Finite Variation Processes

Definition 3.3.1. *A signed measure on a compact interval* $[0,T]$ *is the difference of two finite positive measures on* $[0,T]$.

Definition 3.3.2. *Let* $T \ge 0$. *A continuous function* $a : [0,T] \to \mathbb{R}$ *s.t.* $a(0) = 0$ *is said to have finite variation if there exists a signed measure* μ *on* $[0,T]$ *s.t.* $a(t) = \mu([0,t])$, $\forall t \in [0,T]$.

$\to \mu$ is determined uniquely by a. Since a is continuous and $a(0) = 0$, μ has no atoms.

There exists a unique decomposition $\mu = \mu_+ - \mu_-$ s.t. μ_+, μ_- are supported on disjoint Borel sets.
pf.

- existence

  Started from arbitrary decomposition $\mu = \mu_1 - \mu_2$ and set $\nu = \mu_1 + \mu_2$. Then by the Radon-Nikodym theorem, we can find two nonnegative Borel functions h_1, h_2 on $[0,T]$ s.t. $\begin{matrix} \mu_1(dt) = h_1(t)\nu(dt) \\ \mu_2(dt) = h_2(t)\nu(dt) \end{matrix}$. Then if $h(t) = h_1(t) - h_2(t)$, $\mu(dt) = h(t)\nu(dt) = h(t)^+\nu(dt) - h(t)^-\nu(dt)$ which gives the decomposition $\begin{cases} \mu_+(dt) = h(t)^+\nu(dt) \\ \mu_-(dt) = h(t)^-\nu(dt) \end{cases}$ with disjoint support $\begin{cases} D_+ \equiv \{t : h(t) > 0\} \\ D_- \equiv \{t : h(t) < 0\} \end{cases}$.

- uniqueness

  For every $A \in \mathscr{B}([0,T])$, $\mu_+(A) = \sup\{\mu(C) : C \in \mathscr{B}([0,T]),\ C \subset A\}$ and therefore the uniqueness follows.

Definition 3.3.3. *The measure $|\mu| \equiv \mu_+ + \mu_-$ is called the total variation of a.*

$\to |\mu(A)| \le |\mu|(A)$, $\forall A \in \mathscr{B}([0,T])$ and the Radon-Nokodym derivative of μ w.r.t. $|\mu|$ is

$$\frac{d\mu}{d|\mu|} = \mathbb{1}_{D_+} - \mathbb{1}_{D_-}$$

A finite variation $a(t) = \mu_+([0,t]) - \mu_-([0,t])$ means a is the difference of two monotone non-decreasing continuous functions that vanish at 0. Conversely the difference of two monotone non-decreasing continuous functions that vanish at 0 has finite variation.
pf. the formula $g(t) = \theta([0,t])$, $t \in [0,T]$ induces a bijection between monotone non-decreasing right-continuous functions $g : [0,T] \to \mathbb{R}_{\ge 0}$ and finite positive measures θ on $[0,T]$.

Let $f : [0,T] \to \mathbb{R}$ be a measurable function s.t. $\int_{[0,T]} |f(s)||\mu|(ds) < \infty$. Set
$\begin{cases} \int_0^T f(s)da(s) = \int_{[0,T]} f(s)\mu(ds) \\ \int_0^T f(s)|da(s)| = \int_0^T f(s)|\mu|(ds) \end{cases}$
$\to |\int_0^T f(s)da(s)| \le \int_0^T |f(s)||da(s)|$ holds.

Proposition 3.3.1. $\forall t \in (0,T]$, $\int_0^t |da(s)| = \sup\{\sum_{i=1}^p |a(t_i) - a(t_{i-1})|\}$ *where the supremum is over all subdivisions $0 = t_0 < \cdots < t_p = t$ of $[0,t]$. More precisely, for any increasing sequence $0 = t_0^n < \ldots t_{p_n}^n = t$ of subdivisions of $[0,t]$ whose mesh tends to 0, we have*

$$\lim_{n\to\infty} \sum_{i=1}^{p_n} |a(t_i^n) - a(t_{i=1}^n)| = \int_0^t |da(s)|$$

pf. Consider the case $t = T$. For any subdivision $0 = t_0 < \cdots < t_p = T$ of $[0,T]$,
$\begin{cases} |a(t_i) - a(t_{i-1})| = |\mu((t-i-1, t_i])| \le |\mu|((t_{i-1}, t_i]) \\ \sum_{i=1}^n |\mu|((t_{i-1}, t_i]) = |\mu|([0,t]) = \int_0^T |da(s)| \end{cases} \to \lim \sum |a(t_i^n) - a(t_{i-1}^n)| \le \int_0^t |da(s)|$
follows.
Consider an increasing sequence $0 = t_0^n < \cdots < t_{p_n}^n = T$ of subdivisions of $[0,T]$ whose mesh $\max_i\{t_i^n - t_{i-1}^n\}$ tends to 0. Introduce the probability space $\Omega = [0,T]$, which is equipped with the Borel σ-field $\mathscr{B} = \mathscr{B}([0,T])$ and the probability measure $P(ds) = (|\mu|([0,T]))^{-1}|\mu|(ds)$. On this probability space, consider the discrete filtration $(\mathscr{B}_n)_{n\ge 0}$ s.t. $\forall n \ge 0$, $\mathscr{B}_n$ is the σ-field generated by the intervals $(t_{i-1}^n, t_i^n]$, $1 \le i \le p_n$. Then set $X(s) = \mathbb{1}_{D_+}(s) - \mathbb{1}_{D_-}(s) = \frac{d\mu}{d|\mu|}(s)$ and for every $n \ge 0$, $X_n = E[X|\mathscr{B}_n]$. By the properties of conditional expectation X_n is constant

on every interval $(t^n_{i-1}, t^n_i]$ and takes the value $\frac{\mu((t^n_{i-1},t^n_i])}{|\mu|((t^n_{i-1},t^n_i])} = \frac{a(t^n_i)-a(t^n_{i-1})}{|\mu|((t^n_{i-1},t^n_i])}$ on this interval. On the other hand, the sequence (X_n) is a closed martingale w.r.t. the discrete filtration $(\mathscr{B}_n)$. Since X is measurable w.r.t. $\mathscr{B} = \vee_n \mathscr{B}_n$, this martingale converges to X in L^1, by the convergence theorem for closed discrete martingales. In particular, $\lim E[|X_n|] = E|X| = 1$ since $|X(s)| = 1$, $|\mu|(ds)$ a.e. Since $E|X_n| = (|\mu|([0,T]))^{-1}\sum_{i=1}^{p_n}|(a(t^n_i) - a_{t^n_{i-1}})|$ and $|\mu|([0,T]) = \int_0^T |da(s)|$, the proposition follows.

Lemma 3.3.1. *If $f : [0,T] \to \mathbb{R}$ is a continuous function, and if $0 = t^n_0 < \cdots < t^n_{p_n} = T$ is a sequence of subdivisions of $[0,T]$ whose mesh tends to 0, we have*

$$\int_0^T f(s)da(s) = \lim_{n\to\infty}\sum_{i=1}^{p_m} f(t^n_{i-1})(a(t^m_i) - a(t^n_{i-1}))$$

pf. Let f_n be defined on $[0,T]$ by $f_n(s) = f(t^n_{i-1})$ if $s \in (t^n_{i-1}, t^n_i]$, $\forall i \in [1,p_n]$ and $f_n(0) = f(0)$. Then $\sum_{i=1}^{p_n} f(t^n_{i-1})(a(t^m_i) - a(t^n_{i-1})) = \int_{[0,T]} f_n(s)\mu(ds)$ and by the dominated convergence theorem, the lemma follows.

Definition 3.3.4. *A function $a : \mathbb{R}_{\geq 0} \to \mathbb{R}$ is a finite variation function on $\mathbb{R}_{\geq 0}$ if the restriction of a to $[0,T]$ has finite variation on $[0,T]$, $\forall T > 0$.*

$\to !\exists\sigma$-finite measure on $\mathbb{R}_{\geq 0}$ whose restriction to every interval $[0,T]$ is the total variation measure of the restriction of a to $[0,T] \to \int_0^\infty f(s)|da(s)|$ denote the integral of a nonnegative Borel function f on $\mathbb{R}_{\geq 0}$ w.r.t. this σ-finite measure.

Define

$$\int_0^\infty f(s)da(s) = \lim_{T\to\infty}\int_0^T f(s)da(s) \in (-\infty,\infty)$$

for any real Borel function f on $\mathbb{R}_{\geq 0}$ s.t. $\int_0^\infty |f(s)||da(s)| < \infty$.

3.3.1 Finite Variation Process

Set a filtered probability space $(\Omega, \mathscr{F}, (\mathscr{F}_t), P)$.

Definition 3.3.5. *An adapted process $A = (A_t)_{t\geq 0}$ is called a finite variation process if all its sample paths are finite variation functions on $\mathbb{R}_{\geq 0}$. If in addition the sample paths are non-decreasing functions, the process A is called an increasing process..*

If A is a finite variation process, $V_t = \int_0^r |dA_s|$ is an increasing process. Writing $A_t = \frac{1}{2}(V_t + A_t) - \frac{1}{2}(V_t - A_t)$ shows that any finite variation process can be written as the difference of two increasing process.

Proposition 3.3.2. *Let A be a finite variation process and let H be a progressive process s.t. $\forall t \geq 0$, $\forall \omega \in \Omega$, $\int_0^t |H_s(\omega)||dA_s(\omega)| < \infty$. Then the process $H \cdot A = ((H \cdot A)_t)_{t \geq 0}$ defined by*

$$(H \cdot A)_t = \int_0^t H_s dA_s$$

is also a finite variation process.

pf. We know the sample paths of $H \cdot A$ are finite variation functions by the previous proposition.
fix $t > 0$.

- $\begin{array}{rcl} h : \Omega \times [0,t] & \to & \mathbb{R} \\ (\omega, s) & \mapsto & \mathbb{1}_{(u,v]}(s)\mathbb{1}_\Gamma(\omega) \end{array}$ is a measurable function for the σ-filed $\mathscr{F}_t \otimes \mathscr{B}([0,t])$ s.t. $\int_0^t |h(\omega,s)||dA_s(\omega)| < \infty$, $\forall \omega \in \Omega$. Then $\int_0^t h(\omega,s)dA_s(\omega) = \mathbb{1}_\Gamma(\omega)(A_v(\omega) - A_u(\omega))$ for $(u,v] \subset [0,t]$, $\Gamma \in \mathscr{F}_t$ and the proposition follows.

- $h = \mathbb{1}_G$, $G \in \mathscr{F}_t \otimes \mathscr{B}([0,t]) \to$ by a monotone class argument, the proposition holds.

- write h as a pointwise limit of a sequence of simple functions, i.e. finite linear combinations of indicator functions of measurable sets h_n s.t. $|h_n| \leq |h|$, $\forall n$. Then $\int_0^t h_n(\omega,s)dA_s(\omega) \to \int_0^t h(\omega,s)dA_s(\omega)$ by dominated convergence theorem $\forall \omega \in \Omega$.

$\to$with weaker assumption " a.s. $\forall t \geq 0$, $\int_0^t |H_s(\omega)||dA_s(\omega)| < \infty$", if the filtration is complete, we can still define $H \cdot A$ as a finite variation process by replacing H by the process H' s.t. $H'_t(\omega) = \begin{cases} H_t(\omega) & \int_0^n |H_s(\omega)||dA_s(\omega)| < \infty, \forall n \\ 0 & otherwise \end{cases}$. This is because H' is still progressive, which allows us to define $H \cdot A = H' \cdot A$.
with additional assumptions that H, K two progressive processes s.t.
$\begin{array}{c} \int_0^t |H_s| dA_s| < \infty \\ \int_0^t |H_s K_s||dA_s| < \infty \end{array}$, we have the associativity in the sense that $K \cdot (H \cdot A) = (KH) \cdot A$.

e.g. $A_t = t, H$ a progressive process s.t. $\forall t \geq 0$, $\forall \omega \in \Omega$, $\int_0^t |H_s(\omega)| ds < \infty \to \int_9^t H_s ds$ is a finite variation process.

3.4 Continuous Local Martingales

In the filtered probability space $(\Omega, \mathscr{F}, (\mathscr{F}_t), P)$, if T is a stopping time, and if $X = (X_t)_{t \geq 0}$ is an adapted process with continuous sample paths, write X^T for process

X stopped at T, defined by $X_t^T = X_{t\wedge T}$ for every $t \geq 0$. Then if S is another stopping time, $(X^T)^S = (X^S)^T = X^{S\wedge T}$.

Definition 3.4.1. *An adapted process $M = (M_t)_{t\geq 0}$ with continuous sample paths and s.t. $M_0 = 0$ a.s is called a continuous local martingale if there exists a nondecreasing sequence $(T_n)_{n\geq 0}$ of stopping times s.t. $T_n \uparrow \infty$ and $\forall n$, the stopped process M^{T_n} is a uniformly integrable martingale.*

→ Any martingale with continuous sample paths is a continuous local martingale but not vice versa($(\mathscr{F}_t)$-Brownian motion B started from 0, Z is an $\mathscr{F}_0$-measurable r.v. and the process $M_t = Z + B_t$ is a continuous local martingale, but if $E|Z| = \infty$, it is not a martingale).
The sequence of stopping times (T_n) reduces M if $T_n \uparrow \infty$ and for every n, M^{T_n} is a uniformly integrable martingale.

Properties of Continuous Local Martingales

1. a martingale with continuous sample paths is a continuous local martingale and the sequence $T_n = n$ reduces M

2. if M is a continuous local martingale, for every stopping time T, M^T is a continuous local martingale

3. if (T_n) reduces M and if (S_n) is a sequence of stopping times s.t. $S_n \uparrow \infty$, then the sequence $(T_n \wedge S_n)$ also reduces M

4. the space of all continuous local martingales is a vector space

Proposition 3.4.1. *1. a nonnegative continuous local martingale M s.t. $M_0 \in L^1$ is a supermartingale*

2. a continuous local martingale M s.t. there exists a r.v. $Z \in L^1$ with $|M_t| \leq Z$, $\forall t \geq 0$ is a uniformly integrable martingale

3. if M is a continuous local martingale and $M_0 = 0$, the sequence of stopping times $T_n = \inf\{t \geq 0 : |M_t| \geq n\}$ reduces M.

pf.

- write $M_t = M_0 + N_t$. Then by definition, $\exists (T_n)$ a sequence of stopping times that reduces N. Then if $s \leq t$, $\forall n$, $N_{s\wedge T_n} = E[N_{t\wedge T_n}|\mathscr{F}_s]$. By adding a r.v. M_0 on both sides, we get $M_{s\wedge T_n} = E[M_{t\wedge T_n}|\mathscr{F}_s]$. Since M takes nonnegative values, we can let $n \to \infty$ and by the Fatou's lemma for conditional expectations, $M_s \geq E[M_t|\mathscr{F}_s]$. Taking $s = 0$, $E[M_t] \leq E[M_0] < \infty \to M_t \in L^1$, $\forall t \geq 0$.

- $\forall 0 \le s \le t,\ M_{s_n \wedge T_n} = E[M_{t\wedge T_n}|\mathscr{F}_s]$. Since $|M_{t\wedge T_n}| \le Z$, by the dominated convergence theorem the sequence $M_{t\wedge T_n}$ converges to M_t in L^1. We can pass to the limit $n \to \infty$, and get $M_s = E[M_t|\mathscr{F}_s]$

- suppose $M_0 = 0$. The random times T_n are stopping times. Since M^{T_n} is a continuous local martingale and $|M^{T_n}| \le n$, by the statement 2, we get 3. If we assume $M_0 \in L^1$, M^{T_n} is dominated by $n + |M_0|$ and the result is the same.

Theorem 3.4.1. *Let M be a continuous local martingale. Assume that M is also a finite variation process, in particular $M_0 = 0$. Then $M_t = 0,\ \forall t \ge 0$ a.s.*

pf. Set $\tau_n = \inf\{t \ge 0 : \int_0^t |dM_s| \ge n\}$ for every integer $n \ge 0$, Then τ_n is a stopping time($\int_0^t |dM_s|$ is an increasing sequence if M is a finite variation process). Fix $n \ge 0$, and set $N = M^{\tau_n}$. Then $\forall t \ge 0$, $|N_t| = |M_{t\wedge\tau_n}| \le \int_0^{t\wedge\tau_n} |dM_s| \le n$. By the previous proposition, N is a martingale. Let $t > 0$ and let $0 = t_0 < \cdots < t_p = t$ be any subdivision of $[0,t]$. Then $E[N_t^2] = \sum_{i=1}^p E[(N_{t_i} - N_{t_{i-1}})^2] \le E[(\sup_{1\le i\le p} |N_{t_i} - N_{t_{i-1}}|)\sum_{i=1}^p |N_{t_i} - N_{t_{i-1}}|] \le nE[\sup|N_{t_i} - N_{t_{i-1}}|]$ noting that $\int_0^t |dN_s| \le n$ by the definition of τ_n. Apply the bound we get to a sequence $0 = t_0^k < \cdots < t_{p_k}^k = t$ of subdivisions of $[0,t]$ whose mesh tends to 0. By the continuity of sample paths and by the fact that N is bounded, $\lim_{k\to\infty} E[\sup_{1\le i\le p_k} |N_{t_i^k} - N_{t_{i-1}^k}|] = 0$. So $E[N_t^2] = 0 \to M_{t\wedge\tau_n} = 0$ a.s. Letting $n \to \infty$, $M_t = 0$ a.s.

Theorem 3.4.2. *Let $M = (M_t)_{t\ge0}$ be a continuous local martingale. There exists an increasing process denoted by $(< M,M >_t)_{t\ge0}$, which is unique up to indistinguishability, s.t. $M_t^2 - < M,M >_t$ is a continuous local martingale. Furthermore for every fixed $t \ge 0$, if $0 = t_0^n < \cdots < t_{p_n}^n = t$ is an increasing sequence of subdivisions of $[0,t]$ with mesh tending to 0, we have*

$$< M,M >_t = \lim_{n\to\infty} \sum_{i=1}^{p_n} (M_{t_i^n} - M_{t_{i-1}^n})^2$$

in probability.

pf.

- existence

  - $M_0 = 0$, M is bounded.
    Fix $K > 0$ and an increasing sequence $0 = t_0^n < \cdots < t_{p_n}^n = K$ of subdivisions of $[0,K]$ with meshing tending to 0. Then $\forall 0 \le r < s$, $\forall \mathscr{F}_r$-measurable variable Z bounded, the process $N_t = Z(M_{s\wedge t} - M_{r\wedge t})$ is a

martingale. Then $\forall n$, $X_t^n = \sum_{i=1}^{p_n} M_{t_{i-1}^n}(M_{t_i^n \wedge t} - M_{t_{i-1}^n \wedge t})$ is a bounded martingale. We have an identity that

$$\forall n,\ \forall j \in \{0,1,\dots,p_n\},\ M_{t_j^n}^2 - 2X_{t_j^n}^n = \sum_{i=1}^{j}(M_{t_i^n} - M_{t_{i-1}^n})^2 \tag{3.1}$$

Thanks to the Doob's inequality in L^2 and the lemma below, $\lim_{n,m\to\infty} E[\sup_{t\le K}(X_t^n - X_t^m)^2] = 0$. In particular, $\forall t \in [0,K]$, $(X_t^n)_{n\ge1}$ is a Cauchy in L^2 and thus converges in L^2.

CLAIM : the limit yields a process Y indexed by $[0,K]$ with continuous sample paths.

pf. since $\lim E[\sup(X_t^n - X_t^m)^2] = 0$, $\exists (n_k)_{k\ge1}$ a strictly increasing sequence of positive integers s.t. $\forall k \ge 1$, $E[\sup_{t\le K}(X_t^{n_{k+1}} - X_t^{n_k})^2] \le 2^{-k}$. This implies $E[\sum_{k=1}^{\infty} \sup_{t\le K} |X_t^{n_{k+1}} - X_t^{n_k}|] < \infty \to \sum_{k=1}^{\infty} \sup |X_t^{n_{k+1}} - X_t^{n_k}| < \infty$ a.s. Thus except on the negligible set $\mathcal{N}$ where the series in the last diverges, the sequence of random functions $(X_t^{n_k}, 0 \le t \le K)$ converges uniformly on $[0,K]$ as $k \to \infty$ and the limiting random function is continuous by uniform convergence. Thus we can set $Y_t(\omega) = \begin{cases} \lim X_t^{n_k}(\omega) & \forall t \in [0,K],\ \omega \in \Omega \setminus \mathcal{N} \\ 0 & \forall t \in [0,K],\ \omega \in \mathcal{N} \end{cases}$. Since the filtration is complete $\mathcal{N} \in \mathcal{F}_t$ for every $t \ge 0$ so Y_t is $\mathcal{F}_t$-measurable and it has continuous sample paths. Y_t is the limit of X_t^n since the L^2-limit of $X_t^n)_{n\ge1}$ must coincide with a.s. limit of a subsequence, and we can pass to the limit in the martingale property for X^n to obtain $E[Y_t|\mathcal{F}_s] = Y_s$ for every $0 \le s \le t \le K>$ Thus $(Y_{t\wedge K})_{t\ge0}$ is a martingale with continuous sample paths.

By the identity 2.1, the sample paths of the process $M_t^2 - 2X_t^n$ are nondecreasing along the finite sequence $(t_i^n,\ 0 \le i \le p_n)$. By passing to the limit $n \to \infty$ along the sequence $(n_k)_{k\ge1}$, the sample paths of $M_t^2 - 2Y_t$ are nondecreasing on $[0,K]$ except maybe on the negligible set $\mathcal{N}$. $\forall t \in [0,K]$, set $A_t^{(K)} = \begin{cases} M_t^2 - 2Y_t & \Omega \setminus \mathcal{N} \\ 0 & \mathcal{N} \end{cases}$ Then $A_0^{(K)} = 0$, $A_t^{(K)}$ is $\mathcal{F}_t$-measurable for every $T \in [0,K]$, $A^{(K)}$ has nondecreasing continuous sample paths, and $(M_{t\wedge K}^2 - A_{t\wedge K}^{(K)})_{t\ge0}$ is a martingale. Apply this with $K = l$, $\forall l \ge 1$ and we get a process $(A_t^{(l)})_{0\le t\le l}$. For every $l \ge 1$, $A_{t\wedge l}^{(l+1)} = A_{t\wedge l}^{(k)}$, $\forall t \ge 0$ a.s. by the uniqueness argument below. Thus we can define an increasing process $< M,M >$ s.t. $< M,M >_t = A_t^{(l)}$, $\forall t \in [0,l]$, $\forall l \ge 1$ a.s. and $M_t^2 - < M,M >_t$ is a martingale.

If $K > 0$ and the sequence of subdivisions $0 = t_0^n < \cdots < t_{p_n}^n = K$ are fixed as previously, the process $A_{t\wedge K}^{(K)}$ must be indistinguishable from $< M,M >_{t\wedge K}$ by the uniqueness argument below. In particular $< M,M >_K =$

$A_K^{(K)}$ a.s. Then by the identity 2.1 with $j = p_n$ and the fact that X_K^n converges in L^2 to $Y_K = \frac{1}{2}(M_K^2 - A_K^{(K)})$, $\lim \sum (M_{t_j}^n - M_{t_{j-1}^n})^2 =< M, M >_K$ in L^2.

- general case

  Write $M_t = M_0 + N_t$ so that $N_0 = 0$, N_t is bounded. Then $M_t^2 = M_0^2 + 2M_0N_t + N_t^2$ and M_0N_t is a continuous local martingale. Wlog assume $M_0 = 0$ and set $T_n = \inf\{t \geq 0 : |M_t| \geq n\}$. Then we can apply the bounded case to the stopped martingales M^{T_n}. Set $A^{[n]} =< M^{T_n}, M^{T_n} >$. By the uniqueness the processes $A_{t\wedge T_n}^{[n+1]}$ and $A_t^{[n]}$ are indistinguishable. Thus there exists an increasing process A s.t. $\forall n$, $A_{t\wedge T_n}, A_t^{[n]}$ are indistinguishable. Since $M_{t\wedge T_n}^2 - A_{t\wedge T_n}$ is a martingale $\forall n$, $M_t^2 - A_t$ is a continuous local martingale. Take $< M, M >_t = A_t$ and the general case follows.

- uniqueness

  if A, A' are two increasing processes satisfying the condition given in the theorem, then the process $A_t - A_t' = (M_t^2 - A_t') - (M_t^2 - A_t)$ is both continuous local martingales and a finite process. Then $A - A' = 0$ by the previous theorem.

$\rightarrow < M, M >$ is called the quadratic variation of M.

Proposition 3.4.2. *Let M be a continuous local martingale s.t. $M_0 = 0$. then $< M, M >= 0$ iff $M = 0$.*

pf. Suppose $< M, M >= 0$. Then M_t^2 is a nonnegative continuous local martingale and by the proposition 2.4.1.-1, M_t^2 is a supermartingale, hence $E[M_t^2] \leq E[M_0^2] = 0 \rightarrow M_t = 0$, $\forall t \geq 0$.
the converse is obvious.

Theorem 3.4.3. *Let M be a continuous local martingale with $M_0 \in L^2$.*

1. *TFAE*

   (a) *M is a true martingale bounded in L^2.*

   (b) *$E[< M, M >_\infty] < \infty$.*

   Furthermore, if there properties hold, the process $M_t^2 - < M, M >_t$ is a uniformly integrable martingale and in particular $E[M_\infty^2] = E[M_0^2] + E[< M, M >_\infty$].

2. *TFAE*

   (a) *M is a true martingale and $M_t \in L^2$. $\forall t \geq 0$*

   (b) $E[< M, M >_t] < \infty$, $\forall t \geq 0$.

   Furthermore, if these properties hold, the process $M_t^2 - < M, M >_t$ is a martingale.

pf.

- replacing M by $M - M_0$, we may assume $M_0 = 0$.

  - Assume M is a martingale bounded in L^2. Then by the Doob's inequality in L^2, $\forall T > 0$, $E[\sup_{0 \leq t \leq T} M_t^2] \leq 4E[M_T^2]$. By letting $T \to \infty$, $E[\sup_{t \geq 0} M_t^2] \leq 4 \sup_{t \geq 0} E[M_t^2] \equiv C < \infty$. Set $S_n = \inf\{t \geq 0 :< M, M >_t \geq n\}$. Then the continuous local martingale $M_{t \wedge S_n}^2 - < M, M >_{t \wedge S_n}$ is dominated by the variable $\sup_{s \geq 0} M_s^2 + n$ which is integrable. Then by the previous proposition 2.4.1.-2, this continuous local martingale is a uniformly integrable martingale, hence $E[< M, M >_{t \wedge S_n}] = E[M_{t \wedge S_n}^2]$ $\leq E[\sup_{s \geq 0} M_s^2] \leq C$. By letting $n, t \to \infty$ and by the monotone convergence theorem, $E[< M, M >_\infty] \leq C < \infty$.

  - Assume $E[< M, M >_\infty] < \infty$. Set $T_n = \inf\{t \geq 0 : |M_t| \geq n\}$. Then the continuous local martingale $M_{t \wedge T_n}^2 - < M, M >_{t \wedge T_n}$ is dominated by the variable $n^2 + < M, M >_\infty$ which is integrable. Then by the previous proposition 2.4.1.-2, this continuous local martingale is uniformly integrable martingale, hence $\forall t \geq 0$, $E[M_{t \wedge T_n}^2] = E[< M, M >_{t \wedge T_n}] \leq E[< M, M >_\infty] \equiv C' < \infty$. By letting $n \to \infty$ and by the Fatou's lemma, $E[M_t^2] \leq C'$ so that the collection $(M_t)_{t \geq 0}$ is bounded in L^2. We also know $(M_{t \wedge T_n})_{n \geq 1}$ is uniformly integrable and therefore converges both a.s. and in L^1 to M_t, $\forall t \geq 0$. Since M^{T_n} is a martingale by the previous proposition 2.4.1.-3, the L^1-convergence allows us to pass to the limit $n \to \infty$ in the martingale property $E[M_{t \wedge T_n} | \mathscr{F}_s] = M_{s \wedge T_n}$, $\forall 0 \leq s < t$ and we get M is a martingale.

  - if each of the properties holds, the continuous local martingale M^2 $- < M, M >$ is dominated by the integrable variable $\sup_{t \geq 0} M_t^2 + < M, M >_\infty$ and therefore by the previous proposition 2.4.1-2, is a uniformly integrable martingale.

- by the previous statement 1 with $(M_{t \wedge a})_{t \geq 0}$ for every choice of $a \geq 0$, 2 holds.

Definition 3.4.2. *If M,N are two continuous local martingales, the bracket $<M,N>$ is the finite variation process defined by setting $\forall t \geq 0$*

$$<M,N>_t = \frac{1}{2}(<M+N,M+N>_t - <M,M>_t - <N,N>_t)$$

Proposition 3.4.3. *1. $<M,N>$ is the unique finite variation process s.t. $M_tN_t - <M,N>_t$ is a continuous local martingale.*

2. the mapping $(M,N) \mapsto <M,N>$ is bilinear and symmetric.

3. if $0 = t_0^n < \cdots < t_{p_n}^n = t$ is an increasing sequence of subdivisions of $[0,t]$ with mesh tending to 0,

$$\lim_{n\to\infty} \sum_{i=1}^{p_n} (M_{t_i^n} - M_{t_{i-1}^n})(N_{t_i^n} - N_{t_{i-1}^n}) = <M,N>_t$$

in probability

4. for every stopping time T, $<M^T,N^T>_t = <M^T,N>_t = <M,N>_{t\wedge T}$.

5. if M,N are two martingales with continuous sample paths bounded in L^2, $M_tN_t - <M,N>_t$ is uniformly integrable martingale. Conversely, $<M,N>_\infty$ is well-defined as the a.s. limit of $<M,N>_t$ as $t \to \infty$ is integrable and satisfies $E[M_\infty N_\infty] = E[M_0N_0] + E[<M,N>_\infty]$.

pf.

- 1 follows from the theorem 2.4.1
- 3 is a consequence of the assertion in theorem 2.4.2.
- 2 follows from 3
- 4 is a consequence of 3 given $\forall 0 \leq s \leq t$, a.s.
$$\begin{array}{ll} <M^T,N^T>_t = <M^T,N>_t = <M,N>_t & \{T \geq t\} \\ <M^T,N^T>_t - <M^T,N^T>_s = <M^T,N>_t - <M^T,N>_s & \{T \leq s < t\} \end{array}.$$
- 5 is the consequence of the previous theorem-1

Proposition 3.4.4. *Let B,B' be two independent $(\mathscr{F}_t)$-Brownian motions. Then $<B,B'>_t = 0$,*
$\forall t \geq 0$.

pf. By subtracting initial values, we may assume $B_0 = B'_0 = 0$. Then the process $X_t = \frac{1}{\sqrt{2}}(B_t + B'_t)$ is a martingale, a linear combination of martingales. Also X is a Brownian motion. Thus
$<X,X>_t = t$ and by the bilinearity of the bracket, $<B,B'>_t = 0$.

Definition 3.4.3. *Two continuous local martingales M,N are said to be orthogonal if $< M,N >= 0$ which holds iff MN is a continuous local martingale.*

→ two independent $(\mathscr{F}_t)$-Brownian motions are orthogonal martingale by the previous proposition.
→ if M,N are two orthogonal martingales bounded in L^2, $\begin{cases} E[M_tN_t] = E[M_0N_0] \\ E[M_SN_S] = E[M_0N_0] \end{cases}$ for all stopping time S by the previous proposition-5.

Proposition 3.4.5 (Kunita-Watanabe)**.** *Let M,N be two continuous local martingales and let H,K be two measurable processes. Then a.s.*

$$\int_0^\infty |H_s||K_s||d < M,N >_s| \le (\int_0^\infty H_s^2 d < M,M >_s)^{1/2}(\int_0^\infty K_s^2 d < N,N >_s)^{1/2} \quad (3.2)$$

pf. To simplify the notiation, denote $< M,N >_s^t \equiv < M,N >_t - < M,N >_s$.
CLAIM : $\forall 0 \le s \le t$. **a.s. for every choice of rationals** $s < t$, $| < M,N >_s^t |$ $\le \sqrt{< M,M >_s^t}\sqrt{< N,N >_s^r}$**.**
pf. the approximations of $< M,M >, < M,N >$ given in the theorem 2.4.2. and proposition 2.4.3. respectively

$$\left(\begin{cases} | < M,N >_s^t | = |\lim \sum (M_{t_i^n} - M_{t_{i-1}^n})(N_{t_i^n} - N_{t_{i-1}^n}) - \lim \sum (M_{s_i^n} - M_{s_{i-1}^n})(N_{s_i^n} - N_{s_{i-1}^n})| \\ | < M,M >_s^t |^{1/2} = |\lim \sum (M_{t_i^n} - M_{t_{i-1}^n})^2 - (M_{s_i^n} - M_{s_{i-1}^n})^2|^{1/2} \\ | < N,N >_s^t |^{1/2} = |\lim \sum (N_{t_i^n} - N_{t_{i-1}^n})^2 - (N_{s_i^n} - N_{s_{i-1}^n})^2|^{1/2} \end{cases} \right).$$

Then by the Cauchy-Schwarz inequality the claim follows.
Fix ω s.t. the inequality of the last display holds for all $s < t$. Then $\forall 0 \le s \le t$, $\int_s^t |d < M,N >_u| \le \sqrt{< M,M >_s^t}\sqrt{< N,N >_s^t}$ since for any subdivision $s = t_0 < t_1 < \cdots < t_p = t$, $\sum_{i=1}^p | < M,N >_{t_{i-1}}^{t_i} | \le \sum \sqrt{< M,M >_{t_{i-1}}^{t_i}}\sqrt{< N,N >_{t_{i-1}}^{t_i}}$
$\le (\sup < M,M >_{t_{i-1}}^{t_i})^{1/2}(\sum < N,N >_{t_{i-1}}^{t_i})^{1/2} = \sqrt{< M,M >_s^t}\sqrt{< N,N >_s^t}$. Then for every bounded Borel subset A of $\mathbb{R}_{\ge 0}$,
$\int_A |d < M,N >_u| \le \sqrt{\int_A d < M,M >_u}\sqrt{\int_A d < N,N >_u}$. When $A = [s,t]$ this is the bound above. If A is a finite union of intervals, this follows from the bound above with Cauchy-Schwartz inequality. A monotone class argument shows that the inequality of the last display remains valid for any bounded Borel set A. Let $\begin{cases} h = \sum_{i=1}^p \lambda_i \mathbb{1}_{A_t} \\ k = \sum_{i=1}^p \mu_i \mathbb{1}_{A_i} \end{cases}$ be two nonnegative simple functions on $\mathbb{R}_{\ge 0}$ with bounded support contained i $[0,K]$ for some $K > 0$. Here (A_i) is a measurable partition of $[0,K]$

and λ_i, μ_i are reals. Then

$$\begin{aligned}\int h(s)k(s)|d<M,N>_s| &= \sum_{i=1}^{p} \lambda_i \mu_i \int_{A_t} |d<M,N>_s| \\ &\le (\sum \lambda_i^2 \int_{A_i} d<M,M>_s)^{1/2} (\sum \mu_i \int_{A_i} d<N,N>_s)^{1/2} \\ &= (\int h(s)^2 d<M,M>_s)^{1/2} (\int k(s)^2 d<N,N>_s)^{1/2}\end{aligned}$$

so the proposition for simple functions follows. Since every nonnegative Borel function is a monotone increasing limit of simple functions with bounded support, by the monotone convergence theorem the proposition follows.

Definition 3.4.4. *A process $X = (X_t)_{t\ge 0}$ is a continuous semimartingale if it can be written in the form $X_t = M_t + A_t$ where M is a continuous local martingale and A is a finite variation process.*

$\rightarrow$ the decomposition $X = M + A$ is unique up to indistinguishability and it is called the canonical decomposition of X.

Definition 3.4.5. *Let $X = M + A$ and $Y = M' + A'$ be the canonical decompositions of two continuous semimartingales X and Y. The bracket $< X,Y >$ is the finite variation process defined by $< X,Y >_t = < M,M' >_t$. In particular, $< X,X >_t = < M,M >_t$.*

Proposition 3.4.6. *Let $0 = t_0^n < \cdots < t_{p_n}^n = t$ be an increasing sequence of subdivisions of $[0,t]$ whose mesh tends to 0. Then*

$$\lim_{n\to\infty} \sum_{i=1}^{p_n} (X_{t_i^n} - X_{t_{i-1}^n})(Y_{t_i^n} - Y_{t_{i-1}^n}) = < X,Y >_t$$

in probability.

pf.

- $X = Y$
  $\sum_{i=1}^{p_n} (X_{t_i^n} - X_{t_{i-1}^n})^2 = \sum (M_{t_i^n} - M_{t_{i-1}^n})^2 + \sum (A_{t_i^n} - A_{t_{i-1}^n})^2 + 2\sum (M_{t_i^n} - M_{t_{i-1}^n})(A_{t_i^n} - A_{t_{i-1}^n})$. By the theorem 4.2.1, $\lim \sum (M_{t_i^n} - M_{t_{i-1}^n})^2 = < M,M >_t = < X,X >_t$ in probability. $\sum (A_{t_i^n} - A_{t_{i-1}^n})^2 \le \sup |A_{t_i^n} - A_{t_{i-1}^n}| \sum |A_{t_i^n} - A_{t_{i-1}^n}| \le \int_0^t |dA_s| \sup |A_{t_i^n} - A_{t_{i-1}^n}| \to 0$ as $n \to \infty$ by the continuity of sample paths of A. Similarly, $|\sum (A_{t_i^n} - A_{t_{i-1}^n})(M_{t_i^n} - M_{t_{i-1}^n})| \le \int_0^t |dA_s| \sup |M_{t_i^n} - M_{t_{i-1}^n}| \to 0$ a.s. So the proposition follows.

- $X \neq Y$
  $\sum_{i=1}^{p_n}(X_{t_i^n} - Y_{t_{i-1}^n})^2 = \sum(M_{t_i^n} - M_{t_{i-1}^n})(M'_{t_i^n} - M'_{t_{i-1}^n}) + \sum(A_{t_i^n} - A_{t_{i-1}^n})(A'_{t_i^n} - A'_{t_{i-1}^n}) + \sum(M_{t_i^n} - M_{t_{i-1}^n})(A'_{t_i^n} - A'_{t_{i-1}^n}) + \sum(M'_{t_i^n} - M'_{t_{i-1}^n})(A_{t_i^n} - A_{t_{i-1}^n})$. Then $\lim\sum(M_{t_i^n} - M_{t_{i-1}^n})(M'_{t_i^n} - M'_{t_{i-1}^n}) =< M, M' >_t =< X, Y >_y$ in probability by the proposition 2.4.3. $\sum(A_{t_i^n} - A_{t_{i-1}^n})(A'_{t_i^n} - A'_{t_{i-1}^n}) \leq \sup|A_{t_i^n} - A_{t_{i-1}^n}|\sum|A'_{t_i^n} - A'_{t_{i-1}^n}| \to 0$ as $n \to \infty$ by the continuity of the sample path of A'. Similarly the other component goes to 0 as $n \to \infty$ and the proposition follows.

Let U be an $\mathscr{F}_0$-measurable real random variable, and let M be a continuous local martingale. Then $N_t = UM_t$ is a continuous local martingale.
pf. Since M_t is a continuous local martingale, $\exists[T_n]$ s.t. $M_{t\wedge T_n}$ is uniformly integrable. Let $T_n^A(\omega) = \begin{cases} T_n(\omega) & \omega \in A \\ \infty & \omega \notin A \end{cases}$ where $A = \{U < \infty\}$ is $\mathscr{F}_0$-measurable. Then it is a stopping time given A $\mathscr{F}_T$-measurable and this new sequence reduces N_t and therefore it is a continuous local martingale.

Let M be a true martingale with continuous sample paths s.t. $M_0 = 0$. Assume $(M_t)_{t\geq 0}$ is also a Gaussian process. Then $\forall t \geq 0$, $\forall s > 0$, $M_{t+s} - M_t$ the random variable is independent of $\sigma(M_r, 0 \leq r \leq t)$. Additionally, $\exists$ a continuous monotone decreasing function $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ s.t. $< M, M >_t = f(t)$, $\forall t \geq 0$.
pf.

- Let $\mathscr{F}_t = \sigma(M_r, 0 \leq r \leq t)$ and pick an arbitrary $U \in \mathscr{F}_t$ a random variable. Then $E[U(M_{t+s} - M_t)] = E[E[U(M_{t+s} - M_t)|\mathscr{F}_t]] = E[UM_t - UM_t] = 0 = E[U] \cdot 0 = E[U](M_t - M_t) = E[U]E[M_{t+s} - M_t]$ independent. Since we take an arbitrary U, independence follows.

- Let $t_1 < t_2$. There is an increasing sequence of subdivisions of $\begin{matrix}(0, t_1) \\ (0, t_2)\end{matrix}$ with mesh tending to 0, $\begin{matrix} < M, M >_{t_1} = \lim\sum_{i=1}^{p_n}(M_{t_i^n} - M_{t_{i-1}^n})^2 \\ < M, M >_{t_2} = \lim\sum_{i=1}^{p_m}(M_{t_i^m} - M_{t_{i-1}^m})^2 \end{matrix}$ exists. Since $t_1 < t_2$, there exists a subinterval $[0, t_1] \subset [0, t_2]$. On those interval, we pick the finest subdivision between two and then we get $< M, M >_{t_2} =< M, M >_{t_1} + \lim\sum_{i=t_1+1}^{p_m}(M_{t_i^m} - M_{t_{i-1}^m})^2 \geq < M, M >_{t_1}$ hence nondecreasing.
  Wlog assume $t_1 < t_2$, $t_1, t_2 \in \mathbb{R}_{\geq 0}$. $\forall \epsilon > 0$, suppose $| < M, M >_{t_1} - < M, M >_{t_2} | < \epsilon$. This implies $\lim_{m\to\infty}\sum_{i=t_1+1}^{p_m}(M_{t_i^m} - M_{t_{i-1}^m})^2 = \lim_{n\to\infty}\sum_{i=0}^{p_n}(M_{t_i^n} - M_{t_{i-1}^n})^2 < \epsilon$ for the interval $[0, t_2 - t_1]$. This sum depends on $|t_2 - t_1|$ and it is monotone nondecreasing, and therefore $\exists\delta > 0$ s.t. $|t_2 - t_1| < \delta$.

Let M be a continuous local martingale with $M_0 = 0$. Then $\{\lim_{t\to\infty} M_t$ exists and finite$\} = \{< M, M >_\infty < \infty\}$ a.s.
pf.

- Set $T_n = \inf\{t \geq 0 : |M_t| = n\}$. Then for any event A in $U_{n=1}^{\infty}\{T_n = \infty\}$, $\exists n$ s.t. $A \in \{T_n = \infty\}$, i.e. M_t never reach n and therefore $\sup_t |M_t| < n$ so $\lim M_t$ is bounded. If an event $A \in (U_{n=1}^{\infty}\{T_n = \infty\})^c$, $\forall n \in \mathbb{Z}_{>0}$, $\exists T_n < \infty$ s.t. $M_{T_n} = n$. Then on the event A, $\lim M_t = \infty$ unbounded and therefore the limit does not exists.
  $\rightarrow \{\lim_{t\to\infty} M_t$ exists and finite$\} = \cup_{i=1}^{n}\{T_n = \infty\}$
  By the hypothesis, M is a true martingale. If M is bounded in L^2, $E[< M, M >_\infty] < \infty$ and it exists a.s. by the theorem 2.4.3. So if $< M, M >_\infty = \infty$, then M is not a martingale and/or not bounded in L^2. Since M is a martingale, then M is not bounded in L^2. Then $\infty > var(M_t) = E[M_t^2] - E[M_t]^2$ forces $E[M_t] = \infty$ so on that event $\{\lim M_t$ exists and finite$\}$ cannot happen. Thus $\{\lim M_t$ exists and finite $\} \subset \{< M, M >_\infty < \infty\}$

- Set $S_n = \inf\{t \geq 0 :< M, M >_t = n\}$. Then for any event A in $\cup_{i=1}^{\infty}\{S_n = \infty\}$, $\exists n$ s.t. $A \in \{S_n = \infty\}$. Then, $< M, M >_t$ never reach n and therefore $\forall t$, $< M, M >_t < n$. This implies on the event A, $\{< M, M >_\infty < \infty\}$. Also for any event $A \in (\cup_{i=1}^{\infty}\{S_n = \infty\})^c$, $\forall n \geq 1$, $S_n < \infty$, which implies $\lim < M, M >_t = \infty$. Thus $\{< M, M >_\infty < \infty\} = \cup_{i=1}^{n}\{S_n = \infty\}$. Pick any arbitrary event $A \in \cup_{i=1}^{\infty}\{S_n = \infty\}$. Then $\exists n \in \mathbb{Z}_{\geq 0}$ s.t. $< M, M >_t < n$, $\forall t \geq 0$. This implies $E[< M, M >_t] < \infty$, $\forall t \geq 0$. Then by the theorem 2.4.3., M is a true martingale and $M_t \in L^2$. Then since L^2 convergence implies L^1 convergence, $\lim M_t$ exists and finite.

For every integer $n \geq 1$, let $M^n = (M_t^n)_{t\geq 0}$ be a continuous local martingale with $M_0 = 0$. Assume $\lim_{n\to\infty} < M^n, M^n >_\infty = 0$ in probability. Let $\epsilon > 0$ and $\forall n \geq 1$, let $T_\epsilon^n = \inf\{t \geq 0 :< M^n, M^n >_t \geq \epsilon\}$. Then it is a stopping time and $M_t^{n,\epsilon} = M_{t\wedge T_\epsilon^n}^n$ is a true martingale bounded in L^2.
pf. $\lim_{t\to\infty} M_{t\wedge T_\epsilon^n}^n = M_{T_\epsilon^n}^n$ and $< M^n, M^n >_{T_\epsilon^n} = \epsilon \rightarrow M_{T_\epsilon^n}^n$ is finite. Thus by the previous statement $< M_{t\wedge T_\epsilon^n}^n, M_{t\wedge T_\epsilon^n}^n >_\infty < \infty$. Then by the theorem 2.4.3., it is a true martingale bounded in L^2.
Then by the Doob's inequality in L^2, $E[\sup_{t\geq 0}(M_{t\wedge T_\epsilon^n}^n)^2] = E[\sup_{0\leq t\leq T_\epsilon^n}(M_{t\wedge T_\epsilon^n}^n)^2] \leq 4E[(M_{T_\epsilon^n}^n)^2] \leq 4\sup_{0\leq t\leq T_\epsilon^n} E[(M_{T_\epsilon^t}^n)^2] \leq 4\epsilon^2$. Then $\forall a > 0$,
$P(\sup_{t\geq 0}|M_t^n| \geq a) \leq P(\sup_{t\geq 0}|M_{t\wedge T_\epsilon}^n| > a, T_\epsilon^n < \infty) + P(\sup_{t\geq 0}|M_t^n| > a, T_\epsilon^n < \infty)$
$\leq P(\sup_{t\geq 0}|M_{t\wedge T_\epsilon}^n| > a) + P(T_\epsilon^n < \infty)$
$\Leftrightarrow \lim_{n\to\infty} P(\sup_{t\geq 0}|M_t^n| \geq a) \leq \lim_{n\to\infty}(P(\sup_{t\geq 0}|M_{t\wedge T_\epsilon}^n| > a) + P(T_\epsilon^n < \infty))$. Since $\forall t \geq 0$, $P(t < T_\epsilon) \rightarrow 1$ as $n \rightarrow \infty$, $P(T_\epsilon < \infty) \rightarrow 0$. Also by the maximal inequality, $P(\sup|M_t^{n,\epsilon}| \geq a) \leq \frac{1}{a}(E|M_0^{\epsilon,n}| + 2E|M_t^{\epsilon,n}|) = \frac{2}{a}E|M_t^{\epsilon,n}|$. Since $\lim < M^n, M^n >= 0$,

$\forall t \geq 0$, $\lim M_t^n = 0$. This implies $\lim E|M_t^{c,n}| = 0$. Thus $\lim_{n\to\infty}(\sup_{t\geq 0}|M_t^n|) = 0$ in probability.

Let $p \geq 2$ be a real number s.t. $E[(< M,M >_\infty)^p] < \infty$. Then $E[(< M,M >_\infty)^p] \leq p^p E[|M_\infty|^{2p}]$.

pf. Let A be an increasing process s.t. $A_\infty < \infty$ a.s. and let Z be an integrable random variable. Assume for every stopping time T, $E[A_\infty - A_T] \leq E[Z\mathbb{1}_{\{T<\infty\}}]$. Then by setting $T = \inf\{t \geq 0 : A_t = \lambda\}$, $E[(A_\infty - A_T)\mathbb{1}_{\{T<\infty\}}] = E[(A_\infty - \lambda)\mathbb{1}_{\{A_\infty>\lambda\}}]$ $\leq E[Z\mathbb{1}_{\{T<\infty\}}]$ by the assumption and the definition of T. Let $f : \mathbb{R}_{\geq 0} \to \mathbb{R}$ be a continuously differentiable monotone increasing function s.t. $f(0) = 0$ and set $F(x) = \int_0^x f(t)dt$, $\forall x \geq 0$. If the above hypothesis holds, $E[(A_\infty - \lambda)f(A_\infty)\mathbb{1}_{\{A_\infty>\lambda\}}]$ $\leq E[Zf(A_\infty)\mathbb{1}_{\{A_\infty>\lambda\}}]$. On the contrary, $E[(A_\infty f(A_\infty) - \lambda f(A_\infty))\mathbb{1}_{\{A_\infty>\lambda\}}]$ $\geq E[(A_\infty f(A_\infty) - \int_0^{A_\infty} tf'(t)dt)\mathbb{1}_{\{A_\infty>\lambda\}}] = E[F(A_\infty)\mathbb{1}_{\{A_\infty>\lambda\}}]$. Thus $E[F(A_\infty)] \leq E[Zf(A_\infty)]$. Let M be a true martingale with continuous sample paths and bounded in L^2 s.t. $M_0 = 0$ and let M_∞ be the a.s. limit of M_t as $t \to \infty$. Then if we set $A_t =< M,M >_t$ and $Z = M_\infty^2$, A_t is an increasing process. Since M_t is bounded in L^2, $A_\infty < \infty$ a.s. Additionally, Z is a random variable and and integrable. Thus this satisfies hypothesis of the first statement. Since $\forall q \geq 1$, $f(x) = (q+1)x^q$ is a continuously differentiable monotone increasing function s.t. $f(0) = 0$, $E[F(A_\infty)] \leq E[Zf(A_\infty)]$ holds, i.e.

$$E[(< M,M >_\infty)^{q+1}] \leq (q+1)E[(< M,M >_\infty)^q M_\infty^2]$$

Then $\forall p \geq 2$, $E[(< M,M >_\infty)^p] \leq pE[(< M,M >_\infty)^{p-1}M_\infty^2] = p(E[(< M,M >)^{p-1}]E[M_\infty^2])$ since $\lim\sum(M_{t_i^n} - M_{t_{i-1}^n})^2 \in \sigma(M_r, 0 \leq r \leq \infty)$ and $M_\infty \in \mathscr{F}_\infty$ are independent. Let $\begin{cases} F(A_\infty) = A_\infty^{p-1} \\ Zf(A_\infty) = (p-1)(A_\infty)^{p-2}|M_\infty|^2 \end{cases}$. Then by the previous statement,

$$E[F(A_\infty)] = E[(< M,M >_\infty)^{p-1}] \leq E[(p-1)(A_\infty)^{p-2}|M_\infty|^2] \leq pE[(< M,M >_\infty)^{p-2}]E[|M_\infty|^2]$$

By repeatedly doing this, we get $E[(< M,M >_\infty)^p] \leq p^p E|M_\infty|^{2p}$.

Chapter 4

Stochastic Integration

4.1 The Construction of Stochastic Integration

Set a filtered probability space $(\Omega, \mathscr{F}, (\mathscr{F}_t), P)$ and assume the filtration $(\mathscr{F}_t)$ is complete. Write $\mathbb{H}^w$ for the space of all continuous martingales M which are bounded in L^2 and $M_0 = 0$ with the usual convention that two indistinguishable processes are identified.
$\rightarrow$ if $M \in \mathbb{H}^2$, $M_t = E[M_\infty | \mathscr{F}_t]$ where $M_\infty \in L^2$ is the a.s. limit of M_t as $t \rightarrow \infty$. if $M, N \in \mathbb{H}^2$, $< M, N >_\infty$ is well-defined and $E| < M, N >_\infty | < \infty$. So we can define a symmetric bilinear form on $\mathbb{H}^2$ via $(M,N)_{\mathbb{H}^2} = E[< M, N >_\infty] = E[M_\infty N_\infty]$. This yields a norm on $\mathbb{H}^2$ given by

$$||M||_{\mathbb{H}^2} = (M,M)_{\mathbb{H}^2}^{1/2} = E[< M, M >_\infty]^{1/2} = E[(M_\infty)^2]^{1/2}$$

Proposition 4.1.1. *The space $\mathbb{H}^2$ equipped with the scalar product $(M,N)_{\mathbb{H}^2}$ is a Hilbert space.*

pf. Let $(M^n)_{n \geq 1}$ be a sequence in $\mathbb{H}^2$ which is Cauchy for the norm. Then $\lim_{n,m \rightarrow \infty} E[(M_\infty^m - M_\infty^n)^2] = \lim(M^n - M^m, M^n - M^m)_{\mathbb{H}^2} = 0$. Consequently, it converges in L^2 to a limit, which we denote by Z. By Doob's inequality in L^2, $\forall m, n > 0$, $E[\sup_{t \geq 0}(M_t^n - M_t^m)^2] \leq 4E[(M_\infty^n - M_\infty^m)^2]$. Thus $\lim_{m,n \rightarrow \infty} E[\sup_{t \geq 0}(M_t^n - M_t^m)^2] = 0$. Hence $\forall t \geq 0$, M_t^n converges in L^2. Then find an increasing sequence $n_k \uparrow \infty$ s.t. $E[\sum_{k=1}^\infty \sup_{t \geq 0} |M_t^{n_k} - M_t^{n_{k+1}}|] \leq \sup_{k=1}^\infty E[\sup(M_t^{n_k} - M_t^{n_{k+1}})^2]^{1/2} < \infty$
$\rightarrow \sum_{k=1}^\infty \sup_{t \geq -} |M_t^{n_k} - M_t^{n_{k+1}}| < \infty$ so $(M_t^{n_k})$ converges uniformly in $\mathbb{R}_{\geq 0}$ a.s. to a limit denoted by (M_t). On the zero probability set where the uniform convergence does not hold, take $M_t = 0$, $\forall t \geq 0$. The limiting process M has continuous sample paths and is adapted given the filtration is complete. Furthermore from the L^2-convergence of (M_∞^n) to Z, we immediately get by passing to the limit in the identity $M_t^{n_k} = E[M_\infty^{n_k} | \mathscr{F}_t]$ that $M_t = E[Z | \mathscr{F}_t]$. Hence (M_t) is a continuous martingale and is bounded in L^2, so that $M \in \mathbb{H}^2$. The a.s. uniform convergence of (M^{n_k})

ensures $M_\infty = \lim M_\infty^{n_k} = Z$ a.s. Finally the L^2-convergence of (M_∞^n) to $Z = M_\infty$ shows the sequence (M^n) converges to M in $\mathbb{H}^2$.

Denote the progressive σ-field on $\Omega \times \mathbb{R}_{\geq 0}$ by $\mathscr{P}$ and if $M \in \mathbb{H}^2$, let $L^2(M)$ be the set of all progressive processes H s.t. $E[\int_0^\infty H_s^2 d < M,M >_s] < \infty$ with the convention that two progressive processes H, H' satisfying this integrability condition are identified if $H_s = H'_s = 0$, $d < M,M >_s$ a.e., a.s.
$\rightarrow$ can be viewed as an ordinary L^2 space, namely

$$L^2(M) = L^2(\Omega \times \mathbb{R}_{\geq 0}, \mathscr{P}, dPd < M,M >_s)$$

where $dPd < M,M >_s$ refers to the finite measure on $(\Omega \times \mathbb{R}_{\geq 0}, \mathscr{P})$ that assigns the mass $E[\int_0^\infty \mathbb{1}_A(\omega,s) d < M,M >_s]$ to a set $A \in \mathscr{P}$.

The space $L^2(M)$ is a Hilbert space for the scalar product $(H,K)_{L^2(M)} = E[\int_0^\infty H_s K_s d < M,M >_s]$ and the associated norm is $||H||_{L^2(M)} = (E[\int_0^\infty H_s^2 d < M,M >_s])^{1/2}$.

Definition 4.1.1. *An elementary process is a progressive process of the form*

$$H_s(\omega) = \sum_{i=0}^{p-1} H_{(i)}(\omega) \mathbb{1}_{(t_i,t_{i+1}]}(s)$$

where $0 = t_0 < \cdots < t_p$ *and* $\forall i \in \{0,\ldots,p-1\}$, $H_{(i)}$ *is a bounded* $\mathscr{F}_{t_i}$*-measurable random variable.*

The set $\mathscr{E}$ of all elementary processes forms a linear subspace of $L^2(M)$.

Proposition 4.1.2. $\forall M \in \mathbb{H}^2$, $\mathscr{E}$ *is dense in* $L^2(M)$.

pf. The fact that if $\forall K \in L^2(M)$ is orthogonal to $\mathscr{E}$, then $K = 0$ implies the proposition by the elementary Hilbert space theory.

- **CLAIM : if $K \in L^2(M)$ is orthogonal to $\mathscr{E}$, $K = 0$.**
  pf. Assume $K \in L^2(M)$ is orthogonal to $\mathscr{E}$ and set $\forall t \geq 0$, $X_t = \int_0^t K_u d < M,M >_u$. Then $E[\int_0^t |K_u| d < M,M >_u] \leq (E[\int_0^t (K_u)^2 d < M,M >_u])^{1/2} \times (E[< M,M >_\infty])^{1/2} < \infty$ since $M \in \mathbb{H}^2$, $K \in L^2(M)$. Then the above process $(X_t)_{t\geq 0}$ is well-defined as a finite variation process.
  Let $0 \leq s < t$, let F be a bounded $\mathscr{F}_s$-measurable random variable and let $H \in \mathscr{E}$ be the elementary process defined by $H_r(\omega) = F(\omega)\mathbb{1}_{(s,t]}(r)$. Writing $(H,K)_{L^2(M)} = 0$, we get $E[F \int_s^t K_u d < M,M >_u] = 0$. Thus $E[F(X_t - X_s)] = 0$, $\forall s < t$ and every bounded $\mathscr{F}_s$-measurable variable F. Since X is adapted, and we know that $X_r \in L^1$, $\forall r \geq 0$, X is a continuous martingale. On the other hand, X is also a finite variation process so $X = 0$. Thus $\int_0^t K_u d < M,M >_u = 0$, $\forall t \geq 0$, a.s. which implies a.s. the signed measure having density K_u w.r.t. $d < M,M >_u$ is the zero measure, which is only possible if $K_u = 0$, $d < M,M >_u$ a.e., a.s. or equivalently $K = 0$ in $L^2(M)$.

If $M \in \mathbb{H}^2$, $H^T \in \mathbb{H}^2$. Furthermore if $H \in L^2(M)$, the process $\mathbb{1}_{[0,T]}H$ defined by $(\mathbb{1}_{[0,T]}H)_s(\omega) = \mathbb{1}_{0\le s\le T(\omega)}H_s(\omega)$ also belongs to $L^2(M)$.

Theorem 4.1.1. *Let $M \in \mathbb{H}^2$. $\forall H \in \mathscr{E}$ of the form $H_s(\omega) = \sum_{i=0}^{p-1} H_{(i)}(\omega)\mathbb{1}_{(t_i,t_{i+1}]}(s)$, the formula*

$$(H\cdot M)_t = \sum_{i=0}^{p-1} H_{(i)}(M_{t_{i+1}\wedge t} - M_{t_i\wedge t})$$

defines a process $H\cdot M \in \mathbb{H}^2$. The mapping $H \mapsto H\cdot M$ extends to an isometry from $L^2(M)$ into $\mathbb{H}^2$. Furthermore $H\cdot M$ is the unique martingale of $\mathbb{H}^2$ that satisfies the property $< H\cdot M, N >= H\cdot < M,N >$, $\forall N \in \mathbb{H}^2$. If T is a stopping time,

$$(\mathbb{1}_{[0,T]}H)\cdot M = (H\cdot M)^T = H\cdot M^T$$

pf.

- **the mapping $H \mapsto H\cdot M$ is an isometry from $\mathscr{E}$ into $\mathbb{H}^2$.**

  fix $H \in \mathscr{E}$ of the form given in the theorem, and for every $i \in \{0,\dots,p-1\}$, set $M_t^i = H_{(i)}(M_{t_{i+1}\wedge t} - M_{t_i\wedge t})$, $\forall t \ge 0$. Then M^i is a continuous martingale and this martingale belongs to $\mathbb{H}^2$. Thus $H\cdot M = \sum_{i=0}^{p-1} M^i$ is also a martingale in $\mathbb{H}^2$. Continuous martingale M^i are orthogonal and their respective quadratic variations are given by
  $< M^i, M^i >_t = H_{(i)}^2(< M,M >_{t_{i+1}\wedge t} - < M,M >_{t_i\wedge t})$, so $< H\cdot M, H\cdot M >_t$
  $= \sum_{i=0}^{p-1} H_{(i)}^2(< M,M >_{t_{i+1}\wedge t} - < M,M >_{t_i\wedge t}) = \int_0^t H_s^2 d < M,M >_s$. Consequently
  $||H\cdot M||_{\mathbb{H}^2}^2 = E[< H\cdot M, H\cdot M >_\infty] = E[\int_0^\infty H_s^2 d < M,M >_s] = ||H||_{L^2(M)}^2$. By linearity, this implies $H\cdot M = H'\cdot M$ if H' is another elementary process that is identified with H in $L^2(M)$. Thus the above mapping makse sense form $\mathscr{E}$ viewed as a subspace of $L^2(M)$ into $\mathbb{H}^2$. Since it preserves the norm, it is an isometry from $\mathscr{E}$ into $\mathbb{H}^2$. Since $\mathscr{E}$ is dense in $L^2(M)$, and $\mathbb{H}^2$ is a Hilbert space, this mapping can be extended in a unique way to an isometry from $L^2(M)$ to $\mathbb{H}^2$.

- $< H\cdot M, N >= H\cdot < M,N >$, $\forall N \in \mathbb{H}^2$

  if $H \in L^2(M)$, by the Kunita-Watanabe inequality, $E[\int_0^\infty |H_s||d < M,N >_s|] \le ||H||_{L^2(M)}||N||_{\mathbb{H}^2} < \infty$ and thus the variable $\int_0^\infty H_s d < M,N >_s$
  $= (H\cdot < M,N >)_\infty$ is well-defined and in L^1.

  - H is an elementary process of the form given in the theorem

    let M^i, $0 \le i \le p-1$ continuous martingales. $\forall i \in \{0,\dots,p-1\}$,

$< H \cdot M, N >= \sum_{i=0}^{p-1} < M^i, N >$ where $< M^i, N >_t = H_{(i)}(< M, N >_{t_{i+1} \wedge t}$ $- < M, N >_{t_i \wedge t})$. Then $< H \cdot M, N >_t = \sum_{i=0}^{p-1} H_{(i)}(< M, N >_{t_{i+1} \wedge t} - < M, N >_{t_i \wedge t}$ $) = \int_0^t H_s d < M, N >_s$

– general H and general N

the linear mapping $X \mapsto < X, N >_\infty$ is continuous from $\mathbb{H}^2$ to L^1. By the Kunita-Watanabe inequality, $E[| < X, N >_\infty |] \leq E[< X, X >_\infty]^{1/2} E[< N, N >_\infty]^{1/2} = ||N||_{\mathbb{H}^2} ||X||_{\mathbb{H}^2}$. If (H^n) is a sequence in $\mathscr{E}$ s.t. $H^n \to H \in L^2(M)$, $< H \cdot M, N >_\infty = \lim < H^n \cdot M, N >_\infty = \lim (H^n \cdot < M, N >)_\infty$ $= (H \cdot < M, N >)_\infty$ where the convergence holds in L^1.
Replacing N by the stopped martingale N^t gives $< H \cdot M, N >_t$ $= (H \cdot < M, N >)_t$ so the statement follows.

- **if T is a stopping time, $(\mathbb{1}_{[0,T]} H) \cdot M = (H \cdot M)^T = H \cdot M^T$.**

if $N \in \mathbb{H}^2$, $< (H \cdot M)^T, N >_t = < H \cdot M, N >_{t \wedge T} = (H \cdot < M, N >)_{t \wedge T} = (\mathbb{1}_{[0,T]} H \cdot < M, N >)_t$ so the stopped martingale $(H \cdot M)^T$ satisfies the characteristic property of the stochastic integral $(\mathbb{1}_{[0,T]} H) \cdot M$. Also $< H \cdot M^T, N >= H \cdot < M^T, N >= H \cdot < M, M >^T = \mathbb{1}_{[0,T]} H \cdot < M, N >$.

$\to (H \cdot M)_t = \int_0^t H_s dM_s$ and $H \cdot M$ is called the stochastic integral of H w.r.t. M.
$\to N \mapsto E[(H \cdot < M, N >)_\infty]$ yields a continuous linear form on $\mathbb{H}^2$, and therefore there exists a unique martingale $H \cdot M \in \mathbb{H}^2$ s.t. $E[(H \cdot < M, N >)_\infty] = (H \cdot M, N)_{\mathbb{H}^2}$ $= E[< H \cdot M, N >_\infty]$.
$\to < H \cdot M, N >= H \cdot < M, N > \Leftrightarrow < \int_0^\cdot H_s dM_s, N >_t = \int_0^t H_s d < M, N >_s$ i.e. the stochastic integral commutes with the bracket.

Proposition 4.1.3. *Let $H \in L^2(M)$. If K is a progressive process, $KH \in L^2(M)$ iff $K \in L^2(H \cdot M)$. If the latter properties hold, $(KH) \cdot M = K \cdot (H \cdot M)$.*

pf. $E[\int_0^\infty K_s^2 H_s^2 d < M, M >_s] = E[\int_0^\infty K_s^2 d < H \cdot M, H \cdot M >_s]$ by the previous theorem. $\forall N \in \mathbb{H}^2$,
$< (KH) \cdot M, N >= KH \cdot < M, N >= K \cdot (H \cdot < M, N >) = K \cdot < H \cdot M, N >$. By the uniqueness in the previous theorem $(KH) \cdot M = K \cdot (H \cdot M)$.

Let $M, N \in \mathbb{H}^2, H \in L^2(M)$, $K \in L^2(N)$.Since $H \cdot M$, $K \cdot N$ are martingales in $\mathbb{H}^2$, $\forall t \in [0, \infty]$,

$$E[\int_0^t H_s dM_s] = 0$$

$$E[(\int_0^t H_s dM_s)(\int_0^t K_s dN_s)] = E[\int_0^t H_s K_s d < M, N >_s]$$

Furthermore, since $H \cdot M$ is a martingale, $\forall 0 \le s < t \le \infty$, $E[\int_0^t H_r dM_r | \mathscr{F}_s] = \int_0^s H_r dM_r$.

4.1.1 Stochastic Integrals for Local Martingales

If M is a continuous local martingale, we write $L^2_{loc}(M)$ for the set of all progressive processes H s.t. $\int_0^t H_s^2 d < M, M >_s < \infty$, $\forall t \ge 0$ a.s.

Theorem 4.1.2. *Let M be a continuous local martingale. $\forall H \in L^2_{loc}(M)$, there exists a unique continuous local martingale with initial value 0, which is denoted by $H \cdot M$ s.t. for every continuous local martingale N*

$$< H \cdot M, N >= H \cdot < M, N >$$

If T is a stopping time,

$$(\mathbb{1}_{[0,T]} H) \cdot M = (H \cdot M)^T = H \cdot M^T$$

If $H \in L^2_{loc}(M)$ and K is a progressive process, $K \in L^2_{loc}(H \cdot M)$ iff $HK \in L^2_{loc}(M)$ and then $H \cdot (K \cdot M) = HK \cdot M$. If $M \in \mathbb{H}^2$ and $H \in L^2(M)$, the definition of $H \cdot M$ is consistent with that of the previous theorem.

pf. Wlog assume $M_0 = 0$. Also assume $\int_0^t H_s^2 d < M, M >_s < \infty$, $\forall t \ge 0, \forall \omega \in \Omega$. $\forall n \ge 1$, set $T_n = \int \{t \ge 0 : \int_0^t (1 + H_s^2) d < M, M >_s \ge n\}$ so that (T_n) is a sequence of stopping times that increase to ∞. Since $< M^{T_n}, M^{T_n} >_t = < M, M >_{t \wedge T_n} \le n$, the stopped martingale $M^{T_n} \in \mathbb{H}^2$. Furthermore $\int_0^\infty H_s^2 d < M^{T_n}, M^{T_n} >_s$
$= \int_0^{T_n} H_s^2 d < M, M >_s \le n \to H \in L^2(M^{T_n})$ and the definition of $H \cdot M^{T_n}$ make sense. Also $\forall m > n$, $H \cdot M^{T_n} = (H \cdot M^{T_m})^{T_n}$ by the previous theorem. Thus there exists a unique process denoted by $H \cdot M$ s.t. $\forall n$, $(H \cdot M)^{T_n} = H \cdot M^{T_n}$. This has continuous sample paths and is adapted since $(H \cdot M)_t = \lim (H \cdot M^{T_n})_t$. Since $(H \cdot M)^{T_n}$ are martingales in $\mathbb{H}^2$, $H \cdot M$ is a continuous local martingale.
Assume N is a continuous local martingale s.t. $N_0 = 0$. $\forall n \ge 1$, set $T'_n = \inf\{t \ge 0 : |N_t| \ge n\}$ and $S_t = T_n \wedge T'_n$. Then $N^{T'_n} \in \mathbb{H}^2$, $< H \cdot M, N >^{S_n} = < (H \cdot M)^{T_n}, N^{T'_n} >$
$= < H \cdot M^{T_n}, N^{T'_n} >= H \cdot < M^{T_n}, N^{T'_n} >= H \cdot < M, N >^{S_n} = (H \cdot < M, N >)^{S_n}$ which gives the equality $< H \cdot M, N >= H \cdot < M, N >$ since $S_n \uparrow \infty$ as $n \to \infty$. Then we can get the desired equality.
$(\mathbb{1}_{[0,T]} H) \cdot M = (H \cdot M)^T = H \cdot M^T$ and $H \cdot (K \cdot M) = HK \cdot M$ follows by the previous theorem and proposition.
If $M \in \mathbb{H}^2$, $H \in L^2(M)$, $< H \cdot M, H \cdot M >= H^2 \cdot < M, N >$ by the previous the previous statement. Thus $H \cdot M \in \mathbb{H}^2$. Then the characteristic property of previous theorem shows the definitions are consistent.
$\to (H \cdot M)_t = \int_0^t H_s dM_s$ is the stochastic integral for a continuous local martingale.

e.g. **Wiener Integral**
Suppose B is an $(\mathscr{F}_t)$-Brownian motion and $h \in L^2(\mathbb{R}_{\geq 0}, \mathscr{B}(\mathbb{R}_{\geq 0}, dt)$ is a deterministic square integrable function. Then the Wiener integral is $\int_0^t h(s)dB_s = G(f\mathbb{1}_{[0,t]})$ where G is the Gaussian white noise associated with B. It coincides with the stochastic integral $(h \cdot B)_t$.

4.1.2 Stochastic Integral for Semimartingales

Definition 4.1.2. *A progressive process H is said to be locally bounded if $\forall t \geq 0$, $\sup_{s \leq t} |H_s| < \infty$ a.s.*

→any adapted process with continuous sample paths is a locally bounded progressive process. If H is a locally bounded progressive process, for every finite variation process V, $\forall t \geq 0$, $\int_0^t |H_s||dV_s| < \infty$ a.s. and similarly $H \in L^2_{loc}(M)$ for every continuous local martingale M.

Definition 4.1.3. *Let X be a continuous semimartingale and let $X = M + V$ be its canonical decomposition. If H is a locally bounded progressive process, the stochastic integral $H \cdot X$ is the continuous semimartingale with canonical decomposition*

$$H \cdot X = H \cdot M + H \cdot V$$
$$(H \cdot X)_t = \int_0^t H_s dX_s$$

Properties

- the mapping $(H, X) \mapsto H \cdot X$ is bilinear
- $H \cdot (K \cdot X) = (HK) \cdot X$ if H, K are progressive and locally bounded
- for every stopping time T, $(H \cdot X)^T = H\mathbb{1}_{[0,T]} \cdot X = H \cdot X^T$
- if X is a continuous local martingale(finite variation process resp.), the the same holds for $H \cdot X$
- if H is of the form $H_s(\omega) = \sum_{i=0}^{p-1} H_{(i)}(\omega)\mathbb{1}_{(t_i, t_{i+1}]}(s)$ where $0 = t_0 < \cdots < t_p$ and $\forall i \in \{0, 1, \ldots, p-1\}$, $H_{(i)}$ is $\mathscr{F}_{t_i}$-measurable, then

$$(H \cdot X)_t = \sum_{i=0}^{p-1} H_{(i)}(X_{t_{i+1} \wedge t} - X_{t_i \wedge t})$$

pf. $\forall n \geq 1$, set $T_n = \inf\{t \geq 0 : |H_t| \geq n\} = \inf\{t_i : |H_{(i)}| \geq n\}$. Then it is a stopping time and $T_n \uparrow \infty$ as $n \to \infty$. For every n, $H_s\mathbb{1}_{[0,T_n]}(s) = \sum H^n_{(i)}\mathbb{1}_{(t_i, t_{i+1}]}(s)$

where the random variable $H^n_{(i)} = H_{(i)}\mathbb{1}_{T_n > t_i}$ satisfiy the same properties as the $H_{(i)}$'s and additionally are bounded by n. Hence $H\mathbb{1}_{[0,T_n]}$ is an elementary process and by the definition of the stochastic integral $(H \cdot M)_{t\wedge T_n}$ $= (H\mathbb{1}_{[0,T_n]} \cdot M)_t = \sum H^n_{(i)}(M_{t_{i+1}\wedge t} - M_{t_i \wedge t})$. By letting n tend to infinity, the statement follows.

Proposition 4.1.4 (Dominated Convergence Theorem for Stochastic Integrals)**.** *Let $X = M + V$ be the canonical decomposition of a continuous semimartingale X and let $t > 0$. Let $(H^n)_{n\geq 1}$ and H be locally bounded progressive processes, and let K be a nonnegative progressive process. Assume that the followings hold a.s.:*

1. $H^n_s \to H_s$ *as* $n \to \infty$, $\forall s \in [0,t]$
2. $|H^n_s| \leq K_s$, $\forall n \geq 1$, $\forall s \in [0,t]$
3. $\int_0^t (K_s)^2 d < M,M >_s < \infty$ *and* $\int_0^t K_s |dV_s| < \infty$

Then

$$\int_0^t H^n_s dX_s \to \int_0^t H_s dX_s$$

as $n \to \infty$ in probability.

pf. The a.s. convergence $\int_0^t H^n_s dV_s \to \int_0^t H_s dV_s$ follows from the usual dominated convergence theorem.
For every integer $p \geq 1$, consider the stopping time $T_p \equiv \inf\{r \in [0,t] : \int_0^r (K_s)^2$ $d < M,M >_s \geq p\} \wedge t$ and observe $T_p = t$ for all large enough p a.s. by 3. Then $E[(\int_0^{T_p} H^n_s dM_s - \int_0^{T_p} H_s dM_s)^2] \leq E[\int_0^{T_p} (H^n_s - H_s)^2 d < M,M >_s] \to 0$ as $n \to \infty$ by the usual dominated convergence theorem with 1 and 2 and the fact that $\int_0^{T_p} (K_s)^2$ $d < M,M >_s \leq p$. Since $P(T_p = t) \to 1$ as $p \to \infty$ so the proposition follows.

Proposition 4.1.5. *Let X be a continuous semimartingale, and let H be an adapted process with continuous sample paths. Then $\forall t > 0$, for every sequence $0 = t^n_0 < \cdots < t^n_{p_n} = t$ of subdivisions of $[0,t]$ whose mesh tends to 0, we have*

$$\lim_{n\to\infty} \sum_{i=0}^{p-1} H_{t_i}(X_{t^n_{i+1}} - X_{t^n_i}) = \int_0^t H_s dX_s$$

in probability.

pf. $\forall n \geq 1$, define $H^n = \begin{cases} H_{t^n_i} & t^n_i < s \leq t^n_{i+1} \\ H_0 & s = 0 \\ 0 & s > t \end{cases}$. Then it is progressive. If we take $K_s = \max_{0\leq r\leq s} |H_s|$, all the conditions in the previous proposition hold. It is also a

locally bounded progressive process. Hence $\int_0^t H_s^n s X_s \to \int_0^t H_s dX_s$ in probability. Then by the property of the stochastic integral of semimartingales, $\int_0^t H_s^n dX_s = \sum_{i=0}^{p_n-1} H_{t_i^n}(M_{t_{i+1}^n} - M_{t_i^n})$.
$\to$ the result will fail if we replace $H_{t_i^n}$(the left end) by $H_{t_{i+1}^n}$(the right end).
pf. $H_t = X_t$ and assume the sequence of subdivisions (t_i^n) is increasing. By the proposition $\lim \sum X_{t_i^n}(X_{t_{i+1}^n} - X_{t_i^n}) = \int_0^t X_s dX_s$ in probability. On the other hand, $\sum_{i=0}^{p_n-1} X_{t_{i+1}^n} - X_{t_i^n}) = \sum X_{t_i^n}(X_{t_{i+1}^n} - X_{t_i^n}) + \sum (X_{t_{i+1}^n} - X_{t_i^n})^2 \to \lim \sum X_{t_{i+1}^n}(X_{t_{i+1}^n} - X_{t_i^n}) = \int_0^t X_s dX_s + < X, X >_t \neq \int_0^t X_s dX_s$ in probability unless the martingale part of X is degenerate. .

4.2 Ito's Formula

Theorem 4.2.1 (Ito's Formula)**.** *Let $X^1, \ldots, X^p$ be p continuous semimartingales, and let F be a twice continuously differentiable real function on $\mathbb{R}^p$. Then $\forall t \geq 0$,*

$$F(X_t^1, \ldots, X_t^p) = F(X_0^1, \ldots, X_0^p) + \sum_{i=1}^{p} \int_0^t \frac{\partial F}{\partial x^i}(X_s^1, \ldots, X_s^p) dX_s^i$$
$$+ \frac{1}{2} \sum_{i,j=1}^{p} \int_0^t \frac{\partial^2 F}{\partial x^i \partial x^j}(X_s^1, \ldots, X_s^p) d < X^i, X^j >_s$$

pf.

- $p = 1$
  Write $X = X^1$ for simplicity. Fix $t > 0$ and consider an increasing sequence $0 = t_0^n < \cdots < t_{p_n}^n = t$ of subdivisions of $[0,t]$ whose mesh tends to 0. Then for every n, $F(X_t) = F(X_0) + \sum_{i=0}^{p_n-1}(F(X_{t_{i+1}^n} - F(X_{t_i^n}))$. $\forall i \in \{0, \ldots, p_n - 1\}$, by the Taylor-Lagrange formula to the function $\theta \mapsto F(X_{t_i^n} - \theta(X_{t_{i+1}^n} - X_{t_i^n}))$ between $\theta = 0$ and $\theta = 1$ and we get $F(X_{t_{i+1}^n}) - F(X_{t_i^n}) = F'(X_{t_i^n})(X_{t_{i+1}^n} - X_{t_i^n}) + \frac{1}{2} f_{n,i}(X_{t_{i+1}^n} - X_{t_i^n})^2$ where the quantity $f_{n,i}$ can be written as $F''(X_{t_i^n} + c(X_{t_{i+1}^n} - X_{t_i^n}))$ for some $c \in [0,1]$. By the previous proposition with $H_s = F'(X_s)$, $\lim_{n\to\infty} \sum_{i=0}^{p_n-1} F'(X_{t_i^n})(X_{t_{i+1}^n} - X_{t_i^n}) = \int_0^t F'(X_s) dX_s$ in probability.
  $\sup_{0 \leq i \leq p_n-1} |f_{n,i} - F''(X_{t_i^n})| \leq \sup_{0 \leq i \leq p_n-1}(\sup_{x \in [X_{t_i^n} \wedge X_{t_{i+1}^n} \vee X_{t_{i+1}^n}]} |F''(x) - F''(X_{t_i^n})|)$ $\to 0$ as $n \to \infty$ by the uniform continuity of F'' over a compact interval. Since $\sum_{i=0}^{p_n-1}(X_{t_{i+1}^n} - X_{t_i^n})^2$ converges in probability, $|\sum_{i=0}^{p_n-1} f_{n,i}(X_{t_{i+1}^n} - X_{t_i^n})^2 - \sum_{i=0}^{p_n-1} F''(X_{t_i^n})(X_{t_{i+1}^n} - X_{t_i^n})^2| \to 0$ as $n \to \infty$ in probability.
  $\sum_{i=0}^{p_n-1} F''(X_{t_i^n})(X_{t_{i+1}^n} - X_{t_i^n})^2 = \int_{[0,t]} F''(X_s)\mu_n(ds)$ where μ_n is the random measure on $[0,t]$ defined by $\mu_n(dr) \equiv \sum_{i=0}^{p_n-1}(X_{t_{i+1}^n} - X_{t_i^n})^2 \delta_{t_i^n}(dr)$. Write D for

the dense subset of $[0,t]$ that consists of all t_i^n for $n \geq 1$, $0 \leq i \leq p_n$. Then $\forall r \in D$, $\mu_n([0,r]) \to < X,X >_r$ as $n \to \infty$ in probability. By a diagonal extraction, we can find a subsequence of values of n s.t. along this subsequence we have $\forall r \in D$, $\mu_n([0,r]) \to < X,X >_r$ as $n \to \infty$ a.s. which implies the sequence $\mu_n \to \mathbb{1}_{[0,t]}(r)d < X,X >_r$ as $n \to \infty$ a.s. in the sence of weak convergence of finite measures. Thus $\int_{[0,t]} F''(X_s)\mu_n(ds) \to \int_0^t F''(X_s)d < X,X >_s$ along the chosen subsequence. So the formula follows.

- $p > 1$
  The Taylor-Lagrange formula applied for $\forall n \geq 1$, $\forall i \in \{0,\dots,p_n - 1\}$ to the function $\theta \mapsto F(X^1_{t_i^n} + \theta(X^1_{t_{i+1}^n} - X^1_{t_i^n}),\dots,X^p_{t_i^n} + \theta(X^p_{t_{i+1}^n} - X^p_{t_i^n}))$ gives

$$F(X^1_{t_{i+1}^n},\dots,F^p_{t_{i+1}^n}) - F(X^1_{t_i^n},\dots,F^p_{t_i^n}) = \sum_{k=1}^p \frac{\partial F}{\partial x^k}(X^1_{t_i^n},\dots,X^p_{t_i^n})(X^k_{t_{i+1}^n} - X^k_{t_i^n}) + \sum_{k,l=1}^p \frac{f^{k,l}_{n,i}}{2}(X_{t^k_{i+1}} - X^k_{t_i^n})(X^l_{t_{i+1}^n} - X^l_{t^k_i})$$

  where $\forall k,l \in \{1,\dots,p\}$, $f^{k.l}_{n,i} = \frac{\partial^2 F}{\partial x_k \partial x_l}(X_{t_i^n} + c(X_{t_{i+1}^n} - X_{t_i^n}))$ for some $c \in [0,1]$. By the slight modification of the base case, at least along a suitable sequence of values of n, $\forall k,l \in \{1,\dots,p\}$, $\lim_{n\to\infty} \sum_{i=0}^{p_n-1} f^{k,l}_{n,i}(X^k_{t_{i+1}^n} - X^k_{t_i^n})(X^l_{t_{i+1}^n} - X^l_{t_i^n})$ $= \int_0^t \frac{\partial^2 F}{\partial x_k \partial x_l}(X^1_s,\dots,X^p_s)d < X^k,X^l >_s$ in probability.

e.g.

- $p = 2$, $F(x,y) = xy$. If X,Y are two continuous semimartingales, $X_tY_t = X_0Y_0 + \int_0^t X_s dY_s + \int_0^t Y_s dX_s + < X,Y >_t$. In particular if $X = Y$, $X_t^2 = X_0^2 + 2\int_0^t X_s dX_s + < X,X >_t$. If $X = M$ is a continuous local martingale, $M^2 - < M,M >$ is a continuous local martingale. Thus $M_0^2 + 2\int_0^t M_s dM_s$.

- let B be an $(\mathscr{F}_t)$-Brownian motion. An $(\mathscr{F}_t)$-Brownian motion is a continuous local martingale. Then $F(B_t) = F(B_0) + \int_0^t F'(B_s)dB_s + \frac{1}{2}\int_0^t F''(B_s)ds$. Taking $X_t^1 = t$, $X_t^2 = B_t$, we can get for every twice continuously differentiable function $F(t,B_t) = F(0,B_0) + \int_0^t \frac{\partial F}{\partial x}(s,B_s)dB_s + \int_0^t(\frac{\partial F}{\partial t} + \frac{1}{2}\frac{\partial^2 F}{\partial x^2})(s,B_s)ds$. If $B_t = (B_t^1,\dots,B_t^d)$ be a d-dimensional $(\mathscr{F}_t)$-Brownian motion. Then B^j are $(\mathscr{F}_t)$-Brownian motion. Since $\forall i \neq j$, $< B^i,B^j >= 0$, $F(B_t^1,\dots,B_t^d) = F(B_0^1,\dots,B_0^d) + \sum_{i=1}^d \int_0^t \frac{\partial F}{\partial x_i}(B_s^1,\dots,B_s^d)dB_s^i + \frac{1}{2}\int_0^t \Delta F(B_s^1,\dots,B_s^d)ds$.

Proposition 4.2.1. *Let M be a continuous local martingale and $\forall \lambda \in \mathbb{C}$ let $\mathscr{E}(\lambda M)_t = \exp(\lambda M_t - \frac{\lambda^2}{2} < M,M >_t)$. The process $\mathscr{E}(\lambda M)$ is a complex continuous local martingale, which can be written in the form*

$$\mathscr{E}(\lambda M)_t = e^{\lambda M_0} + \lambda \int_0^t \mathscr{E}(\lambda M)_s dM_s$$

pf. If $F(r,x)$ is a twice continuously differentiable function on $\mathbb{R}^2$, Ito's formula gives $F(<M,M>_t,M_t) = F(0,M_0)+\int_0^t \frac{\partial F}{\partial x}(<M,M>_s,M_s)dM_s+\int_0^t(\frac{\partial F}{\partial r}+\frac{1}{2}\frac{\partial^2 F}{\partial x^2})(<M,M>_s$ $,M_s)d<M,M>_s$. Hence $F(<M,M>_t,M_t)$ is a continuous local martingale as soon as F satisfies the equation

$$\frac{\partial F}{\partial r}+\frac{1}{2}\frac{\partial^2 F}{\partial x^2}=0$$

This holds for $F(r,x) = \exp(\lambda x - \frac{\lambda^2}{2}r)$. For this choice of F, we have $\frac{\partial F}{\partial x} = \lambda F$, which leads to the proposition.

4.2.1 Applications of Ito's Formula

- **Levy's Representation of the Brownian Motion**

  Theorem 4.2.2 (Levy's Theorem). *Let $X = (X^1,\dots,X^d)$ be an adapted process with continuous sample paths. TFAE*

  1. *X is a d-dimensional $(\mathscr{F}_t)$-Brownian motion*
  2. *The process $X^1,\dots,X^d$ are continuous local martingales and $<X^i,X^j>_t$*

     $= \delta_{ij}t$ for every $i,j \in \{1,\dots,d\}$.

  In particular, a continuous local martingale M is an $(\mathscr{F}_t)$-Brownian motion iff $<M,M>_t = t$, $\forall t \geq 0$ or equivalently iff $M_t^2 - t$ is a continuous local martingale.

  pf.

  - $1 \to 2$ follows by previous proposition.
  - $2 \to 1$

    let $\xi = (\xi_1,\dots,\xi_d) \in \mathbb{R}^d$. Then $\xi \cdot X_t = \sum_{j=1}^d \xi_j X_t^j$ is a continuous local martingale with quadratic variation $\sum_{j=1}^d\sum_{k=1}^d \xi_j\xi_k <X^j,X^k>_t = |\xi|^2 t$. By the previous proposition, $\exp(i\xi\cdot X_t + \frac{1}{2}|\xi|^2 t)$ is a complex continuous local martingale. This is bounded on every interval $[0,a]$, $a>0$ and is therefore a martingale in the sense that its real and imaginary parts are both martingales. Hence $\forall 0 \leq s < t$, $E[\exp(i\xi\cdot X_t + \frac{1}{2}|\xi|^2 t)|\mathscr{F}_s] = \exp(i\xi\cdot X_s + \frac{1}{2}|\xi|^2 s)$ and thus $E[\exp(i\xi\cdot(X_t - X_s))|\mathscr{F}_s]$ $= \exp(-\frac{1}{2}|\xi|^2(t-s))$. This implies $\forall A \in \mathscr{F}_s$,

$$E[\mathbb{1}_A \exp(i\xi\cdot(X_t - X_s))] = P(A)\exp(-\frac{1}{2}|\xi|^2(t-s))$$

Taking $A = \Omega$, we get $X_t - X_s$ is a centered Gaussian vector with covariance matrix $(t-s)I$. Furthermore, fix $A \in \mathscr{F}_s$ with $P(A) > 0$ and write P_A for the conditional probability measure $P_A(\cdot) = P(A)^{-1}P(\cdot \cap A)$. Then $P_A[\exp(i\xi \cdot (X_t - X_s))] = \exp(-\frac{1}{2}|\xi|^2(t-s))$ which means the law of $X_t - X_s$ under P_A is the same as its law under P. Thus for any nonnegative measurable function f on $\mathbb{R}^d$,

$$P_A[f(X_t - X_s)] = E[f(X_t - X_s)] \Leftrightarrow E[\mathbb{1}_A f(X_t - X_s)] = P(A)E[f(X_t - X_s)]$$

This holds for any $A \in \mathscr{F}_s$ and therefore $X_t - X_s$ is independent of $\mathscr{F}_s$. Then if $t_0 = 0 < t_1 < \cdots < t_p$, $X_{t_1} - X_{t_0}, \ldots, X_{t_p} - X_{t_{p-1}}$ are independent. Since the components of each of these vectors are independent r.v.s, all variables $X^j_{t_k} - X^j_{t_{k-1}}$ are independent and it is distributed according to $\mathscr{N}(0, t_k - t_{k-1})$. This implies $X - X_0$ is a d-dimensional Brownian motion started from 0. Since we also know $X_t - X_0$ is independent of X_0, X is a d-dimensional Brownian motion. Since X is adapted and has independent increments w.r.t. $(\mathscr{F}_t)$, X is a d-dimensional $(\mathscr{F}_t)$-Brownian motion

– **Continuous Martingales as Time-Changed Brownian Motions**

Lemma 4.2.1. *a.s.* $\forall 0 \le a < b$, $M_t = M_a$, $\forall t \in [a,b] \Leftrightarrow < M,M >_b = < M,M >_a$

pf. By the continuity of the sample paths of M and $< M,M >$, if for any fixed a,b s.t. $0 \le a < b$, $\{M_t = M_a,\ \forall t \in [a,b] = \{< M,M >_b = < M,M >_a\}$ a.s. the lemma follows.

* $\subseteq$

  by the approximations of $< M,M >$ the containment follows.

* $\supseteq$

  consider the continuous local martingale $N_t = M_t - M_{t\wedge a}$. Then

  $< N,N >_t = < M,M >_t - < M,M >_{t\wedge a}$. $\forall \epsilon > 0$, introduce the stopping time $T_\epsilon = \inf\{t \ge 0 :< N,N >_t \ge \epsilon\}$. Then N^{T_ϵ} is a martingale in $\mathbb{H}^2$. Fix $t \in [a,b]$. Then $E[N^2_{t\wedge T_\epsilon}] = E[< N,N >_{t\wedge T_\epsilon}] \le \epsilon$. Hence considering the event $A \equiv \{< M,M >_b = < M,M >_a\} \subseteq \{T_\epsilon \ge b\}$, $E[\mathbb{1}_A N_t^2] = E[\mathbb{1}_A N^2_{t\wedge T_\epsilon}]$
  $\le E[N^2_{t\wedge T_\epsilon}] \le \epsilon$. By letting $\epsilon \to 0$, $E[\mathbb{1}_A N_t^2]0 \to N_t = 0$ a.s. on A.

Theorem 4.2.3 (Dambis-Dubins-Schwarz)**.** *Let M be a continuous local martingales s.t. $< M,M >_\infty = \infty$ a.s. There exists a Brownian morion $(\beta_s)_{s\ge 0}$ s.t. $\forall t \ge 0$, $M_t = \beta_{<M,M>_t}$ a.s.*

pf.

- assume $M_0 = 0$.

  $\forall r \geq 0$, $\tau_r = \inf\{t \geq 0 :< M,M >_t \geq r\}$. Then it is a stopping time and $\tau_r < \infty$, $\forall r \geq 0$ on the event $\{< M,M >_\infty = \infty\}$. So redefine τ_r on the negligible event $\mathcal{N} = \{< M,M >_\infty < \infty\}$ by taking $\tau_r(\omega) = 0$, $\forall r \geq 0$ if $\omega \in \mathcal{N}$. Since the filtration is complete, τ_r remains a stopping time after the modification. By construction $\forall \omega \in \Omega$, the function $r \mapsto \tau_r(\omega)$ is nondecreasing and left-continuous and therefore has a right limit at $\forall r \geq 0$. This right limit is denoted by $\tau_{r+} = \inf\{t \geq 0 :< M,M >_t > r\}$ except on the negligible set $\mathcal{N}$ where $\tau_{r+} = 0$. Set $\beta_r = M_{\tau_r}$, $\forall r \geq 0$. Then it is adapted w.r.t. the filtration $(\mathcal{G}_r)$ defined by $\mathcal{G}_r = \mathcal{F}_{\tau_r}$, $\forall r \geq 0$ and $\mathcal{G}_\infty = \mathcal{F}_\infty$. Then the filtration $(\mathcal{G}_r)$ is complete since the property holds for $(\mathcal{F}_t)$. The sample paths $r \mapsto \beta_r(\omega)$ are left-continuous and have right limits given $\forall r \geq 0$ by $\beta_{r+} = \lim_{s \downarrow \downarrow r} \beta_s = M_{\tau_{r+}}$. Since $< M,M >_{\tau_r}$ $=< M,M >_{\tau_{r+}}$, $\forall r \geq 0$, by the previous lemma $M_\tau = M_{\tau_{r+}}$, $\forall r \geq 0$ a.s. Hence the sample paths of β are continuous.
  $\forall n \in \mathbb{N}$, the stopped continuous local martingales M^{τ_n} and $(M^{\tau_n})^2 -$ $< M,M >^{\tau_n}$ are uniformly integrable martingales. By the optional stopping theorem, $\forall 0 \leq r \leq s \leq n$, $E[\beta_s|\mathcal{G}_r] = E[M^{\tau_n}_{\tau_s}|\mathcal{F}_{\tau_r}] = M^{\tau_n}_{\tau_r} = \beta_r$ and $E[\beta_s^2 - s|\mathcal{G}_r] = E[(M^{\tau_n}_{\tau_s})^2 - < M^{\tau_n}, M^{\tau_n} >_{\tau_s} |\mathcal{F}_{\tau_r}] = (M^{\tau_n}_{\tau_r})^2 - < M^{\tau_n}, M^{\tau_n} >_{\tau_r}$ $= \beta_r^2 - r$. Then when $d = 1$, by the Levy's theorem, β is a $(\mathcal{G}_r)$-Brownian motion. By the definition of β, a.s. $\forall t \geq 0$, $\beta_{<M,M>_t} = M_{\tau_{<M,M>_t}}$. But since $\tau_{<M,M>_t} \leq t \leq \tau_{<M,M>_t+}$ and since $< M,M >$ takes the same value at $\tau_{<M,M>_t}$ and at $\tau_{<M,M>_t+}$, by the previous lemma $M_t = M_{\tau_{<M,M>_t}}$, $\forall t \geq$ 0 a.s. Thus $M_t = \beta_{<M,M>_t}$, $\forall t \geq 0$ a.s.

- $M_0 \neq 0$
  $M_t = M_0 + M'_t$ and apply the previous argument to M' and get $M'_t =$ $\beta'_{<M',M'>_t}$ for every $t \geq 0$ a.s. Since β' is a $(\mathcal{G}_r)$-Brownian motion, β' is independent of $\mathcal{G}_0 = \mathcal{F}_0$ hence of M_0. Thus $\beta_s = M_0 + \beta'_s$ is also a Brownian motion and we get the theorem.

Proposition 4.2.2. *Let M,N be two continuous local martingales s.t. $M_0 =$ $N_0 = 0$. Assume that*

1. $< M,M >_t =< N,N >_t$, $\forall t \geq 0$ *a.s.*
2. *M,N are orthogonal ($< M,N >_t = 0$, $\forall t \geq 0$, a.s.)*
3. $< M,M >_\infty =< N,N >_\infty = \infty$ *a.s.*

Let $\begin{cases} \beta = (\beta_t)_{t\geq 0} \\ \gamma = (\gamma_t)_{t\geq 0} \end{cases}$ *be the real Brownian motion s.t.* $\begin{cases} M_t = \beta_{<M,M>_t} \\ N_t = \gamma_{<N,N>_t} \end{cases}$ *for every* $t \geq 0$ *a.s. Then* β *and* γ *are independent.*

pf. Let $\begin{cases} \beta_r = M_{\tau_r} \\ \gamma_r = N_{\tau_r} \end{cases}$ where $\tau_r = \inf\{t \geq 0 :< M,M >_t \geq r\} = \inf\{t \geq 0 :< N,N >_t \geq r\}$. Since β,γ are $(\mathscr{G}_r)$-Brownian motions and M,N are orthogonal martingales, M_tN_t is a local martingale. $\forall n \geq 1, M_t^{\tau_n}N_t^{\tau_n}$ is a uniformly integrable martingale and by the optional stopping theorem, $\forall r \leq s \leq n$, $E[\beta_s\gamma_s|\mathscr{G}_r] = E[M_{\tau_s}^{\tau_r}N_{\tau_s}^{\tau_n}|\mathscr{F}_{\tau_r}] = M_{\tau_r}^{\tau_n}N_{\tau_r}^{\tau_n} = \beta_r\gamma_r$ so that $\beta_r\gamma_r$ is a $(\mathscr{G}_r)$-martingale and the bracket $< \beta,\gamma >$ is identically 0. By the Levy's theorem (β,γ) is a two-dimensional Brownian motion and since $\beta_0 = \gamma_0 = 0$, $\beta \perp \gamma$.

- **The Burkholder-Davis-Gundy Inequalities**

If M is a continuous local martingale, set $M_t^* = \sup_{s\leq t}|M_s|$, $\forall t \geq 0$.

Theorem 4.2.4 (Burkholder-Davis-Gundy Inequalities)**.** *For every real* $p > 0$*, there exists two constants* $c_p, C_p > 0$ *depending only on* p *s.t. for every continuous local martingales* M *with* $M_0 = 0$ *and every stopping time* T

$$c_pE[< M,M >_T^{p/2}] \leq E[(M_T^*)^p] \leq C_pE[< M,M >_T^{p/2}]$$

pf. Replacing M by the stopping martingale M^T, we can reduce the case into the one where $T = \infty$.

- M is bounded.
  - When $p \geq 2$, by the Ito's formula to the function $|x|^p$, $|M_t|^p = \int_0^t p|M_s|^{p-1}sgn(M_s)dM_s + \frac{1}{2}\int_0^t p(p-1)|M_s|^{p-2}d < M,M >_s$. Since M is bounded, $M \in \mathbb{H}^2$ and $\int_0^t p|M_s|^{p-1}sgn(M_s)dM_s$ is a martingale in $\mathbb{H}^2$. Thus

$$E[|M_t|^p] = \frac{p(p-1)}{2}E[\int_0^t |M_s|^{p-2}d < M,M >_s] \leq \frac{p(p-1)}{2}E[(M_t^*)^{p-2} < M,M >_t]$$
$$\leq \frac{p(p-1)}{2}(E[(M_t^*)^p])^{\frac{(p-2)}{p}}(E[< M,M >_t^{p/2}])^{2/p}$$

by Holder's inequality. Also by the Doob's inequality in L^p, $E[(M_t^*)^p] \leq (\frac{p}{p-1})^pE[|M_t|^p]$. By combining two bounds,

$$E[(M_t^*)^p] \leq ((\frac{p}{p-1})^p\frac{p(p-1)}{2})^{p/2}E[< M,M >_t^{p/2}]$$

By letting $t \to \infty$, the right hand side inequality follows.

* When $p < 2$, since $M \in \mathbb{H}^2$, $M^2 - < M, M >$ is a uniformly integrable martingale and for every stopping time T, $E[(M_T)^2]$ $= E[< M, M >_T]$. Let $x > 0$ and consider the stopping time $T_x \equiv \inf\{t \geq 0 : (M_t)^2 \geq x\}$. Then if T is any bounded stopping time, $P((M_T^*)^2 \geq x) = P(T_x \leq T) = P((M_{T_x \wedge T})^2 \geq x) \leq \frac{1}{x} E[(M_{T_x \wedge T})^2]$ $= \frac{1}{x} E[< M, M >_{T_x \wedge T}] \leq \frac{1}{x} E[< M, M >_T]$. Consider the stopping time $S_x \equiv \inf\{t \geq 0 :< M, M >_t \geq x\}$. Then $\forall t \geq 0$, $\{(M_t^*)^2 \geq x\} \subset (\{(M_{S_x \wedge t}^*)^2 \geq x\} \cup \{S_x \leq t\})$. By the previous bound with $T = S_x \wedge t$,

$$\begin{aligned} P((M_t^*)^2 \geq x) &\leq P((M_{S_x \wedge t}^*)^2 \geq x) + P(S_x \leq t) \\ &\leq \frac{1}{x} E[< M, M >_{S_x \wedge t}] + P(< M, M >_t \geq x) \\ &= \frac{1}{x} E[< M, M >_t \wedge x] + P(< M, M >_t \geq x) \\ &= \frac{1}{x} E[< M, M >_t \mathbb{1}_{\{< M, M >_t < x\}}] + 2P(< M, M >_t \geq x) \end{aligned}$$

Set $q = p/2 \in (0,1)$ and integrate each side of the last bound w.r.t. the measure $qx^{q-1}dx$. Then $\int_0^\infty P((M_t^*)^2 \geq x) q x^{q-1} dx = E[\int_0^{(M_t^*)^2} q x^{q-1} dx]$ $= E[(M_t^*)^{2q}]$ and similarly, $\int_0^\infty P(< M, M >_t \geq x) q x^{q-1} dx = E[< M, M >_t^q$]. Furthermore $\int_0^\infty \frac{1}{x} E[< M, M >_t \mathbb{1}_{\{< M, M >_t < x\}}] q x^{q-1} dx = E[< M, M >_t$ $\int_{< M, M >_t}^\infty q x^{q-2} dx] = \frac{q}{1-q} E[< M, M >_t^q]$. Summarizing, we obtain the bound

$$E[(M_t^*)^{2q}] \leq (2 + \frac{q}{1-q}) E[< M, M >_t^q]$$

and we just have to let $t \to \infty$ to get the theorem.

- M is unbounded.
  replace M by M^{T_n} where $T_n = \inf\{t \geq 0 : |M_t| = n\}$ and by the previous argument, the theorem follows. By letting $n \to \infty$, the unbounded case follows.

Corollary 4.2.1. *Let M be a continuous local martingale s.t. $M_0 = 0$. The condition*

$$E[< M, M >_\infty^{1/2}] < \infty$$

implies M is a uniformly integrable martingale.

pf. by the case $p = 1$ of the previous theorem, the condition $E[< M, M >_\infty^{1/2}]$ $< \infty$ imlies $E[M_\infty^*] < \infty$. Then the continuous local martingale M, which is dominated by the variable M_∞^* is a uniformly integrable martingale.

4.2.2 The Representation of Martingales as Stochastic Integrals

Lemma 4.2.2. *Assume that the filtration $(\mathscr{F}_t)$ on Ω is the completed canonical filtration of a real Brownian motion B started from 0. The vector space generated by the random variables* $\exp(i\sum_{j=1}^n \lambda_j(B_{t_j}-B_{t_{j-1}})$, *for any choice of* $0=t_0<\cdots<t_n,\ \lambda_1,\ldots,\lambda_n\in\mathbb{R}$ *is dense in the space* $L^2_{\mathbb{C}}(\Omega,\mathscr{F}_\infty,P)$ *of all square integrable complex valued $\mathscr{F}_\infty$-measurable random variables.*

pf. Let $Z\in L^2_{\mathbb{C}}(\Omega,\mathscr{F}_\infty,P)$ s.t. $E[Z\exp(i\sum_j\lambda_j(B_{t_j}-B_{t_{j-1}})]=0$ for any choice of $0=t_0<\cdots<t_n,\ \lambda_1,\ldots,\lambda_m\in\mathbb{R}$. Fix $0=t_0<\cdots<t_n$ and consider the complex measure μ on $\mathbb{R}^n$ defined by $\mu(F)=E[Z\mathbb{1}_F(B_{t_1},B_{t_2}-B_{t_1},\ldots,B_{t_n}-B_{t_{n-1}})]$ for any Boreal subset F of $\mathbb{R}^n$. Then $E[Z\exp(i\sum\lambda_j(B_{t_j}-B_{t_{j-1}}))]=0$ shows that the Fourier transform of μ is identically 0. By the injectivity of the Fourier transform on complex measure on $\mathbb{R}^d$, $\mu=0$. Thus $E[Z\mathbb{1}_A]=0,\ \forall A\in\sigma(B_{t_1},\ldots,B_{t_n})$. A monotone class argument then shows the identity $E[Z\mathbb{1}_A]=0$ remains valid for any $A\in\sigma(B_t,t\geq 0)$ and then by completion for any $A\in\mathscr{F}_\infty$. Thus $Z=0$ and the lemma follows.

Theorem 4.2.5. *Assume that the filtration $(\mathscr{F}_t)$ on Ω is the completed canonical filtration of a real Brownian motion B started from 0. Then for every random variable $Z\in L^2(\Omega,\mathscr{F}_\infty,P)$, there exists a unique progressive process $h\in L^2(B)$ s.t.*

$$Z=E[Z]+\int_0^\infty h_s dB_s$$

Consequently for every martingale M or continuous local martingale M that is bounded in L^2, there exists a unique process $h\in L^2(B)$ or $h\in L^2_{loc}(B)$ and a constant $C\in\mathbb{R}$ s.t. $M_t=C+\int_0^t h_s dB_s$.

pf.

- Uniqueness
  pf. if the representation of a given variable Z holds with two processes $h.h'\in L^2(B)$, $E[\int_0^\infty(h_s-h'_s)^2ds]=E[(\int_0^\infty h_s dB_s-\int_0^\infty h'_s dB_s)^2]=0$. Hence $h=h'$ in $L^2(B)$.

- Existence
  pf. Let $\mathscr{H}$ be the vector space of all variables $Z\in L^2(\Omega,\mathscr{F}_\infty,P)$ for which the property of the statement holds. Then if $Z\in\mathscr{H}$ and h is the associated process in $L^2(B)$, then $E[Z^2]=(E[Z])^2+E[\int_0^\infty(h_s)^2ds]$. Thus $\mathscr{H}$ is a closed subspace of $L^2(\Omega,\mathscr{F}_\infty,P)$. Indeed if (Z_n) is a sequence in $\mathscr{H}$ that converges to Z in $L^2(\Omega,\mathscr{F}_\infty,P)$, the processes $h^{(n)}$ corresponding respectively to the variables Z_n form a Cauchy sequence in $L^2(B)$, hence converges in $L^2(B)$ to

a certain process $h \in L^2(B)$. Thus $Z = E[Z] + \int_0^\infty h_s dB_s$.
Let $0 = t_0 < \cdots < t_n$, $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$ and set $f(s) = \sum_{j=1}^n \lambda_j \mathbb{1}_{(t_{j-1},t_j]}(s)$. Write $\mathscr{E}_t^f$ for the exponential martingale $\mathscr{E}(i\int_0^\cdot f(s)dB_s)$. Then $\exp(i\sum_{j=1}^n \lambda_j(B_{t_j} - B_{t_{j-1}}) + \frac{1}{2}\sum_{j=1}^n \lambda_j^2(t_j - t_{j-1})) = \mathscr{E}_\infty^f = 1 + i\int_0^\infty \mathscr{E}_s^f f(s)dB_s$ by the proposition 3.2.1. and it follows that both real and imaginary part of variables of the form $\exp(i\sum_j \lambda_j(B_{t_j} - B_{t_{j-1}}))$ are in $\mathscr{H}$. By the previous lemma, linear combinations of such random variables are dense in $L^2(\Omega, \mathscr{F}_\infty, P)$. So $\mathscr{H}$ is a complete space, containing the dense open set for $L^2(\Omega, \mathscr{F}_\infty, P)$ and contained in $L^2(\Omega, \mathscr{F}_\infty, P)$, $\mathscr{H} = L^2(\Omega, \mathscr{F}_\infty, P)$

- if M is a matingale that is bounded in L^2, $M_\infty \in L^2(\Omega, \mathscr{F}_\infty, P)$ and therefore it can be written in the form $M_\infty = E[M_\infty] + \int_0^\infty h_s dB_s$ where $h \in L^2(B)$. Then since $E[\int_0^t H_r dM_r | \mathscr{F}_s] = \int_0^s H_r dM_r$, $M_t = E[M_\infty | \mathscr{F}_t] = E[M_\infty] + \int_0^t h_s dB_s$.

- if M is a continuous local martingale, $M_0 = C \in \mathbb{R}$ because the σ-field $\mathscr{F}_0$ contains only events of probability zero or one. If $T_n = \inf\{t \geq 0 : |M_t| \geq n\}$, by the case of martingales bounded in L^2 to M^{T_n} and we get a process $h^{(n)} \in L^2(B)$ s.t. $M_t^{T_n} = C + \int_0^t h_s^{(n)} dB_s$. By the uniqueness, we get $h_s^{(m)} = \mathbb{1}_{[0,T_m]}(s) h_s^{(n)}$ if $m < n$ ds a.e., a.s. Then we can construct a process $h \in L^2_{loc}(B)$ s.t. $\forall m$, $h_s^{(m)} = \mathbb{1}_{[0,T_m]}(s) h_s ds$ a.e., a.s.

→ the filtration $(\mathscr{F}_t)_{t\geq 0}$ is right-continuous and all martingales of the filtration $(\mathscr{F}_t)_{t\geq 0}$ have a modification with continuous sample paths.
pf. For a martingale that is bounded in L^2, by the representation formula, all martingales of the filtration $(\mathscr{F}_t)$ have a modification with continuous sample paths. For a uniformly integrable martingale M, $\forall t \geq 0$, $M_t = E[M_\infty | \mathscr{F}_t]$. Then M_t has a modification with cadlag sample paths and we consider this modification. Let $M_\infty^{(n)}$ be a sequence of bounded random variables s.t. $M_\infty^{(n)} \to M_\infty$ in L^1 as $n \to \infty$. Introduce the martingales $M_t^{(n)} = E[M_\infty^{(n)} | \mathscr{F}_t]$ which are bounded in L^2. Then we can assume $\forall n$, the sample paths of $M^{(n)}$ are continuous. On the other hand, by the Doob's maximal inequality, $\forall \lambda > 0$, $P(\sup_{t\geq 0} |M_t^{(n)} - M_t| > \lambda) \leq \frac{3}{\lambda} E[|M_\infty^{(n)} - M_\infty|] \to$ $\exists (n_k) \uparrow \infty$ s.t. $\forall k \geq 1$, $P(\sup_{t\geq 0} |M_t^{(n_k)} - M_t| > 2^{-k}) \leq 2^{-k}$. By the Borel-Cantelli lemma, $\sup_{t\geq 0} |M_t^{(n_k)} - M_t| \to 0$ a.s. and we get the sample paths of M are continuous as uniform limits of the continuous functions.

4.3 Girsanov's Theorem

Proposition 4.3.1. *Assume Q is a probability measure on $(\Omega, \mathscr{F})$, which is absolutely continuous w.r.t. P on the σ-field $\mathscr{F}_\infty$. For every $t \in [0, \infty]$, let $D_t = \frac{dQ}{dP}|_{\mathscr{F}_t}$ be the Radon-Nikodym derivative of Q w.r.t. P on the σ-field $\mathscr{F}_t$. The process $(D_t)_{t\geq 0}$*

is a uniformly integrable martingale. Consequently $(D_t)_{t\geq 0}$ has a cadlag modification. Keeping the same notation $(D_t)_{t\geq 0}$ for this modification, we have for every stopping time T,

$$D_T = \frac{dQ}{dP}|_{\mathscr{F}_T}$$

Finally, if we assume further that P,Q are mutually absolutely continuous on $\mathscr{F}_\infty$, $\inf_{t\geq 0} D_t > 0$, P a.s.

pf.

- If $A \in \mathscr{F}_t$, $Q(A) = E_Q[\mathbb{1}_A] = E_P[\mathbb{1}_A D_\infty] = E_P[\mathbb{1}_A E_P[D_\infty|\mathscr{F}_t]]$ and by the uniqueness of the Radon-Nikodym derivative on $\mathscr{F}_t$, $D_t = E_P[D_\infty|\mathscr{F}_t]$ a.s. Hence D is a uniformly integrable martingale, which is closed by D_∞. Then we can find a cadlag modification of $(D_t)_{t\geq 0}$.

- If T is a stopping time, by the optional stopping theorem, $\forall A \in \mathscr{F}_T$, $Q(A) = E_Q[\mathbb{1}_A] = E_P[\mathbb{1}_A D_\infty] = E_P[\mathbb{1}_A E_P[D_\infty|\mathscr{F}_T]] = E_P[\mathbb{1}_A D_T]$ and since D_T is $\mathscr{F}_T$-measurable, it follows that $D_T = \frac{dQ}{dP}|_{\mathscr{F}_T}$.

- $\forall \epsilon > 0$, set $T_\epsilon = \inf\{t \geq 0 : D_t < \epsilon\}$. Then it is a stopping time. Then since the event $\{T_\epsilon < \infty\}$ is $\mathscr{F}_{T_\epsilon}$-measurable and $D_{T_\epsilon} \leq \epsilon$ on $\{T_\epsilon < \infty\}$ by the right-continuity of sample paths, $Q(T_\epsilon < \infty) = E_P[\mathbb{1}_{\{T_\epsilon<\infty\}} D_{T_\epsilon}] \leq \epsilon$. Thus $Q(\cap_{n=1}^\infty \{T_{1/n} < \infty\}) = 0$ and since P is absolutely continuous w.r.t. Q we also have $P(\cap_{n=1}^\infty \{T_{1/n} < \infty\}) = 0$. This means P a.s., there exists an integer $n \geq 1$ s.t. $T_{1/n} = \infty$ so the last statement follows.

Proposition 4.3.2. *Let D be a continuous local martingale taking positive values. There exists a unique continuous local martingale L s.t.*

$$D_t = \exp(L_t - \frac{1}{2} < L,L >_t) = \mathscr{E}(L)_t$$

Moreover, $L_t = \log D_0 + \int_0^t D_s^{-1} dD_s$.

pf. Uniqueness follows from the previous theorem 2.4.1. Since D takes positive values, by Ito's formula, we get $\log D_t = \log D_0 + \int_0^t \frac{dD_s}{D_s} - \frac{1}{2}\int_0^t \frac{d<D,D>_s}{D_s^2}$
$= L_t - \frac{1}{2} < L,L >_t$ where L is as in the statement.

Theorem 4.3.1 (Girsanov)**.** *Assume that the probability measure P,Q are mutually absolutely continuous on $\mathscr{F}_\infty$. Let $(D_t)_{t\geq 0}$ be the martingale with cadlag sample paths s.t. $\forall t \geq 0$,*

$$D_t = \frac{dQ}{dP}|_{\mathscr{F}_t}$$

Assume D has continuous sample paths, and let L be the unique continuous local martingale s.t. $D_t = \mathscr{E}(L)_t$. Then if M is a continuous local martingale under P, the process $\tilde{M} = M - < M,L >$ is a continuous local martingale under Q.

pf. By the previous proposition, $D_t = \mathscr{E}(L)_t$. Then let T be a stopping time and let X be an adapted process with continuous sample paths.

- **CLAIM : if $(XD)^T$ is a martingale under P, then X^T is a martingale under Q.**
  pf. By the proposition 3.3.1, $E_Q[|X_{T\wedge t}|] = E_P[|X_{t\wedge T}D_{T\wedge t}|] < \infty$ and therefore $X_t^T \in L^1(Q)$. Let $A \in \mathscr{F}_s$ and $s < t$. Since $A \cap \{T > s\} \in \mathscr{F}_s$, $E_P[\mathbb{1}_{A\cap\{T>s\}}X_{T\wedge t}D_{T\wedge t}]$ $= E_P[\mathbb{1}_{A\cap\{T>s\}}X_{T\wedge s}D_{T\wedge s}]$, given $(XD)^T$ a martingale under P. Then by the proposition 3.3.1, $D_{T\wedge t} = \frac{dQ}{dP}|_{\mathscr{F}_{T\wedge t}}$, $D_{T\wedge s} = \frac{dQ}{dP}|_{\mathscr{F}_{T\wedge s}}$ and therefore $E_Q[\mathbb{1}_{A\cap\{T>s\}}X_{T\wedge t}]$ $= E_Q[\mathbb{1}_{A\cap\{T>s\}}X_{T\wedge s}]$, given $A \cap \{T > s\} \in \mathscr{F}_{T\wedge s} \subset \mathscr{F}_{T\wedge t}$. On the other hand, $E_Q[\mathbb{1}_{A\cap\{T\le s\}}X_{T\wedge t}] = E_Q[\mathbb{1}_{A\cap\{T\le s\}}X_{T\wedge s}]$ trivially. Thus we have $E_Q[\mathbb{1}_A X_{T\wedge t}] = E_Q[\mathbb{1}_A X_{T\wedge s}]$.

Let M be a continuous local martingale under P and let $\tilde{M}$ be as in the theorem. With $X = \tilde{M}$, by Ito's formula, $\tilde{M}_t D_t = M_0 D_0 + \int_0^t \tilde{M}_s dD_s + \int_0^t D_s dM_s - \int_0^t D_s d < M,L >_s + < M,D >_t = M_0 D_0 + \int_0^t \tilde{M}_s dD_s + \int_0^t D_s dM_s$ since $d < M,L >_s = D_s^{-1} d < M,D >_s$ by the previous proposition. So $\tilde{M}D$ is a continuous local martingale under P and by the claim above, $\tilde{M}$ is a continuous local martingale under Q.

$\rightarrow$ a process M which is a continuous local martingale under P remains a semimartingale under Q and its canonical decomposition under Q is $M = \tilde{M} + < M,L >$. So the class of semimartingales under P is contained in the class of semimartingales under Q.

$\rightarrow$Let X,Y be two semimartingales under P or Q. The bracket $< X,Y >$ is the same under P and under Q. Similarly, if H is a locally bounded progressive process, the stochastic integral $H \cdot X$ is the same under P and Q. $((H \cdot \tilde{M})_P = (H \cdot M)_P - H \cdot < M,L >= (H \cdot M)_P - < (H \cdot M)_P, L >$ and by the previous theorem $(H \cdot \tilde{M})_P$ is a continuous martingale under Q).
$\rightarrow \tilde{M} = \mathscr{G}_Q^P(M)$ implies $\mathscr{G}_Q^P$ is a functor from the set of all P-continuous local martingales to the set of all Q-continuous local martingales($H \cdot \mathscr{G}_Q^P(M) = \mathscr{G}_Q^P(H \cdot M)$ given H a locally bounded progressive process).

e.g. Suppose $M = B$ is an $(\mathscr{F}_t)$-Brownian motion under P. Then $\tilde{B} = B - < B,L >$ is a continuous local martingale under Q with quadratic variation $< \tilde{B},\tilde{B} >_t$ $=< B,B >_t = t$. So $\tilde{B}_t$ is an $(\mathscr{F}_t)$-Brownian motion under Q.

Theorem 4.3.2. *Let L be a continuous local martingale s.t. $L_0 = 0$. Consider the following properties;*

1. ***Novikov's Criterion*** *: $E[\exp(\frac{1}{2} < L,L >_\infty)] < \infty$*
2. ***Kazamaki's Criterion*** *: L is a uniformly integrable martingale and $E[\exp(\frac{1}{2}L_\infty)] < \infty$*

3. *$\mathscr{E}(L)$ is a uniformly integrable martingale.*

Then $1 \to 2 \to 3$.

pf.

- $1 \to 2$

  Property 1 implies $E[< L,L >_\infty] < \infty$ hence also L is a continuous martingale bounded in L^2. Then $\exp(\frac{1}{2}L_\infty) = (\mathscr{E}(L)_\infty)^{1/2}(\exp(\frac{1}{2} < L,L >_\infty))^{1/2}$ so that by the Cauchy-Schwarz inequality,
  $E[\exp(\frac{1}{2}L_\infty)] \le (E[\mathscr{E}(L)_\infty)^{1/2}(E[\exp(\frac{1}{2} < L,L >_\infty)])^{1/2} \le (E[\exp(\frac{1}{2} < L,L >_\infty)])^{1/2}$ $< \infty$.

- $2 \to 3$

  Since L is a uniformly integrable martingale, for any stopping time T, $L_T = E[L_\infty|\mathscr{F}_T]$. By Jensen' inequality, $\exp(\frac{1}{2}L_T) \le E[\exp(\frac{1}{2}L_\infty|\mathscr{F}_T] < \infty$. So the collection of all variables of the form $E[\exp\frac{1}{2}L_\infty|\mathscr{F}_T]$ for any stopping time T, is uniformly integrable. Then the collection of all variables $\exp\frac{1}{2}L_T$ for any stopping time T is also uniformly integrable. For $0 < a < 1$, set $Z_t^{(a)} = \exp(\frac{aL_t}{1+a})$. Then $\mathscr{E}(aL)_t = \mathscr{E}(L)_t)^{a^2}(Z_t^{(a)})^{1-a^2}$. If $\Gamma \in \mathscr{F}$ and T is a stopping time, by the Holder's inequality,
  $$\begin{aligned} E[\mathbb{1}_\Gamma \mathscr{E}(aL)_T] &\le E[\mathscr{E}(L)_T]^{a^2} E[\mathbb{1}_\Gamma Z_T^{(a)}]^{1-a^2} \\ &\le E[\mathbb{1}_\Gamma Z_T^{(a)}]^{1-a^2} && E[\mathscr{E}(L_T)] \le 1 \\ &\le E\mathbb{1}_\Gamma \exp\tfrac{1}{2}L_T]^{2a(1-a)} && \text{jensen's inequality} \end{aligned}$$
  . Since the collection of all variables of the form $\exp\frac{1}{2}L_T$ for any stopping time T is uniformly integrable, by the previous equation show so is the collection of all variables $\mathscr{E}(aL)_T$ for any stopping time T. By the definition of a continuous local martingale, there is an increasing sequence $T_n \uparrow \infty$ of stopping times s.t. $\forall n$, $\mathscr{E}(aL)_{t\wedge T_n}$ is a martingale. If $0 \le s \le t$, we can use U.I to pass to the limit $n \to \infty$ in the equality $E[\mathscr{E}(aL)_{t\wedge T_n}|\mathscr{F}_s] = \mathscr{E}(aL)_{s\wedge T_n}$ and we get $\mathscr{E}(aL)$ is a UI martingale. Thus

  $$1 = E[\mathscr{E}(aL)_\infty] \le E[\mathscr{E}(L)_\infty]^{a^2} E[Z_\infty^{(a)}]^{1-a^2} \le E[\mathscr{E}(L)_\infty]^{a^2} E[\exp\frac{1}{2}L_\infty]^{2a(1-a)}$$

  When $a \to 1$, $E[\mathscr{E}(L_\infty] \ge 1 \to E[\mathscr{E}(L)_\infty] = 1$.

Applications of the Girsanov's Theorem

1. Let b be a bounded measurable function on $\mathbb{R}_{\ge 0} \times \mathbb{R}$. Assume there exists a function $g \in L^2(\mathbb{R}_{\ge 0}, \mathscr{B}(\mathbb{R}_{\ge 0}), dt)$ s.t. $|b(t,x)| \le g(t)$, $\forall (t,x) \in \mathbb{R}_{\ge 0} \times \mathbb{R}$. Let

B be an $(\mathscr{F}_t)$-Brownian motion. Consider the continuous local martingale $L_t = \int_0^t b(s,B_s)dB_s$ and the associated exponential martingale $D_t = \mathscr{E}(L)_t = \exp(\int_0^t b(s,B_s)dB_s - \frac{1}{2}\int_0^t b(s,B_s)^2 ds)$. By the assumption on b, the condition 1 of the previous theorem holds and thus D is uniformly integrable martingale. Set $Q = D_\infty \cdot P$. By the Girsanov's theorem, $\beta_t \equiv B_t - \int_0^t b(s,B_s)ds$ is an $(\mathscr{F}_t)$-Brownian motion under Q. In other words, under the probability measure Q, $\exists(\mathscr{F}_t)$-Brownian motion β s.t. the process $X = B$ solves the stochastic differential equation $dX_t = d\beta_t + b(t,X_t)dt$.

2. Assume $b(t,x) = g(t)$, where $g \in L^2(\mathbb{R}_{\geq 0}, \mathscr{B}(\mathbb{R}_{\geq 0}), dt)$ and set $\forall t \geq 0$, $h(t) = \int_0^t g(s)ds$. The set $\mathscr{H}$ of all functions h that can be written in this form is called the Cameron-Martin space. If $h \in \mathscr{H}$, $\dot{h} = g$ for the associated function in $L^2(\mathbb{R}_{\geq 0}, \mathscr{B}(\mathbb{R}_{\geq 0}), dt)$. Then under the probability measure $Q \equiv D_\infty \cdot P$ $= \exp(\int_0^\infty g(s)dB_s - \frac{1}{2}\int_0^\infty g(s)^2 ds) \cdot P$, the process $\beta_t \equiv B_t - h(t)$ is a Brownian motion by the previous argument. So for every nonnegative measurable function Φ on $C(\mathbb{R}_{\geq 0}, \mathbb{R})$, $E_P[D_\infty \Phi((B_t)_{t\geq 0})] = E_Q[\Phi((B_t)_{t\geq 0})] = E_Q[\Phi((\beta_t + h(t))_{t\geq 0})]$ $= E_P[\Phi((B_t + h(t))_{t\geq 0})]$.

Proposition 4.3.3 (Cameron-Martin Formula)**.** *Let $W(dw)$ be the Wiener measure on $C(\mathbb{R}_{\geq 0}, \mathbb{R})$ and let h be a function in the Cameron-Martin space $\mathscr{H}$. Then for every nonnegative measurable function Φ on $C(\mathbb{R}_{\geq 0}, \mathbb{R})$,*

$$\int W(dw)\Phi(w+h) = \int W(dw)\exp(\int_0^\infty \dot{h}(s)dw(s) - \frac{1}{2}\int_0^\infty \dot{h}(s)^2 ds)\Phi(w)$$

$\rightarrow$the integral $\int_0^\infty \dot{h}(s)dw(s)$ is a stochastic integral w.r.t. $w(s)$ which is a Brownian motion under $W(dw)$, but it can also be viewed as a Wiener integral since the function $\dot{h}(s)$ is deterministic.
$\rightarrow$the image of Wiener measure $W(dw)$ under the mapping $w \mapsto w + h$ has a density w.r.t. $W(dw)$ and this density is the terminal value of the exponential martingale associated with the martingale $\int_0^t \dot{h}(s)dw(s)$.

Chapter 5

Markov Process

Let $(E,\mathscr{E})$ be a measurable space. A Markovian transition kernel from E to E is a mapping $Q : E \times \mathscr{E} \to [0,1]$ satisfying the following two properties:

1. $\forall x \in E$, the mapping $A \mapsto Q(x,A)$ is a probability measure on $(E,\mathscr{E})$
2. $\forall A \in \mathscr{E}$, the mapping $x \mapsto Q(x,A)$ is $\mathscr{E}$-measurable.

$\to$ when E is finite or countable, Q is characterised by the matrix $(Q(x,\{y\}))_{x,y\in E}$.

If $f : E \to \mathbb{R}$ is bounded and measurable, the function Qf defined by $Qf(x) = \int Q(x,dy)f(y)$ is also bounded and measurable on E(if f is the indicator function, by the property 2, the measurability of $Q_t f$ follows. for the general case follows from the standard argument).

Definition 5.0.1. *A collection $(Q_t)_{t\geq 0}$ of transition kernels on E is called a transition semigroup if the following three properties hold:*

1. $\forall x \in E,\ Q_0(x,dy) = \delta_x(dy)$
2. ***Chapman-Kolmogorov Identity*** *:* $\forall s,t \geq 0,\ \forall A \in \mathscr{E},\ \ Q_{t+s}(x,A) = \int_E Q_t(x,dy)Q_s(y,A)$
3. $\forall A \in \mathscr{E}$, *the function* $(t,x) \mapsto Q_t(x,A)$ *is measurable w.r.t. the σ-field* $\mathscr{B}(\mathbb{R}_{\geq 0}) \otimes \mathscr{E}$.

$\to$Let $B(E)$ be the vector space of all bounded measurable real functions on E which is equipped with the norm $||f|| = \sup\{|f(x)| : x \in E\}$. Then the linear mapping $f \mapsto Q_t f$ is a contraction of $B(E)$. From this perspective, the Chapman-Kolmogorov identity is equivalent to the relation $Q_{t+s} = Q_t Q_s,\ \forall s,t \geq 0$. This allows one to view $(Q_t)_{t\geq 0}$ as a semigroup of contractions of $B(E)$.

Consider a filtered probability space $(\Omega, \mathscr{F}_t, (\mathscr{F}_t)_{t\in[0,\infty]}, P)$.

Definition 5.0.2. *Let $(Q_t)_{t\geq 0}$ be a transition semigroup on E. A Markov process with transition semigroup $(Q_t)_{t\geq 0}$ is an $(\mathscr{F}_t)$-adapted process $(X_t)_{t\geq 0}$ with values in E s.t. $\forall s,t \geq 0$, $\forall f \in B(E)$,*

$$E[f(X_{s+t})|\mathscr{F}_s] = Q_t f(X_s)$$

$\rightarrow$ if $f = \mathbb{1}_A$, $P(X_{s+t} \in A|\mathscr{F}_s) = Q_t(X_s, A)$ and in particular, $P(X_{s+t} \in A|X_r, 0 \leq r \leq s) = Q_t(X_s, A)$. Hence the conditional distribution of X_{s+t} knowing the past $(X_r : 0 \leq r \leq s)$ before time s is given by $Q_t(X_s, \cdot)$ and this only depends on the present state X_s. This is called the Markov property.

Let $\gamma(dx)$ be the law of X_0. Then if $0 < t_1 < \cdots < t_p$ and $A_0, A_1, \ldots, A_p \in \mathscr{E}$, $P(X_0 \in A_0, \ldots, X_{t_p} \in A_p) = \int_{A_0} \gamma(dx_0) \cdots \int_{A_p} Q_{t_p - t_{p-1}}(x_{p-1}, dx_p)$. More generally, if $f_0, \ldots, f_p \in B(E)$, $E[f_0(X_0) \ldots f_p(X_{t_p}) = \int \gamma(dx_0) f_0(x_0) \times \cdots \times \int Q_{t_p - t_{p-1}}(x_{p-1}, dx_p) f_p(x_p)$

e.g. $E = \mathbb{R}^d$, $\forall t > 0$, $\forall x \in \mathbb{R}^d$, $Q_t(x, dy) = p_t(y-x)dy$, where $p_t(z) = (2\pi t)^{-d/2} \exp(-\frac{|z|^2}{2t})$ is the density of the Gaussian vector in $\mathbb{R}^d$ with covariance tId.

Let $\Omega^* = E_{\mathbb{R}_{\geq 0}}$ be the space of all mappings $\omega : \mathbb{R}_{\geq 0} \rightarrow E$. Equip Ω^* with the σ-field $\mathscr{F}^*$ generated by the coordinate mappings $\omega \mapsto \omega(t)$, $\forall t \in \mathbb{R}_{\geq 0}$. Let $F(\mathbb{R}_{\geq 0})$ be the collection of all finite subsets of $\mathbb{R}_{\geq 0}$ and for every $U \in F(\mathbb{R}_{\geq 0})$, let $\begin{array}{rcl} \pi_U : \Omega^* & \rightarrow & E^U \\ \omega & \mapsto & \omega|_U \end{array}$, $\omega : \mathbb{R}_{\geq 0} \rightarrow E$ restricted to U. If $U, V \in F(\mathbb{R}_{\geq 0})$ and $U \subset V$, $\pi_U^V : E^V \rightarrow E^U$ for the similar restriction mapping.

Definition 5.0.3. *A topological space is Polish if its topology is separable and can be defined by a complete metric.*

Theorem 5.0.1 (Kolmogorov Extension-Special)**.** *Assume E is a Polish space equipped with its Borel σ-field $\mathscr{E}$. $\forall U \in F(\mathbb{R}_{\geq 0}$, let μ_U be a probability measure on E^U. Assume the collection $(\mu_U, U) \in F(\mathbb{R}_{\geq 0})$ is consistent in the following sense: if $U \subset V$, μ_U is the image of μ_V under π_U^V. Then there exists a unique probability measure μ on $(\Omega^*, \mathscr{F}^*)$ s.t. $\pi_U(\mu) = \mu_U$, $\forall U \in F(\mathbb{R}_{\geq 0}$.*

$\rightarrow$ we can construct random processes having prescribed finite-dimensional marginals ;
let $(X_t)_{t\geq 0}$ be the canonical process on Ω^* : $X_t(\omega) = \omega(t)$, $\forall t \geq 0$. If μ is a probability measure on Ω^*, $U = \{t_1, \ldots, t_p\} \in F(\mathbb{R}_{\geq 0})$ with $t_1 < \cdots < t_p$, then $(X_{t_1}, \ldots, X_{t_p})$ can be

viewed as a r.v. with values in E^U, provided we identify E^U with E^p via the mapping $\omega \mapsto (\omega(t_1),\dots,\omega(t_p))$. Furthermore, the distribution of $(X_{t_1},\dots,X_{t_p})$ under μ is $\pi_U(\mu)$. By the Kolmogorov extension theorem, given a collection $(\mu_U, U) \in F(\mathbb{R}_{\geq 0})$ of finite dimensional marginal distributions, which satisfies the consistency condition, one can construct a probability measure μ on the space Ω^* under which the finite dimensional marginals of the canonical process X are the measures μ_U, $U \in F(\mathbb{R}_{\geq 0})$.

Corollary 5.0.1. *We assume E satisfies the assumptions of the previous theorem and $(Q_t)_{t\geq 0}$ is a transition semigroup on E. Let γ be a probability measure on E. Then there exists a unique probability measure P on Ω^* under which the canonical process $(X_t)_{t\geq 0}$ is a Markov process with transition semigroup $(Q_t)_{t\geq 0}$ and the law of X_0 is γ.*

pf. Let $U = \{t_1,\dots,t_p\} \in F(\mathbb{R}_{\geq 0})$ with $0 \leq t_1 < \cdots < t_p$. Define a probability measure P^U on E^U by setting

$$\int P^U(dx_1,\dots,dx_p)\mathbb{1}_A(x_1,\dots,x_p) = \int \gamma(dx_0)\int Q_{t_1}(x_0,dx_1)\cdots\int Q_{t_p-t_{p-1}}(x_{p-1},dx_p)\mathbb{1}_A(x_1,\dots,x_p)$$

for any measurable subset $A \subseteq E^U$. By the Chapman-Kolmogorov relation, the measure P^U satisfy the consistency condition. The Kolmogorov extension theorem then gives the existence and uniqueness of a probability measure P on Ω^* whose finite-dimensional marginals are the measures P_U, $U \in F(\mathbb{R}_{\geq 0})$. This implies $(X_t)_{t\geq 0}$ is under P a Markov process with semigroup $(Q_t)_{t\geq 0}$ w.r.t. the canonical filtration.

$\forall x \in E$, let P_x be the measure given by the previous corollary when $\gamma = \delta_x$. Then the mapping $x \mapsto P_x$ is measurable in the sense that $x \mapsto P_x(A)$ is measurable, $\forall A \in \mathscr{F}^*$. Moreover, for any probability measure γ on E, the measure defined by $P_{(\gamma)}(A) = \int \gamma(dx)P_x(A)$ is the unique probability measure on Ω^* under which the canonical process $(X_t)_{t\geq 0}$ is a Markov process with semigroup $(Q_t)_{t\geq 0}$ and the law of X_0 is γ.
$\rightarrow$ we can get a measurable collection of probability measures $(P_x)_{x\in E}$ s.t. the Markov process X starts from x under P_x. But there is no guarantee for the regularity conditions(e.g. continuity of the sample paths).

Definition 5.0.4. *Let $\lambda > 0$. The λ-resolvent of the transition semigroup $(Q_t)_{t\geq 0}$ is the linear operator* $\begin{array}{rcl} R_\lambda : B(E) & \rightarrow & B(E) \\ f(x) & \mapsto & \int_0^\infty e^{-\lambda t} Q_t f(x)dt \end{array}$ *for $f \in B(E)$, $\forall x \in E$.*

Properties

1. $||R_\lambda f|| \leq \frac{1}{\lambda}||f||$

2. if $0 \le f \le 1,\ 0 \le \lambda R_\lambda f \le 1$
3. **the Resolvent Equation** : if $\lambda, \mu > 0,\ R_\lambda - R_\mu + (\lambda - \mu) R_\lambda R_\mu) = 0$
   pf. Assume wlog $\lambda \neq \mu$. Then

$$\begin{aligned} R_\lambda(R_\mu f)(x) &= \int_0^\infty e^{-\lambda s} Q_s(\int_0^\infty e^{-\mu t} Q_t f dt)(x) ds \\ &= \int_0^\infty e^{-\lambda s}(\int Q_s(x, dy) \int_0^\infty e^{-\mu t} Q_t f(y) dt) ds \\ = \int_0^\infty e^{-\lambda s}(\int_0^\infty e^{-\mu t} Q_{s+t} f(y) dt) ds &= \int_0^\infty e^{-(\lambda-\mu)s}(\int_0^\infty e^{-\mu(s+t)} Q_{s+t} f(y) dt) ds \\ = \int_0^\infty e^{-(\lambda-\mu)s}(\int_s^\infty e^{-\mu r} Q_r f(y) dr) ds &= \int_0^\infty Q_r f(y) e^{-\mu r}(\int_0^r e^{-(\lambda-\mu)s} ds) dr \\ &= \int_0^\infty Q_r f(y)(\frac{e^{-\mu r} - e^{-\lambda r}}{\lambda - \mu}) dr = (\mu - \lambda)(R_\lambda - R_\mu)(x) \end{aligned}$$

e.g. if we are dealing with real Brownian motion, $R_\lambda f(x) = \int r_\lambda(y-x) f(y) dy$ where $r_\lambda(y-x) = \int_0^\infty (2\pi t)^{-1/2} \exp(-\frac{|y-x|^2}{2t} - \lambda t) dt = \frac{1}{\sqrt{2\lambda}} \exp(-|y-x|\sqrt{2\lambda})$.

Lemma 5.0.1. *Let X be a Markov process with semigroup $(Q_t)_{t\ge 0}$ w.r.t. the filtration $(\mathscr{F}_t)$. Let $h \in B(E)$ be nonnegative and let $\lambda > 0$. Then the proces*

$$e^{-\lambda t} R_\lambda h(X_t)$$

is an $(\mathscr{F}_t)$-supermartingale.

pf. The r.v. $e^{-\lambda t} R_\lambda h(X_t)$ are bounded and therefore in L^1. Then $\forall s \ge 0$, $Q_s R_\lambda h = \int_0^\infty e^{-\lambda t} Q_{s+t} h dt$
$\to e^{-\lambda s} Q_s R_\lambda h = \int_0^\infty e^{-\lambda(s+t)} Q_{s+t} h dt = \int_s^\infty e^{-\lambda t} Q_t h dt \le R_\lambda h$. Thus $\forall s, t \ge 0$, $E[e^{-\lambda(t+s)} R_\lambda h(X_{t+s}) | \mathscr{F}_t] = e^{-\lambda(t+s)} Q_s R_\lambda h(X_t) \le e^{-\lambda t} R_\lambda h(X_t)$.

5.1 Feller Semigroups

Assume E is a metrisable locally compact topological space, and countable at infinity, i.e. E is a countable union of compact sets. Also assume E is equipped with its Borel σ-field.
$\to E$ is Polish and $\exists (K_n)_{n\ge 1}$ an increasing sequence of compact subsets of E s.t. any compact subset of E is contained in K_n for some n.

A function $f : E \to \mathbb{R}$ tends to 0 at infinity if $\forall \epsilon > 0$, $\exists K \subset E$ a compact subset s.t. $|f(x)| \le \epsilon\ \forall x \in E \setminus K \Leftrightarrow \sup_{x \in E \setminus K_n} |f(x)| \to 0$ as $n \to \infty$. Let $C_0(E)$ stands for the set of all continuous real functions on E that tends to 0 at infinity. The space $C_0(E)$ is a Banach space for the supremum norm $||f|| = \sup_{x\in E} |f(x)|$.

Definition 5.1.1. *Let $(Q_t)_{t\geq 0}$ be a transition semigroup on E. We say $(Q_t)_{t\geq 0}$ is a Feller semigroup if*

1. $\forall f \in C_0(E),\ Q_t f \in C_0(E)$
2. $\forall f \in C_0(E),\ ||Q_t f - f|| \to 0$ *as* $t \to 0$

A Markov process with values in E is a Feller process if its semigroup is Feller.

$\to \forall s \geq 0,\ \lim_{t\downarrow 0} ||Q_{s+t}f - Q_s f|| = \lim ||Q_s(Q_t f - f)|| = 0$ since Q_s is a contraction of $C_0(E)$.

Fix a Feller semigroup $(Q_t)_{t\geq 0}$ on E. Then $R_\lambda f \in C_0(E),\ \forall f \in C_0(E),\ \lambda > 0$.

Proposition 5.1.1. *Let $\lambda > 0$ and set $\mathscr{R} = \{R_\lambda f : f \in C_0(E)\}$. Then $\mathscr{R}$ does not depend on the choice $\lambda > 0$. Furthermore $\mathscr{R}$ is a dense subspace of $C_0(E)$.*

pf. if $\lambda \neq \mu$, $R_\lambda f = R_\mu(f + (\mu - \lambda)R_\lambda f)$ by the resolvent equation. Thus any function of the form $R_\lambda f$ with $f \in C_0(E)$ is also of the form $R_\mu g$ for some $g \in C_0(E)$. So the first statement follows.
$\mathscr{R}$ is a linear subspace of $C_0(E)$ trivially. $\forall f \in C_0(E),\ \lambda R_\lambda f = \lambda \int_0^\infty e^{-\lambda t} Q_t f\, dt = \int_0^\infty e^{-t} Q_{t/\lambda} f\, dt \to f \in C_0(E)$ as $\lambda \to \infty$ by property 2 of the definition and dominated convergence.

Definition 5.1.2. *Set $D(L) = \{f \in C_0(E) : \frac{Q_t f - f}{t}$ converges in $C_0(E)$ when $t \downarrow 0\}$ and $\forall f \in D(L)$, set*

$$Lf = \lim_{t\downarrow 0} \frac{Q_t f - f}{t}$$

Then $D(L)$ is a linear subspace of $C_0(E)$ and $L : D(L) \to C_0(E)$ is a linear operator called the generator of the semigroup $(Q_t)_{t\geq 0}$. The subspace $D(L)$ is called the domain of L.

Proposition 5.1.2. *Let $f \in D(L),\ s \geq 0$. Then $Q_s f \in D(L),\ L(Q_s f) = Q_s(Lf)$.*

pf. Writing $\frac{Q_r(Q_s f) - Q_s f}{t} = Q_s(\frac{Q_t f - f}{t})$. Then by the fact that Q_s is a contraction of $C_0(E)$, $t^{-1}(Q_t(Q_s f) - Q_s f)$ converges to $Q_s(Lf)$, so the proposition follows.

Proposition 5.1.3. *If $f \in D(L),\ \forall t \geq 0$*

$$Q_t f = f + \int_0^t Q_s(Lf)ds = f + \int_0^t L(Q_s f)ds$$

pf. $\forall f \in D(L)$, $\forall t \geq 0$, $\epsilon^{-1}(Q_{t+\epsilon}f - Q_t f) = Q_t(\epsilon^{-1}(Q_\epsilon f - f)) \to Q_t(Lf)$ as $\epsilon \downarrow 0$. Moreover the convergence is uniform when t varies over $\mathbb{R}_{\geq 0}$.This implies $\forall x \in E$, $t \mapsto Q_t f(x)$ is differentiable on $\mathbb{R}_{\geq 0}$ and its derivative is $Q_t(Lf)(x)$, which is a continuous function of t. Then by the previous proposition, the current proposition follows.

Proposition 5.1.4. *Let $\lambda > 0$.*

1. $\forall g \in C_0(E)$, $R_\lambda g \in D(L)$ *and* $(\lambda - L)R_\lambda g = g$
2. *if* $f \in D(L)$, $R_\lambda(\lambda - L)f = f$

Consequently, $D(L) = \mathscr{R}$ and the operators $R_\lambda : C_0(E) \to \mathscr{R}$ and $\lambda - L : D(L) \to C_0(E)$ are the inverse of each other.

pf.

- if $g \in C_0(E)$, $\forall \epsilon > 0$, $\epsilon^{-1}(Q_\epsilon R_\lambda g - R_\lambda g) = \epsilon^{-1}(\int_0^\infty e^{-\lambda t} Q_{\epsilon+t} g dt - \int_0^\infty e^{-\lambda t} Q_t g dt)$ $= \epsilon^{-1}((1 - e^{-\lambda\epsilon}) \int_0^\infty e^{-\lambda t} Q_{\epsilon+t} g dt - \int_0^\epsilon e^{-\lambda t} Q_t g dt) \to \lambda R_\lambda g - g$ as $\epsilon \to 0$ by the property 2 of the definition of a Feller semigroup. Thus $R_\lambda g \in D(L)$, $L(R_\lambda g) = \lambda R_\lambda g - g$.
- let $f \in D(L)$. Then by the previous proposition $Q_t f = f + \int_0^t Q_s(Lf) ds$. Thus $\int_0^\infty e^{-\lambda t} Q_t f(x) dt = \frac{f(x)}{\lambda} + \int_0^\infty e^{-\lambda t}(\int_0^t Q_s(Lf)(x) ds) dt = \frac{f(x)}{\lambda} + \int_0^\infty \frac{e^{-\lambda s}}{\lambda} Q_s(Lf)(x) ds$ $\to \lambda R_\lambda f = f + R_\lambda L f$.

$\to \begin{cases} \mathscr{R} \subseteq D(L) \\ D(L) \subseteq \mathscr{R} \end{cases}$ by the previous identities and therefore the last assertion follows.

Corollary 5.1.1. *The semigroup $(Q_t)_{t \geq 0}$ is determined by the generator L.*

pf. Let f be a nonnegative function in $C_0(E)$. Then R_λ is the unique element of $D(L)$ s.t. $(\lambda - L)R_\lambda f = f$. On the other hand, knowing $R_\lambda f(x) = \int_0^\infty e^{-\lambda t} Q_t f(x) dt$, $\forall \lambda > 0$ determines the continuous function $t \mapsto Q_t f(x)$. Since Q_t is characterised by the values of $Q_t f$ for every nonnegative function f in $C_0(E)$, the corollary follows.

e.g. **Real Brownian Motion**
$\forall \lambda > 0$, $\forall f \in C_0(\mathbb{R})$, $R_\lambda f(x) = \frac{1}{\sqrt{2\lambda}} \exp(-\sqrt{2\lambda}|y - x|) f(y) dy$. If $h \in D(L)$, $\exists f \in C_0(\mathbb{R})$ s.t. $h = R_\lambda f$. Taking $\lambda = \frac{1}{2}$, $h(x) = \int \exp(-|y - x|) f(y) dy$. By differentiating under integral sign, $h'(x) = \int sgn(y - x) \exp(-|y - x|) f(y) dy$ with the notation $sgn(z)$ $= \mathbb{1}_{\{z>0\}} - \mathbb{1}_{\{z \leq 0\}}$. Let $x_0 \in \mathbb{R}$. Then $\forall x > x_0$, $h'(x) - h'(x_0) = \int sgn(y - x) \exp(-|y - x|) - sgn(y - x_0) \exp(-|y - x_0|)) f(d) dy = \int_{x_0}^x (-\exp(-|y - x|) - \exp(|y - x_0|)) f(y) dy +$

$\int_{\mathbb{R}\setminus[x_0,x]} sgn(y-x_0)(\exp(-|y-x|)-\exp(-|y-x_0|))f(y)dy \to \frac{h'(x)-h'(x_0)}{x-x_0} \to -2f(x_0)+h(x_0)$ as $x \downarrow x_0$. Also when $x \uparrow x_0$ we get the same limit and therefore h is twice differentiable and $h'' = -2f+h$. Since $h = R_{1/2}f$, $(\frac{1}{2}-L)h = f$ by the previous proposition, and therefore $Lh = -f+\frac{1}{2}h = \frac{1}{2}h''$. Thus we get $D(L) \subset \{h \in C^2(\mathbb{R}) : h, h'' \in C_0(\mathbb{R})\}$ and if $h \in D(L)$, $Lh = \frac{1}{2}h''$.

Consider a general Feller semigroup $(Q_t)_{t\geq 0}$. Assume on some probability space, $\forall x \in E$, $\exists (X_t^x)_{t\geq 0}$, a Markov process with semigroup $(Q_t)_{t\geq 0}$ w.r.t. the filtration $(\mathscr{F}_t)_{t\geq 0}$ and s.t. $P(X_0^x = x) = 1$. Also assume the sample paths of $(X_t^x)_{t\geq 0}$ is cadlag.

Theorem 5.1.1. *Let $h, g \in C_0(E)$. TFAE*

1. *$h \in D(L)$, $Lh = g$*
2. *$\forall x \in E$, $h(X_t^x) - \int_0^t g(X_s^x)ds$ is a martingale w.r.t. the filtration $(\mathscr{F}_t)$.*

pf.

- $1 \to 2$

  let $h \in D(L)$, $g = Lh$. Then by the proposition 4.1.3, $\forall s \geq 0$, $Q_s h = h + \int_0^s Q_r g dr$. Thus $\forall t, s \geq 0$, $E[h(X_{t+s}^x)|\mathscr{F}_t] = Q_s h(X_t^x) = h(X_t^x) + \int_0^s Q_r g(X_t^r)dr$. On the other hand, $E[\int_t^{t+s} g(X_r^x)dr|\mathscr{F}_t] = \int_t^{t+s} E[g(X_r^x)|\mathscr{F}_t]dr$ $= \int_t^{t+s} Q_{r-t} g(X_t^x)dr = \int_0^s Q_r g(X_t^x)dr$ by the fact that the conditional expectation and the integral can be interchanged. Thus we get $E[h(X_{t+s}^x) - \int_0^{t+s} g(X_r^x)dr|\mathscr{F}_t]$ $= h(X_t^x) - \int_0^t g(X_r^x)dr$.

- $2 \to 1$

  $\forall t \geq 0$, $E[h(X_t^x) - \int_0^t g(X_r^x)dr] = h(x)$ and by the definition of Markov process, $E[h(X_t^x) - \int_0^t g(X_r^x)dr] = Q_t h(x) - \int_0^t Q_r g(x)dr$. Thus $\frac{Q_t h - h}{t} = \frac{1}{t}\int_0^t Q_r g dr \to g$ as $t \downarrow 0$ in $C_0(E)$ by the property 2 of the definition of Feller semigroup. Thus $h \in D(L)$, $Lh = g$.

e.g. **d-dimensional Brownian Motion**
if $h \in C^2(\mathbb{R}^d)$, $h(X_t) - \frac{1}{2}\int_0^t \Delta h(X_s)ds$ is a continuous local martingale by the Ito's formula. This is a martingale if we assume $h, \Delta h$ are in $C_0(\mathbb{R}^d)$. So by the previous theorem, $h \in D(L)$, $Lh = \frac{1}{2}\Delta h$.

Theorem 5.1.2. *Let $(X_t)_{t\geq 0}$ be a Markov process with semigroup $(Q_t)_{t\geq 0}$, w.r.t. the filtration $(\mathscr{F}_t)_{t\in[0,\infty]}$. Set $\tilde{\mathscr{F}}_t = \begin{cases} \mathscr{F}_\infty & t = \infty \\ \mathscr{F}_{t+} \vee \sigma(\mathscr{N}) & t \in [0,\infty) \end{cases}$ where $\mathscr{N}$ denotes the class*

of all $\mathscr{F}_\infty$-measurable sets that have zero probability. Then the process $(X_t)_{t\geq 0}$ has a cadlag modification $(\tilde{X}_t)_{t\geq 0}$ which is adapted to the filtration $(\tilde{\mathscr{F}}_t)$. Moreover $(\tilde{X}_t)_{t\geq 0}$ is a Markov process with semigroup $(Q_t)_{t\geq 0}$ w.r.t. the filtration $(\tilde{\mathscr{F}}_t)_{t\in[0,\infty]}$.

pf.

- Construction of cadlag $\tilde{X}$ in E_Δ

  let $E_\Delta = E \cup \{\Delta\}$ be the Alexandroff compactification of E, which is obtained by adding the point at infinity Δ to E. Then $\forall f \in C_0(E)$ can be extended to a continuous function on E_Δ by setting $f(\Delta) = 0$. Write $C_0^+(E)$ for the set of all nonnegative functions in $C_0(E)$. Then $\exists (f_n)_{n\geq 0}$ in $C_0^+(E)$ which separates the points of E_Δ, in the sense that $\forall x, y \in E_\Delta$ with $x \neq y$, $\exists n \in \mathbb{Z}_{\geq 0}$ s.t. $f_n(x) \neq f_n(y)$. Then $\mathscr{H} = \{R_p f_n : p \geq 1,\ n \geq 0\}$ is also a countable subset of $C_0^+(E)$ which separates the points of E_Δ. If $h \in \mathscr{H}$, by the previous lemma $\exists p \geq 1$ s.t. $e^{-pt}h(X_t)$ is a supermartingale. Let D be a countable dense subset of $\mathbb{R}_{\geq 0}$. Then $\lim_{s\downarrow\downarrow t,\, s\in D} h(X_s)$, $\lim_{s\uparrow\uparrow t, s\in D} h(X_s)$ exists simultaneously for all $t \in \mathbb{R}_{\geq 0}$ outside an $\mathscr{F}_\infty$-measurable event N_h of zero probability. Then set $N = U_{h\in\mathscr{H}} N_h$ in such a way that we still have $N \in \mathscr{N}$. Then if $\omega \notin N$, $\lim_{s\downarrow\downarrow t,\, s\in D} h(X_s)$, $\lim_{s\uparrow\uparrow t, s\in D} h(X_s)$ exists $\forall t \geq 0$ in E_Δ(in fact if we assume $X_s(\omega)$ has two distinct accumulation points in E_Δ when $s \downarrow\downarrow t$, $\exists h \in \mathscr{H}$ that separates these two points, a contradiction). Set $\tilde{X}_t(\omega) = \begin{cases} \lim_{s\downarrow\downarrow t, s\in D} X_s(\omega) & \forall \omega \in \Omega \setminus N,\ \forall t \geq 0 \\ x_0 & \omega \in N,\ \forall t \geq 0 \end{cases}$. where x_0 is a fixed point in E. Then $\forall t \geq 0$, $\tilde{X}_t$ is an $\tilde{\mathscr{F}}_t$-measurable r.v. with values in E_Δ. Furthermore, $\forall \omega \in \Omega$, $t \mapsto \tilde{X}_t(\omega)$ viewed as a mapping with values in E_Δ, is cadlag by the lemma 2.2.1.

- $\tilde{X}$ is a modificaton of X

  $\forall t \geq 0$, $\forall f, g \in C_0(E)$, let (t_n) be a sequence in D that decreases to t. Then $E[f(X_t)g(\tilde{X}_t)] = \lim E[f(X_t)g(X_{t_n})] = \lim E[f(X_t)Q_{t_n - t}g(X_t)] = E[f(X_t)g(X_t)]$ since $Q_{t_n-t}g \to g$ by the definition of Feller semigroup. Thus $(X_t, \tilde{X}_t)$, (X_t, X_t) have the same distribution and therefore $P(X_t = \tilde{X}_t) = 1$.

- $\tilde{X}$ is a Markov proces

  $\forall s \geq 0$, $t > 0$, $\forall A \in \tilde{\mathscr{F}}_s$, $\forall f \in C_0(E)$, $\begin{matrix} E[\mathbb{1}_A f(\tilde{X}_{s+t})] = E[\mathbb{1}_A (X_{s+t})] \\ E[\mathbb{1}_A Q_t f(\tilde{X}_s)] = E[\mathbb{1}_A Q_t f(X_s)] \end{matrix}$ since $\tilde{X}_s = X_s$, $\tilde{X}_{s+t} = X_{s+t}$ a.s. Because A is equal a.s. to an element of $\mathscr{F}_{s+}$, we may assume $A \in \mathscr{F}_{s+}$. Let (s_n) be a sequence in D that decreases to s so that $A \in \mathscr{F}_{s_n}$, $\forall n$. Then as soon as $s_n \leq s + t$, $E[\mathbb{1}_A f(X_{s+t})] = E[\mathbb{1}_A E[f(X_{s+t})|\mathscr{F}_{s_n}]]$

$= E[\mathbb{1}_A Q_{s+t-s_n} f(X_{s_n})]$. Since (Q_t) is a Feller process, $Q_{s+t-s_n} f \to Q_t f$ and since $X_{s_n} = \tilde{X}_{s_n}$ a.s. we get the Markov property by passing them to the limit.

- $t \mapsto \tilde{X}_t(\omega)$ are cadlag as E-valued mappings

  fix a function $g \in C_0^+(E)$ s.t. $g(x) > 0$, $\forall x \in E$. The function $h = R_1 g$ then satisfies the same property. Set $\forall t \geq 0$, $Y_t = e^{-t} h(\tilde{X}_t)$. Then by the previous lemma, $(Y_t)_{t\geq 0}$ is a nonnegative supermartingale w.r.t. the filtration $(\tilde{\mathscr{F}}_t)$. Also the sample paths of $(Y_t)_{t\geq 0}$ are cadlag. $\forall n \geq 1$, set $T_{(n)} = \inf\{t \geq 0 : Y_t < \frac{1}{n}\}$. Then it is a stopping time of the filtration $(\tilde{\mathscr{F}}_t)$. Consequently, $T = \lim_{n\to\infty} \uparrow T_{(n)}$ is a stopping time. $\forall t \in [0, T_{(n)})$, $\tilde{X}_t \in E$, $\tilde{X}_{t-} \in E$ and we may redefine $\tilde{X}_t(\omega) = x_0$, $\forall \omega \in \{T < \infty\} \in \mathscr{N}$. Let q be a rational number. Then $E[Y_{T+q} \mathbb{1}_{\{T<\infty\}}] \leq E[Y_{T_{(n)}} \mathbb{1}_{\{T_{(n)}<\infty\}}] \leq \frac{1}{n}$ by the theorem 2.2.6. By letting n tend to ∞, $E[Y_{T+q} \mathbb{1}_{\{T<\infty\}}] = 0$ hence $Y_{T+q} = 0$ a.s. on $\{T < \infty\}$. By the right-continuity of sample paths of Y, $Y_t = 0$ $\forall t \in [T, \infty)$ a.s. on $\{T < \infty\}$. Then for every integer $k \geq 0$, $Y_k = e^{-k} h(\tilde{X}_k) > 0$ a.s. since $\tilde{X}_k \in E$ a.s. Thus $P(T < \infty) = 0$.

Let $(Q_t)_{t\geq 0}$ be a transition semigroup on E, where E is a metric space with its Borel σ-field. Also assume $\forall x \in E, \exists (X_t^x)_{t\geq 0}$ a Markov process with semigroup $(Q_t)_{t\geq 0}$ s.t. $X_0^x = x$ a.s. and the sample paths of X are cadlag. If semigroup is Feller, its existence is proved by the previous theorem. The space of all caldag paths $f : \mathbb{R}_{\geq 0} \to E$ is denoted by $\mathbb{D}(E)$. This space is equipped with the σ-field $\mathscr{D}$ generated by the coordinate mappings $f \mapsto f(t)$. $\forall x \in E$, $\mathbb{P}_x$ is the probability measure on $\mathbb{D}(E)$, which is the law of the random paths $(X_t^x)_{t\geq 0}$.

Theorem 5.1.3 (Simple Markov Property)**.** *Let $(Y_t)_{t\geq 0}$ be a Markov process with semigroup $(Q_t)_{t\geq 0}$ w.r.t. the filtration $(\mathscr{F}_t)_{t\geq}$. Assume the sample paths of Y are cadlag. Let $s \geq 0$ and $\Phi : \mathbb{D}(E) \to \mathbb{R}_{\geq 0}$ be a measurable function. Then*

$$E[\Phi((Y_{s+t})_{t\geq 0}) | \mathscr{F}_s] = \mathbb{E}_{Y_s}[\Phi].$$

pf.Consider the case where $\Phi = \mathbb{1}_A$ where $A = \{f \in \mathbb{D}(E) : f(t_1) \in B_1, \ldots, f(t_p) \in B_p\}$ where $0 \leq t_1 < \cdots < t_p$, $B_1, \ldots, B_p$ are measurable subsets of E.
CLAIM : $\phi_1, \ldots, \phi_p \in B(E)$**,**

$$E[\phi_1(Y_{s+t_1}) \ldots \phi_s(Y_{s+t_p}) | \mathscr{F}_s] = \int Q_{t_1}(Y_s, dx_1)\phi_1(x_1) \cdots \int Q_{t_p - t_{p-1}}(x_{p-1}, dx_p)\phi_p(x_p)$$

pf. if $p = 1$, the claim means the Markov property so it holds trivially.
if $p > 1$, $E[\phi_1(Y_{s+t_1}) \ldots \phi_p(Y_{s+t_p}) | \mathscr{F}_s] = E[\phi_1(Y_{s+t_1}) \ldots \phi_{p-1}(Y_{s+t_{p-1}}) E[\phi_p(Y_{s+t_p}) | \mathscr{F}_{s+t_{p-1}}] | \mathscr{F}_s]$
$= E[\phi_1(Y_{s+t_1}) \ldots \phi_{p-1}(Y_{s+t_{p-1}}) Q_{t_p - t_{p-1}} \phi_p(Y_{s+t_p}) | \mathscr{F}_s]$. By induction, the theorem follows.

Theorem 5.1.4 (Strong Markov Property)**.** *Retain the assumptions of the previous theorem and suppose in addition that $(Q_t)_{t\geq 0}$ is a Feller semigroup. Let T be a stopping time of the filtration $(\mathscr{F}_{t+})$ and let $\Phi : \mathbb{D}(E) \to \mathbb{R}_{\geq 0}$ be a measurable function. Then $\forall x \in E$*

$$E[\mathbb{1}_{\{T<\infty\}}\Phi((Y_{T+t})_{t\geq 0})|\mathscr{F}_T] = \mathbb{1}_{\{T<\infty\}}\mathbb{E}_{Y_T}[\Phi]$$

pf. $\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{Y_T}[\Phi]$ is $\mathscr{F}_T$-measurable because $\forall \omega \in \{T<\infty\}$, $\omega \mapsto Y_T(\omega)$ is $\mathscr{F}_T$-measurable and the function $y \mapsto \mathbb{E}_y[\Phi]$ is measurable. Fix $A \in \mathscr{F}_T$ and consider the case where $\Phi(f) = \phi_1(f(t_1))\dots\phi_p(f(t_p))$ where $0 \leq t_1 < \dots < t_p$, $\phi_1,\dots,\phi_p \in B(E)$. When $p = 1$, we get the equation above trivially. When $p > 1$,
$E[\mathbb{1}_{A\cap\{T<\infty\}}\phi_1(Y_{T+t_1})\dots\phi_p(Y_{T+t_p})]$
$= E[\mathbb{1}_{A\cap\{T<\infty\}}\phi_1(Y_{T+t_1})\dots\phi_{p-1}(Y_{T+t_{p-1}})E[\phi_p(Y_{T+t_p})|\mathscr{F}_{T+t_{p-1}}]]$
$= E[\mathbb{1}_{A\cap\{T<\infty\}}\phi_1(Y_{T+t_1})\dots\phi_{p-1}(Y_{T+t_{p-1}})Q_{t_p-t_{p-1}}\phi_p(Y_{T+t_p})]$. Fix $t \geq 0$, $\phi \in B(E)$. Then if $E[\mathbb{1}_{A\cap\{T<\infty\}}\phi(Y_{T+t})] = E[\mathbb{1}_{A\cap\{T<\infty\}}Q_t\phi(Y_T)]$, the statement follows by induction.

- $E[\mathbb{1}_{A\cap\{T<\infty\}}\phi(Y_{T+t})] = E[\mathbb{1}_{A\cap\{T<\infty\}}Q_t\phi(Y_T)]$
  pf. Assume $\phi \in C_0(E)$. On the event $T<\infty$, write $[T]_n$ for the smallest real number of the form $i2^{-n}$, $i \in \mathbb{Z}_{\geq 0}$ which is strictly greater than T. Then
  $$\begin{aligned} E[\mathbb{1}_{A\cap\{T<\infty\}}\phi(Y_{T+t})] &= \lim_{n\to\infty} E[\mathbb{1}_{A\cap\{T<\infty\}}\phi(Y_{[T]_n+t})] && \text{right-continuity of sample paths} \\ &= \lim \textstyle\sum_{i=1}^{\infty} E[\mathbb{1}_{A\cap\{(i-1)2^{-n}\leq T<i2^{-n}\}}\phi(Y_{i2^{-n}+t})] \\ &= \lim \textstyle\sum E[\mathbb{1}_{A\cap\{(i-1)2^{-n}\leq T<iw^{-n}\}}Q_t\phi(Y_{i2^{-n}})] && A\cap\{(i-1)2^{-n}\leq T<i2^{-n}\}\in\mathscr{F}_{i2^{-n}} \\ &= \lim E[\mathbb{1}_{A\cap\{T<\infty\}}Q_t\phi(Y_{[T]_n})] = E[\mathbb{1}_{A\cap\{T<\infty\}}Q_t\phi(Y_T)] && \text{continuity of } Q_t\phi \end{aligned}$$

$\to$for Brownian motions, the above theorem is equivalent to the strong Markov property in the Brownian Motion chapter.

e.g.

- **Jump Processes on a Finite State Space**
  Assume the state space E is finite and equipped with the discrete topology. Any cadlag function $f \in \mathbb{D}(E)$ must be the following : $\exists t_1 \in (0,\infty]$, a real number, s.t. $f(t) = f(0)$, $\forall t \in [0,t_1)$, then if $t_1 < \infty$, $\exists t_2 \in (t_1,\infty]$ s.t. $f(t) = f(t_1) \neq f(0)$ $\forall t \in [t_1,t_2)$ and so on. Consider a Feller semigroup $(Q_t)_{t\geq 0}$ on E. On a probability space Ω, equipped with a right-continuous filtration $(\mathscr{F}_t)_{t\in[0,\infty]}$, we can construct a collection $(P_x)_{x\in E}$ of probability measures and a process $(X_t)_{t\geq 0}$ with cadlag sample paths s.t. under P_x, X is Markov with semigroup $(Q_t)_{t\geq 0}$ w.r.t. the filtraton $(\mathscr{F}_t)$, $P_x(X_0 = x) = 1$. Then $\forall \omega \in \Omega$, $\exists T_0(\omega) = 0 < T_1(\omega) \leq \dots \leq \infty$ a sequence s.t. $X_t(\omega) = X_0(\omega)$, $\forall t \in [0,T_1(\omega))$, and $\forall i \geq 1$,
  $$T_i(\omega) < \infty \text{ implies } \begin{array}{c} T_i(\omega) < T_{i+1}(\omega) \\ X_{T_i(\omega)} \neq X_{T_{i-1}(\omega)}(\omega) \\ X_t(\omega) = X_{T_i(\omega)}(\omega),\ \forall t \in [T_i(\omega),T_{i+1}(\omega)) \end{array} \text{. Moreover } T_n(\omega) \uparrow$$

∞ as $n \to \infty$. Then $T_0, T_1, \ldots$ are stopping times, given $\{T_1 < t\} = \cup_{q \in [0,t) \cap \mathbb{Q}} \{X_q \neq X_0\} \in \mathscr{F}_t$. For $\lambda > 0$, a positive random variable U is exponentially distributed with parameter λ if $P(U > r) = e^{-\lambda r}$, $\forall r \geq 0$.

Lemma 5.1.1. *Let $x \in E$. There exists a real number $q(x) \geq 0$ s.t. the random variable T_1 is exponentially distributed with parameter $q(x)$ under P_x. Furthermore, if $q(x) > 0$, T_1, X_{T_1} are independent under P_x.*

pf. Let $s, t \geq 0$. Then $P_x(T_1 > s + t) = E_x[\mathbb{1}_{\{T_1 > s\}} \Phi((X_{s+r})_{r \geq 0})]$ where $\Phi(f) = \mathbb{1}_{\{f(r) = f(0), \forall r \in [0,t]}$ for $f \in \mathbb{D}(E)$. By the simple Markov property, $P_x(T_1 > s + t) = E_x[\mathbb{1}_{\{T_1 > s\}} E_{X_s}[\Phi((X_r)_{r \geq 0})]] = E_x[\mathbb{1}_{\{T_1 > s\}} P_x(T > t)] = P_x(T_1 > s) P_x(T_1 > t)$ which implies T_1 is exponentially distributed under P_x. Assume $q(x) > 0$ so that $T_1 < \infty$, P_x a.s. Then $\forall t \geq 0$, $\forall y \in E, P_x(T_1 > t, X_{T_1} = y) = E_x[\mathbb{1}_{\{T_1 > t\}} \Psi((X_{r+t})_{r \geq 0})]$ where for $f \in \mathbb{D}(E)$, $\Psi(f) = \begin{cases} 0 & f \text{ is constant} \\ \mathbb{1}_{\{\gamma_1(f) = y\}} & \text{otherwise} \end{cases}$ where $\gamma_1(f)$ is the value of the f after its first jump. Thus $P_x(T_1 > t, X_{T_1} = y) = E_x[\mathbb{1}_{\{T_1 > t\}} E_{X_t}[\Psi((X_r)_{r \geq 0})]]$ $= E_x[\mathbb{1}_{\{T_1 > x\}} P_x(X_{T_1} = y)] = P_x(T_1 > t) P_x(X_{T_1} = y)$ which gives independency.

Points x s.t. $q(x) = 0$ are absorbing states for the Markov process, in the sense that $P_x(X_t = x, \forall t \geq 0) = 1$.

$\forall x \in E$ s.t. $g(x) > 0$, $\forall y \in E$, set $\Pi(x, y) = P_x(X_{T_1} = y)$. Then $\Pi(x, \cdot)$ is a probability measure on E and $\Pi(x, x) = 0$.

Proposition 5.1.5. *Let L denote the generator of $(Q_t)_{t \geq 0}$. Then $D(L) = C_0(E) = B(E)$ and $\forall \phi \in B(E)$, $\forall x \in E$,*

- *if $q(x) = 0$, $L\phi(x) = 0$*
- *if $q(x) > 0$, $L\phi(x) = q(x) \sum_{y \in E, y \neq x} \Pi(x, y)(\phi(y) - \phi(x)) = \sum_{y \in E} L(x, y) \phi(y)$*

where $L(x, y) = \begin{cases} q(x)\Pi(x, y) & y \neq x \\ -q(x) & y = x \end{cases}$

pf. Let $\phi \in B(E)$, $x \in E$. If $q(x) = 0$, $Q_t \phi(x) = \phi(x)$ and therefore $\lim_{t \downarrow 0} \frac{Q_t \phi(x) - \phi(x)}{t} = 0$. By the Markov property at T_1, $P_x(T_2 \leq t) \leq P_x(T_1 \leq t, T_2 \leq T_1 + t)$ $= E_x[\mathbb{1}_{\{T_1 \leq t\}} P_{X_{T_1}}(T_1 \leq t)] \leq E_x[\mathbb{1}_{\{T_1 \leq t\}} \sup_{y \in E} P_y(T_1 \leq t)] \leq E_x[\mathbb{1}_{\{T_1 \leq t\}} t \sup_{y \in E} q(y)]$ $\to$If $q(x) > 0$, $P_x(T_2 \leq t) = O(t^2)$ as $t \to 0$. Then $Q_t \phi(x) = E_x[\phi(X_t)] = E_x[\phi(X_t) \mathbb{1}_{\{t < T_1\}}] + E_x[\phi(X_{T_1}) \mathbb{1}_{\{T_1 \leq t\}}] + O(t^2) = \phi(x) e^{-q(x)t} + (1 - e^{-q(x)t}) \sum_{y \in E, y \neq x} \Pi(x, y) \phi(y) + O(t^2)$ by the independence of T_1 and X_{T_1}. Then

$$\frac{Q_t \phi(x) - \phi(x)}{t} \to -q(x)\phi(x) + q(x) \sum_{y \in E, x \neq y} \Pi(x, y) \phi(y)$$

$\rightarrow$ if $\phi(y) = \mathbb{1}_{\{y\}}$, $y \neq x$, $L(x,y) = \frac{d}{dt}P_x(X_t = y)|_{t=0}$ so $L(x,y)$ can be interpreted as the instantaneous rate of transition from x to y.

Proposition 5.1.6. *We assume $q(y) > 0$, $\forall y \in E$. Let $x \in E$. Then P_x a.s. the jump times $T_1 < T_2 < \dots$ are all finite and the sequence $X_0, X_{T_1}, \dots$, is under P_x a discrete Markov chain with transition kernel Π started from x. Furthermore, conditionally on $(X_0, X_{T_1}, \dots)$, the random variables $T_1 - T_0, T_2 - T_1, \dots$ are independent and $\forall i \geq 0$, the conditional distribution of $T_{i+1} - T_i$ is exponential with parameter $q(X_{T_i})$.*

pf. By the strong Markov property, all stopping times $T_1, T_2, \dots$ are finite P_x a.s. Then let $y, z \in E$ and $f_1, f_2 \in B(\mathbb{R}_{\geq 0})$. By the strong Markov property at T_1,
$E_x[\mathbb{1}_{X_{T_1}=y} f_1(T_1) \mathbb{1}_{\{X_{T_2}=z\}} f_2(T_2 - T_1)] = E_x[\mathbb{1}_{\{X_{T_1}=x\}} f_1(T_1) E_{X_{T_1}}[\mathbb{1}_{\{X_{t_1}=x\}} f_2(T_1)]]$
$= \Pi(x,y)\Pi(y,z) \int_0^\infty ds_1 e^{-q(x)s_1} f_1(s_1) \int_0^\infty ds_2 e^{-q(y)s_2} f_2(s_2)$. By induction, $\forall y_1, \dots, y_p \in E$, $f_1, \dots, f_p \in B(\mathbb{R}_{\geq 0})$, $E_x[\mathbb{1}_{X_{T_1}=y_1} \dots \mathbb{1}_{\{X_{T_p}=y_p\}} f_1(T_1) f_2(T_2 - T_1) \dots f_p(T_p - T_{p-1})]$
$= \Pi(x, y_1) \dots \Pi(y_{p-1}, y_p) \Pi_{i=1}^p (\int_0^\infty ds e^{-q(y_{i-1})s} f_i(s))$ where $y_0 = x$ by convention. Then the proposition follows.

- **Feynman-Kac Formula**
  Let v be a nonnegative function in $C_0(E)$. $\forall x \in E$, $\forall t \geq 0$, set

$$\forall \phi \in B(E),\ Q_t^* \phi(x) = E_x[\phi(X_t) \exp(-\int_0^t v(X_s) ds)]$$

  Then $\forall \phi \in B(E)$, $\forall s, t \geq 0$, $Q_{t+s}^* \phi = Q_t^*(Q_s^* \phi)$ and $Q_t \phi - Q_t^* \phi = \int_0^t Q_s(v Q_{t-s}^* \phi) ds$.
  Also if $\phi \in D(L)$, $\frac{d}{dt} Q_t^* \phi|_{t=0} = L\phi - v\phi$.
  pf.

  - $E_x[\phi(X_{t+s}) \exp(-\int_0^{s+t} v(X_r) dr)]$
    $= E_x[\phi(X_{t+s}) \exp(-\int_0^s v(X_r) dr) \exp(-\int_s^{s+t} v(X_r) dr)]$
    $= E_x[E_x[\phi(X_{s+t}) | \mathscr{F}_s] \exp(-\int_0^s v(X_r) dr) \exp(-\int_0^t Q_s v(X_{r-s}) dr)]$
    $= E_x[Q_s \phi(X_t) \exp(-\int_0^s v(X_r) dr) \exp(-\int_0^t Q_s v(X_r) dr)] = Q_t^*(Q_s^* \phi)$
  - $1 - \exp(-\int_0^t v(X_s) ds) = \int_0^t v(X_s) \exp(-\int_s^t v(X_r) dr) ds$
    $\Leftrightarrow E_x[Q_t \phi] - E_x[Q_t \phi \exp(-\int_0^t v(X_s) ds)] = E_x[Q_t \phi \int_0^t v(X_s) \exp(-\int_s^t v(X_r) dr) ds]$
    $\int_0^t Q_s(v Q_{t-s}^* \phi(x)) ds = \int_0^t Q_s v E_x[Q_{t-s} \phi \exp(-\int_0^{t-s} v(X_r) dr)] ds$
    $= E_x[\int_0^t Q_s v Q_{t-s} \phi \exp(-\int_0^{t-s} v(X_r) dr) ds] = E_x[\int_0^t v Q_t \phi \exp(-\int_0^{t-s} v(X_r) dr) ds]$
    $= E_x[Q_t \phi \int_0^t v \exp(-\int_0^{t-s} v(X_r) dr) ds] = E_x[Q_t \phi \int_0^t v \exp(-\int_s^t v(X_r) dr) ds]$
    Thus two are equivalent.
  - By the previous statement with differentiation,

$$\frac{d}{dt} Q_t \phi|_{t=0} - \frac{d}{dt} Q_t^* \phi|_{t=0} = \frac{d}{dt} \int_0^t Q_s(v Q_{t-s}^* \phi) ds|_{t=0}$$

Then, $\frac{d}{dt}Q_t^*\phi|_{t=0} = \frac{d}{dt}(Q_t\phi) - Q_0(vQ_0^*\phi) = L\phi - v\phi$ and the last statement follows.

Chapter 6

Partial Differential Equations

6.1 the Heat Equation

Let B stand for a d-dimensional Brownian motion that starts from x under the probability measure $P_x,\ x \in \mathbb{R}^d$. Then $(B_t)_{t\geq 0}$ is a Feller process with semigroup

$$Q_t\phi(x) = \int_{\mathbb{R}^d} (2\pi t)^{-d/2} \exp(-\frac{|y-x|^2}{2t})\phi(y)dy,\ \phi \in B(\mathbb{R}^d)$$

Write L for the generator of this Feller process. If ψ is a twice continuously differentiable function on $\mathbb{R}^d$ s.t. both $\psi,\ \Delta\psi \in C_0(\mathbb{R}^d)$, then $\psi \in D(L)$ and $L\psi = \frac{1}{2}\Delta\psi$. If $\phi \in B(\mathbb{R}^d),\ \forall t > 0,\ Q_t\phi$ can be viewed as the convolution of ϕ with the C^∞ function $p_t(x) = (2\pi t)^{-d/2}\exp(-\frac{|x|^2}{2t})$. Thus $Q_t\phi$ is also a C^∞ function and all derivatives of $Q_t\phi$ also belong to $C_0(\mathbb{R}^d)$. Thus $Q_t\phi \in D(L),\ L(Q_t\phi) = \frac{1}{2}\Delta(Q_t\phi),\ \forall t > 0$.

Theorem 6.1.1. *Let $\phi \in C_0(\mathbb{R}^d)$. $\forall t > 0,\ x \in \mathbb{R}^d$, set*

$$u_t(x) = Q_t\phi(x) = E_x[\phi(B_t)]$$

Then the function $(u_t(x))_{t\geq 0, x\in\mathbb{R}^d}$ solves the partial differential equation $\frac{\partial u_t}{\partial t} = \frac{1}{2}\Delta u_t$ on $(0,\infty)\times\mathbb{R}^d$. Furthermore,

$$\forall x \in \mathbb{R}^d,\ \lim_{s\downarrow 0, y\to x} u_s(y) = \phi(x)$$

pf. $\forall t > 0,\ u_t$ is a C^∞ function, $u_t \in D(L),\ Lu_t = \frac{1}{2}\Delta u_t$. Let $\epsilon > 0$. By the proposition 4.1.3, $\forall t \geq \epsilon,\ u_t = u_\epsilon + \int_0^{t-\epsilon} L(Q_s u_\epsilon)ds = u_\epsilon + \int_\epsilon^t Lu_t ds$. Since $Lu_s = Q_{s-\epsilon}(Lu_\epsilon)$ depends continuously on $s \in [\epsilon,\infty),\ \forall t \geq \epsilon,\ \frac{\partial u_t}{\partial t} = Lu_t = \frac{1}{2}\Delta u_t$. Since $Q_s\phi \to \phi$ as $s \to 0$, the last statement follows.

Brownian motion can be used to represent solutions for many parabolic partial differential equations. In particular, solutions of equations of the form $\frac{\partial u}{\partial t} = \frac{1}{2}\Delta u - vu$ where v is a nonnegative function on $\mathbb{R}^d$, are expressed via the Feynman-Kac formula.

e.g. **Feynman-Kac Formula for Brownian Motion**

Assume $E = \mathbb{R}^d$, $X_t = B_t$. Let v be a nonnegative function in $C_0(\mathbb{R}^d)$ and assume v is continuously differentiable with bounded first derivatives. Set $\forall \phi \in B(\mathbb{R}^d)$, $Q_t^*\phi(x) = E_x[\phi(X_t)\exp(-\int_0^t v(X_s)ds)]$. Then $\forall t > 0$, $\forall \phi \in C_0(\mathbb{R}^d)$, $Q_t^*\phi$ is twice continuously differentiable on $\mathbb{R}^d$ and $Q_t^*\phi$ and its partial derivatives up to order 2 belong to $C_0(\mathbb{R}^d)$.

pf. $\frac{\partial}{\partial x}Q_t^*\phi(x) = \frac{\partial}{\partial x}Q_t\phi(x) - \frac{\partial}{\partial x}\int_0^t Q_s v Q_{t-s}^*\phi ds$

Let $\phi \in C_0(\mathbb{R}^d)$, and set $u_t(x) = Q_t^*\phi(x)$, $\forall t > 0$, $\forall x \in \mathbb{R}^d$. Then $\forall x \in \mathbb{R}^d$, the function $t \mapsto u_t(x)$ is continuously differentiable on $(0,\infty)$ and

$$\frac{\partial u_t}{\partial t} = \frac{1}{2}\Delta u_t - vu_t$$

pf. $\frac{\partial}{\partial t}Q_t^*\phi = \lim_{dt\to 0}\frac{Q_{t+dt}^*\phi - Q_t^*\phi}{dt} = Q_t^*\lim_{dt\to 0}(\frac{Q_{dt}^*\phi-\phi}{dt}) = Q_t^*(L\phi - v\phi)$ where $L\phi, v\phi \in C_0(E)$

$u_t(x) = Q_t^*\phi(x) \to du = \frac{\partial}{\partial t}udt + \frac{\partial}{\partial x}udx = \lim_{dx,dt\to 0}(\frac{\partial}{\partial t}udt + \frac{\partial}{\partial x}udx + \frac{1}{2}(\frac{\partial^2}{\partial t^2}u(dt)^2 + \frac{\partial^2}{\partial x^2}u(dx)^2) + \ldots) = \lim_{dt\to 0}((\frac{\partial}{\partial t}u + \frac{1}{2}\frac{\partial^2}{\partial x^2}u)dt + \frac{\partial}{\partial x}udX_t) = (Q_t^*L\phi - Q_t^*v\phi + \frac{1}{2}\Delta u)dt + \frac{\partial}{\partial x}E_x[)dX_t$ where $Q_t^*L\phi - X_t^*v\phi + \frac{1}{2}\Delta u = 0 \Leftrightarrow LQ_t^*\phi - vQ_t^*\phi + \frac{1}{2}\Delta u = 0$ which gives the above equation.

6.2 Harmonic Functions

Definition 6.2.1. *Let D be a domain of $\mathbb{R}^d$. A function $u : D \to \mathbb{R}$ is said to be harmonic on D if it is twice continuously differentiable and $\Delta u = 0$ on D.*

Definition 6.2.2 (Classical Dirichlet Problem)**.** *Let g be a continuous function on ∂D. A function $u : D \to \mathbb{R}$ solves the Dirichlet problem in D with boundary condition g if u is harmonic on D and has boundary condition g in the sense that $\forall y \in \partial D$, $u(x) \to g(y)$ as $x \to y$, $x \in D$.*

If u solves the Dirichlet problem with boundary condition g, the function $\tilde{u}$ defined on $\bar{D}$ by $\tilde{u}(x) = \begin{cases} u(x) & x \in D \\ g(x) & x \in \partial D \end{cases}$ is continuous, hence bounded on $\bar{D}$.

Proposition 6.2.1. *Let D be a bounded domain and write $T = \inf\{t \geq 0 : B_t \notin D\}$ for the exit time of Brownian motion from D.*

1. *Let g be a continuous function on ∂D and let u be a solution of the Dirichlet problem in D with boundary condition g. Then $\forall x \in D$,*

$$u(x) = E_x[g(B_T)]$$

2. *Let g be a bounded measurable function on ∂D. Then the function $u(x) = E_x[g(B_T)]$ is harmonic on D.*

pf.

- i
  fix $x \in D$, $\epsilon_0 > 0$ s.t. the ball of radius ϵ_0 centered at x is contained in D. $\forall \epsilon \in (0, \epsilon_0)$, let D_ϵ be the connected component containing x of the open set consisting of all points of D whose distance to D^c is greater than ϵ. If $T_\epsilon = \inf\{t \geq 0 : B_t \notin D_\epsilon\}$, by the previous proposition, $u(x) = E_x[u(B_{T_\epsilon})]$. Since $T_\epsilon \uparrow T$ as $\epsilon \downarrow 0$, by the dominated convergence, $E_x[u(B_{T_\epsilon})]$ converges to $E_x[g(B_T)]$ as $\epsilon \to 0$.

- ii
  Set $T_{x,r} = \inf\{t \geq 0 : |B_t - x| = r\}$, $\forall x \in \mathbb{R}^d$, $r > 0$. Fix $x \in D$, $r > 0$ s.t. the closed ball of radius r centered at x is contained in D. By the strong Markov property, $g(B_T) = \Phi(B_t, t \geq 0) = \Phi(B_{T_{x,r}+t}, t \geq 0)$, P_x a.s. given $\Phi(w)$ is the mapping s.t. $\Phi(w) =$ the first value of g at the first exit point of $w \in C(\mathbb{R}_{\geq 0}, \mathbb{R}^d)$ with $w(0) \in D$ from D. The above holds because the paths $(B_t, t \geq 0)$ and $(B_{T_{x,r}+t}, t \geq 0)$ have the same exit point from D. Then $u(x) = E_x[g(B_T)] = E_x[\Phi(B_{T_{x,r}+t}, t \geq 0)] = E_x[E_{B_{T_{\epsilon,r}}}[\Phi(B_t, t \geq 0)]] = E_x[u(B_{T_{\epsilon,r}})]$. Since we know the law of $B_{T_{\epsilon,r}}$ under P_x is $\sigma_{x,r}$, it has the mean value property. Then by the previous lemma, it is harmonic on D.

Definition 6.2.3. *If $y \in \partial D$, we say D satisfies the exterior cone condition at y if there exists a nonempty open cone $\mathscr{C}$ with apex y and a real $r > 0$ s.t. the intersection of $\mathscr{C}$ with the open ball of radius r centered at y is contained in D^c.*

$\rightarrow$ a convex domain satisfies the exterior cone condition at every point of its boundary.

Theorem 6.2.1 (Solution of the Dirichlet Problem)**.** *Let D be a bounded domain in $\mathbb{R}^d$. Assume that D satisfies the exterior cone condition at every $y \in \partial D$. Then for every continuous function g on ∂D,*

$$u(x) = E_x[g(B_T)], \;\; T = \inf\{t \geq 0 : \; B_t \notin D\}$$

gives the unique solution of the Dirichlet problem with boundary condition g.

pf. By the previous proposition ii, we only need to verify $\forall y \in \partial D$, $\lim_{x\to y,\, x\in D} u(x) = g(x)$. Let $\epsilon > 0$. Since g is continuous, $\exists \delta > 0$ s.t. $|g(z) - g(y)| \le \epsilon/3$ whenever $z \in \partial D$, $|z - y| < \delta$. Let $M > 0$ be s.t. $|g(z)| \le M$, $\forall z \in \partial D$. Then $\forall \eta > 0$,

$$\begin{aligned}|u(x) - g(y)| &\le E_x[|g(B_T) - g(y)|\mathbb{1}_{\{T\le\eta\}}] + E_x[|g(B_T) - g(y)|\mathbb{1}_{\{T>\eta\}}] \\ &\le E_x[|g(B_T) - g(y)|\mathbb{1}_{\{T\le\eta\}}\mathbb{1}_{\{\sup\{|B_t - x| : t\le\eta\}\le\delta/2\}}] \\ &\quad + 2MP_x(\sup_{t\le\eta}|B_t - x| > \delta/2) + 2MP_x(T > \eta)\end{aligned}$$

- $|B_T - y| \le |B_T - x| + |y - x| < \delta$ on the event $\{T \le \eta\} \cap \sup\{|B_t - x| : t \le \eta\} \le \delta/2\}$. Given our choice of $\delta, E_x[|g(B_T) - g(y)|\mathbb{1}_{\{T\le\eta\}}\mathbb{1}_{\{\sup\{|B_t - x| : t\le\eta\}\le\delta/2\}}] \le \epsilon/3$

- $2MP_x(\sup_{t\le\eta}|B_t - x| > \delta/2) = 2MP_0(\sup_{t\le\eta}|B_t| > \delta/2)$ by the translation invariance. Then it tends to 0 when $\eta > 0$ by the continuity of the sample paths. So we can fix $\eta > 0$ so that $2MP_x)\sup_{t\le\eta}|B_t - x| > \delta/2) < \epsilon/3$

-

  Lemma 6.2.1. *Under the exterior cone condition,* $\forall y \in \partial D$, $\forall \eta > 0$,

  $$\lim_{x\to y,\, x\in D} P_x(T > \eta) = 0$$

  pf. $\forall u \in \mathbb{R}^d$ with $|u| = 1$, $\forall \gamma \in (0,1)$, consider the circular cone $\mathscr{C}(u,\gamma) \equiv \{z \in \mathbb{R}^d : z \cdot u > (1-\gamma)|z|\}$ where $z \cdot u$ stands for the usual scalar product. If $y \in \partial D$ is given, the exterior cone condition means we can fix $r > 0$ and γ s.t. $y + (\mathscr{C}(u,\gamma) \cap \mathscr{B}_r) \subset D^c$ where $\mathscr{B}_r$ denotes the open ball of radius r centered at 0. Let $\begin{matrix}\mathscr{C} = \mathscr{C}(u,\gamma) \cap \mathscr{B}_r \\ \mathscr{C}' = \mathscr{C}(u, \frac{\gamma}{2}) \cap \mathscr{B}_{r/2}\end{matrix}$ which is the intersection of a smaller cone with $\mathscr{B}_{r/2}$. For every open subset $F \subset \mathbb{R}^d$, write $T_F = \inf\{t \ge 0 : B_t \in F\}$. By the Blumenthal's zero-one law $P_0(T_{\mathscr{C}(u,\gamma/2)} = 0) = 1$ and therefore $P_0(T_{\mathscr{C}'} = 0) = 1$ by the continuity of sample paths. On the other hand, set $\mathscr{C}'_a = \{z \in \mathscr{C}' : |z| > a\}$, $\forall a \in (0,r)$. The sets $\mathscr{C}'_a$ increases to $\mathscr{C}'$ as $a \downarrow 0$ and therefore $T_{\mathscr{C}'_a} \downarrow T_{\mathscr{C}'} = 0$ as $a \downarrow 0$, P_0a.s. Hence given any $\beta > 0$ we can fix a small enough so that $P_0(T_{\mathscr{C}'_a} \le \eta) \ge 1 - \beta$. Since $y + \mathscr{C} \subset D^c$, $P_x(T \le \eta) \ge P_x(T_{y+\mathscr{C}} \le \eta) = P_0(T_{y-x+\mathscr{C}} \le \eta) \ge P_0(T_{\mathscr{C}'_a} \le \eta) \ge 1 - \beta$, given $|y - x|$ small enough, the shifted cone $y - x + \mathscr{C}$ contains $\mathscr{C}'_a$. Since β was arbitrary, the lemma follows.

  Then by the previous lemma, we can choose $\alpha \in (0, \delta/2]$ small enough so that $2MP_x(T > \eta) < \epsilon/3$

$\to |u(x) - g(y)| < \epsilon$ whenever $|x - y| < \alpha$ and the theorem follows.

6.3 the Poisson Kernel

Consider a bounded domain D and a continuous function g on ∂D. Let $T = \inf\{t \geq 0 : B_t \notin D\}$ be the exit time of D by Brownian motion. Then by the previous proposition i, the solution of the Dirichlet problem in D with boundary condition g, if it exists, is given by $u(x) = E_x[g(B_T)] = \int_{\partial D} \omega(x,dy)g(y)$ where $\forall x \in D$, $\omega(x,dy)$ denotes the distribution of B_T under P_x. The measure $\omega(x,dy)$ is called the harmonic measure for D relative to x. Suppose $D = \mathscr{B}_1$ is the open unit ball in $\mathbb{R}^d$. Assume $d \geq 2$. The boundary $\partial\mathscr{B}_1$ is the unit sphere in $\mathbb{R}^d$.

Definition 6.3.1. *The Poisson kernel of the unit ball is the function K defined on $\mathscr{B}_1 \times \partial\mathscr{B}_1$ by*

$$K(x,y) = \frac{1-|x|^2}{|y-x|^d}, \quad \forall x \in \mathscr{B}_1, \ \forall y \in \partial\mathscr{B}_1$$

Lemma 6.3.1. *For every fixed $y \in \partial\mathscr{B}_1$, the function $x \mapsto K(x,y)$ is harmonic on $\mathscr{B}_1$.*

pf. Set $K_y(x) = K(x,y)$, $\forall x \in \mathscr{B}_1$. Then K_y is a C^∞ function on $\mathscr{B}_1$. Moreover $\Delta K_y = 0$ on $\mathscr{B}_1$. So the lemma follows.

Lemma 6.3.2. *Let $0 \leq r_1 < r_2$ be two real numbers and let $h : (r_1, r_2) \to \mathbb{R}$ be a measurable function. The function $u(x) = h(|x|)$ is harmonic on the domain $\{x \in \mathbb{R}^d : r_1 < |x| < r_2\}$ iff there exist two constants a, b s.t. $h(r) = \begin{cases} a + b\log r & d = 2 \\ a + br^{2-d} & d \geq 3 \end{cases}$.*

pf. Suppose $u(x) = h(|x|)$ is harmonic on $\{x \in \mathbb{R}^d : r_1 < |x| < r_2\}$. Then u is twice continuously differentiable and so is h. By the Laplacian of a radial function, $\Delta u = 0$ iff $h''(r) + \frac{d-a}{r}h'(r) = 0$, $r \in (r_1, r_2)$. The solutions of the ODE are the functions of the form given in the statement. Thus the lemma follows.

Lemma 6.3.3. $\forall x \in \mathscr{B}_1$,

$$\int_{\partial\mathscr{B}_1} K(x,y)\sigma_1(dy) = 1$$

pf. $\forall x \in \mathscr{B}_1$, set $F(x) = \int_{\partial\mathscr{B}_1} K(x,y)\sigma_1(dy)$. Then by the previous lemma F is harmonic on $\mathscr{B}_1$. If $\begin{cases} x \in \mathscr{B}_1 \\ r < 1 - |x| \end{cases}$, by the mean value property for every $y \in \partial\mathscr{B}_1$, $K(x,y) = \int K(z,y)\sigma_{x,r}(dz)$. Hence by Fubini's theorem,
$\int F(z)\sigma_{x,r}(dz) = \int(\int K(z,y)\sigma_1(dy))\sigma_{x,r}(dz) = \int(\int K(z,y)\sigma_{x,r}(dz))\sigma_1(dy) = \int K(x,y)\sigma_1(dy)$
$= F(x) \to$ the mean value property holds for F. If Ψ is a vector isometry of $\mathbb{R}^d$, $K(\psi(x), \psi(y))$
$= K(x,y)$, $\forall x \in \mathscr{B}_1$, $y \in \partial\mathscr{B}_1$. Also $\sigma_1(dy)$ is invariant under Ψ and therefore $F(\Psi(x)) = F(x)$, $\forall x \in \mathscr{B}_1$. Hence F is a radial harmonic function and by the previous lemma, F is constant. Since $F(0) = 1$, the lemma follows.

Theorem 6.3.1. *Let g be a continuous function on $\partial\mathscr{B}_1$. The unique solution of the Dirichlet problem in $\mathscr{B}_1$ with boundary condition g is given by*

$$u(x) = \int_{\partial\mathscr{B}_1} g(y)K(x,y)\sigma_1(dy),\ \forall x \in \mathscr{B}_1$$

pf. Fix $y_0 \in \partial\mathscr{B}_1$. $\forall\delta > 0$, if $x \in \mathscr{B}_1$, $y \in \partial\mathscr{B}_1$ are s.t. $\begin{matrix} |x - y_0 < \delta/2 \\ |y - y_0 > \delta \end{matrix}$, then $K(x,y) \leq (\frac{2}{\delta})^d(1-|x|^2)$. Thus $\forall\delta > 0$, $\lim_{x\to y_0,\, x\in\mathscr{B}_1} \int_{\{|y-y_0|>\delta\}} K(x,y)\sigma_1(dy) = 0$. Given $\epsilon > 0$, $\exists\delta > 0$ sufficiently small so that the condition $\left\{ \begin{matrix} y \in \partial\mathscr{B}_1 \\ |y - y_0| \leq \delta \end{matrix} \right.$ imply $|g(y) - g(y_0)| \leq \epsilon$. If $M = \sup\{|g(y)| : y \in \partial\mathscr{B}_1\}$, $,|u(x) - g(y_0)| = |\int K(x,y)(g(y) - g(y_0))\sigma_1(dy)| \leq 2M \int_{\{|y-y_0|>\delta\}} K(x,y)\sigma_1(dy) + \epsilon$ by the previous lemma and the choice of δ. Since $\lim_{x\to y_0,\, x\in\mathscr{B}_1} \int_{\{|y-y_0|>\delta\}} K(x,y)\sigma_1(dy) = 0$, $\lim_{x\to y_0,\, x\in\mathscr{B}_1} \sup|u(x) - g(y_0)| \leq \epsilon$. So the boundary condition is satisfied. u is harmonic by the previous lemma.

Corollary 6.3.1. *Let $T = \inf\{t \geq 0 : B_t \notin \mathscr{B}_1\}$. $\forall x \in \mathscr{B}_1$, the distribution of B_T under P_x has density $K(x,y)$ w.r.t. $\sigma_1(dy)$.*

pf. immediate since by the previous theorem and previous proposition i, we get $E_x[g(B_T)] = \int_{\partial\mathscr{B}_1} g(y)K(x,y)\sigma_1(dy)$, $\forall x \in \mathscr{B}_1$.

6.4 Transience and Recurrence of Brownian Motion

Consider a d-dimensional Brownian motion $(B_t)_{t\geq 0}$ that starts from x under the probability measure P_x. Suppose $d \geq 2$. For every $a \geq 0$, $U_a = \inf\{t \geq 0 : |B_t| = 1\}$.

Proposition 6.4.1. *Suppose $x \neq 0$ and let ϵ, R be s.t. $0 < \epsilon < |x| < R$. Then*

$$P_x(U_\epsilon < U_R) = \begin{cases} \frac{\log R - \log|x|}{\log R - \log\epsilon} & d = 2 \\ \frac{R^{2-d} - |x|^{2-d}}{R^{2-d} - \epsilon^{2-d}} & d \geq 3 \end{cases}$$

Consequently we have $P_x(U_0 < \infty) = 0$ and $\forall\epsilon(0,|x|)$, $P_x(U_\epsilon < \infty) = \left\{ \begin{matrix} 1 & d = 2 \\ (\frac{\epsilon}{|x|})^{d-2} & d \geq 3 \end{matrix} \right.$

pf. Write $D_{\epsilon,R}$ for the annulus $\{y \in \mathbb{R}^d : \epsilon < |y| < R\}$. Let $u(x)$ be the function defined for $x \in D_{\epsilon,R}$ that appears in the RHS of the first equation. By the lemma 5.3.2., u is harmonic on $D_{\epsilon,R}$ and it is clear that u solves the Dirichlet problem in $D_{\epsilon,R}$ with boundary condition $\left\{ \begin{matrix} g(y) = 0 & |y| = R \\ g(y) = 1 & |y| = \epsilon \end{matrix} \right.$ If T_ϵ denotes the first exist time

from $D_{\epsilon,R}$, by the proposition 5.2.1., we need to have $u(x) = E_x[g(B_{T_{\epsilon,R}})]$ for every $x \in D_{\epsilon,R}$. The first equation follows since $E_x[g(B_{T_{\epsilon,R}})] = P_x(U_\epsilon < U_R)$. If $R > |x|$ is fixed, the event $\{U_0 < U_R\}$ is contained in $\{U_\epsilon < U_R\}$, $\forall 0 < \epsilon < |x|$. By passing to the limit $\epsilon \to 0$ in the RHS of the first equation, we get $P_x(U_0 < U_R) = 0$. Since $U_R \uparrow \infty$ as $R \uparrow \infty$, $P_x(U_0 < \infty) = 0$.
$P_x(U_\epsilon < \infty) = \lim_{R\to\infty} P_x(U_\epsilon < U_R)$. By passing the limit $R \to \infty$ in the RHS of the first equation, we get $P_x(U_\epsilon < \infty) = \begin{cases} 1 & d = 2 \\ (\frac{\epsilon}{|x|})^{d-2} & d \geq 3 \end{cases}$.

$\forall y \in \mathbb{R}^d$, $\tau_y \equiv \inf\{t \geq 0 : B_t = y\}$ so that $\tau_0 = U_0$. The property $P_x(\tau_0 < \infty) = 0$, $\forall x \neq 0$ implies $P_x(\tau_y < \infty) = 0$, $y \neq x$ by translation invariance. This means the probability for Brownian motion to visit a fixed point other than its starting point is 0: points are polar for d-dimensional Brownian motion with $d \geq 2$.

If $\mathbf{m}$ denotes Lebesgue measure on $\mathbb{R}^d$, by the Fubini's theorem,

$$E_x[\mathbf{m}(\{B_r, t \geq 0\})] = E_x[\int_{\mathbb{R}^d} dy_{\{\tau_y < \infty\}}] = \int_{\mathbb{R}^d} dy P_x(\tau_y < \infty) = 0$$

and therefore $\mathbf{m}(\{B_t, t \geq 0\}) = 0$, P_x a.s.

Theorem 6.4.1. *1. In dimension $d = 2$, Brownian motion is recurrent, meaning that a.s. for every nonempty open subset O of $\mathbb{R}^d$, the set $\{t \geq 0 : B_t \in O\}$ is unbounded.*

2. In dimension $d \geq 3$, Brownian motion is transient, meaning that $\lim_{t\to\infty} |B_t| = \infty$ a.s.

pf.

- i
  Assume wlog O is an open ball of rational radius centered at a point with rational coordinates. Then it suffices to consider a fixed open ball $\mathscr{B}$ and we may assume $\mathscr{B}$ is centered at 0 and the starting point of B is $x \neq 0$. Then by the previous proposition, Brownian motion will never hit 0, but still will hit any open ball centered at 0. Thus B must visit $\mathscr{B}$ at arbitrarily large times, a.s.

- ii
  Assume wlog the starting point of B is $x \neq 0$. Since the function $y \mapsto |y|^{2-d}$ is harmonic on $\mathbb{R}^d \setminus \{0\}$ and since B does not hit 0, we get that $|B_t|^{2-d}$ is a local martingale and hence a supermartingale. Since a positive supermartingale is automatically bounded in L^1, $|B_t|^{2-d}$ converges a.s. as $t \to \infty$. The a.s. limit must be zero, since otherwise the curve $\{B_t : t \geq 0\}$ would be bounded. Therefore $|B_t|$ converges to ∞ as $t \to \infty$

Chapter 7

Stochastic Differential Equations

7.1 The Definition of SDE Problem

Consider an ordinary differential equation of the form $\begin{cases} y'(t) = b(y(t)) \\ dy_t = b(y_t)dt \end{cases}$. If we take random perturbations of the system into account, we add a noise term, which is typically of the from σdB_t where B denotes a Brownian motion, and σ is a constant corresponding to the intensity of the noise. Since the random perturbations affecting disjoint time intervals are assumed to be independent, we can use the Brownian motion as a perturbation. In this way, we arrive at a stochastic differential equation of the form

$$\begin{cases} dy_t = b(y_t)dt + \sigma dB_t \\ y_t = y_0 + \int_0^t b(y_s)ds + \sigma B_t \end{cases}$$

We can generalise the above equation by allowing σ to depend on the state of the system at time t

$$\begin{cases} dy_t = b(y_t)dt + \sigma(y_t)dB_t \\ y_t = y_0 + \int_0^t b(y_s)ds + \int_0^t \sigma(y_s)dB_s \end{cases}$$

Definition 7.1.1. *Let d, m be positive integers and let σ, b be locally bounded measurable functions defined on $\mathbb{R}_{\geq 0} \times \mathbb{R}^d$ and taking values in $M_{d\times m}(\mathbb{R})$ and in $\mathbb{R}^d$ respectively, where $M_{d\times m}(\mathbb{R})$ is the set of all $d \times m$ matrices with real coefficients. Write $\begin{cases} \sigma = (\sigma_{ij})_{1\leq i\leq d, 1\leq j\leq m} \\ b = (b_i)_{1\leq i\leq d} \end{cases}$. Then a solution of the stochastic differential equation*

$$E(\sigma, b) \ : \ dX_t = \sigma(t, X_t)dB_t + b(t, X_t)dt$$

consists of

- *a filtered probability space $(\Omega, \mathscr{F}, (\mathscr{F}_t)_{t\in[0,\infty]}, P)$*

- *an m-dimensional* $(\mathscr{F}_t)$*-Brownian motion* $B = (B^1, \dots, B^m)$ *started from 0*
- *an* $(\mathscr{F}_t)$*-adapted process* $X = (X^1, \dots, X^d)$ *with values in* $\mathbb{R}^d$ *with continuous sample paths s.t.* $X_t = X_0 + \int_0^t \sigma(s, X_s) dB_s + \int_0^t b(s, X_s) ds$*, meaning that* $\forall i \in \{1, \dots, d\}$
$$X_t^i = X_0^t + \sum_{j=1}^m \int_0^t \sigma_{ij}(s, X_s) dB_s^j + \int_0^t b_i(s, X_s) ds$$

If additionally $X_0 = x \in \mathbb{R}^d$*, we say* X *is a solution of* $E_x(\sigma, b)$*.*

Definition 7.1.2. *For the equation* $E(\sigma, b)$*, we say that there is*

- *weak existence if* $\forall x \in \mathbb{R}^d$*,* $\exists$ *a solution of* $E_x(\sigma, b)$
- *weak existence and weak uniqueness if in addition* $\forall x \in \mathbb{R}^d$*, all solutions of* $E_x(\sigma, b)$ *have the same law*
- *pathwise uniqueness if whenever the filtered probability space* $(\Omega, \mathscr{F}, (\mathscr{F}_t), P)$ *and the* $(\mathscr{F}_t)$*-Brownian motion* B *are fixed, two solutions* X, X' *s.t.* $X_0 = X_0'$ *a.s. are indistinguishable.*

Furthermore, we say a solution X of $E_x(\sigma, b)$ is a strong solution if X is adapted w.r.t. the complete canonical filtration of B.

e.g. Consider a real Brownian motion β started from $\beta_0 = y$, and set $B_t = \int_0^t sgn(\beta_s) d\beta_s$ where $sgn(x) = \begin{cases} 1 & x > 0 \\ -1 & x \le 0 \end{cases}$. Then by the associativity of stochastic integrals $\beta_t = y + \int_0^t sgn(\beta_s) dB_s$. Moreover B is a continuous martingale with quadratic variation $< B, B >_t = t$ and B is a Brownian motion started from 0. Thus β solves the SDE $dX_t = sgn(X_t) dB_t$, $X_0 = y$ and weak existence holds for this equation. Also any other solution of this must be a Brownian motion, which gives weak uniqueness. But pathwise uniqueness fails. If we take $y = 0$, $\beta, -\beta$ solve the SDE with the same Brownian motion B with initial value 0.

Theorem 7.1.1 (Yamada-Watanabe)**.** *If both weak existence and pathwise uniqueness hold, then weak uniqueness also holds. Moreover, for any choice of the filtered probability space* $(\Omega, \mathscr{F}, (\mathscr{F}_t), P)$ *and of the* $(\mathscr{F}_t)$*-Brownian motion* B*, there exists* $\forall x \in \mathbb{R}^d$ *a unique strong solution of* $E_x(\sigma, b)$*.*

7.2 The Lipschitz Case

Assumptions : The functions σ, b **are continuous on** $\mathbb{R}_{\ge 0} \times \mathbb{R}^d$ **and Lipschitz in the variable** $x \to \exists K$ **a constant s.t.** $\forall t \ge 0,\ \forall x, y \in \mathbb{R}^d,\ \begin{array}{l} |\sigma(t,x) - \sigma(t,y)| \le K|x-y| \\ |b(t,x) - b(t,y)| \le K|x-y| \end{array}$.

Lemma 7.2.1 (Grownwall's Lemma). *Let $T > -$ and let g be a nonnegative bounded measurable function on $[0,T]$. Assume there exist two constants $a,b \geq 0$ s.t. $\forall t \in [0,T]$, $g(t) \leq a + b\int_0^t g(s)ds$. Then $\forall t \in [0,T]$*

$$g(t) \leq a\exp(bt)$$

pf. By iterating the condition on g, we get $g(t) \leq a + a(bt) + b^2\int_0^t ds\int_0^s drg(r) \leq a + a(bt) + a\frac{(bt)^2}{2} + \cdots + a\frac{(bt)^n}{n!} + b^{n+1}\int_0^t ds_1\int_0^{s_1} ds_2 \cdots \int_0^{s_n} ds_{n+1}g(s_{n+1})$, $\forall n \geq 1$. If A is a constant s.t. $0 \leq g \leq A$, the last term in the RHS is bounded by $A(bt)^{n+1}/(n+1)!$ hence tends to 0 as $n \to \infty$. So the desired result follows.

Theorem 7.2.1. *Under the preceding assumptions, pathwise uniqueness holds for $E(\sigma,b)$ and for every choice of the filtered probability space $(\Omega,\mathscr{F},(\mathscr{F}_t,P)$ and of the $(\mathscr{F}_t)$-Brownian motion B, $\forall x \in \mathbb{R}^d$, there exists a unique strong solution of $E_x(\sigma,b)$.*

pf. Consider the case $d = m = 1$.

- pathwise uniqueness

  suppose there are two solutions X,X' s.t. $X_0 = X'_0$. Fix $M > 0$ and set $\tau = \inf\{t \geq 0 : |X_t| \geq M \text{ or } |X'_t| \geq M\}$. Then $\forall t \geq 0$, $X_{t\wedge\tau} = X_0 + \int_0^{t\wedge\tau}\sigma(s,X_s)dB_s + \int_0^{t\wedge\tau} b(s,X_s)ds$ and an analogous equation holds for $X'_{t\wedge\tau}$. Fix a constant $T > 0$. By considering the difference between two equations, $\forall t \in [0,T]$

$$\begin{aligned}
E[(X_{t\wedge\tau} - X'_{t\wedge\tau})^2] &\leq 2E[(\int_0^{t\wedge\tau}(\sigma(s,X_s) - \sigma(s,X'_s))dB_s)^2] + 2E[(\int_0^{t\wedge\tau}(b(s,X_s) - b(s,X'_s))ds)^2] \\
&\leq 2(E[\int_0^{t\wedge\tau}(\sigma(s,X_s) - \sigma(s,X'_s))^2 ds] + TE[\int_0^{t\wedge\tau}(b(s,X_s) - b(s,X'_s))^2 ds]) \\
&\leq 2K^2(1+T)E[\int_0^{t\wedge\tau}(X_s - X'_s)^2 ds] \leq 2K(1+T)E[\int_0^t (X_{s\wedge\tau} - X'_{s\wedge\tau})^2 ds]
\end{aligned}$$

  since $E[(\int_0^t H_s dM_s)^2] \leq E[\int_0^t H_s^2 d < M,M >_s]$. Hence $h(t) = E[(X_{t\wedge\tau} - X'_{t\wedge\tau})^2]$ satisfies $h(t) \leq C\int_0^t h(s)ds$, $\forall t \in [0,T]$ with $C = 2K^2(1+T)$. The function h is bounded above by $4M^2$ and the hypothesis of the lemma holds with $a = 0, b = C$. Thus we get $h = 0$ so that $X_{t\wedge\tau} = X'_{t\wedge\tau}$. By letting $M \to \infty$, $X_t = X'_t$.

- existence of the strong solution

  define by induction $\begin{array}{rcl} X_t^0 &=& x \\ X_t^1 &=& x + \int_0^t \sigma(s,x)dB_s + \int_0^t b(s,x)ds \\ X_t^n &=& x + \int_0^t \sigma(s,X_s^{n-1})dB_s + \int_0^t b(s,X_s^{n-1})ds \end{array}$. The stochastic integrals are well-defined since for every n the process X^n is adapted and has continuous samle paths. Fix any $T > 0$ and for every $n \geq 1$, $\forall t \in [0,T]$, we

set $g_n(t) = E[\sup_{0\le s\le t}|X_s^n - X_s^{n-1}|^2]$. The fact that $\sigma(\cdot,x), b(\cdot,x)$ are continuous, hence bounded over $[0,T]$ implies there exists a constant C'_T s.t. $g_1(t) \le C'_T$, $\forall t \in [0,T]$ by the Doob's inequality in L^2. Then $X_t^{n+1} - X_t^n = \int_0^t(\sigma(s,X_s^n) - \sigma(s,X_s^{n-1}))dB_s + \int_0^t(b(s,X_s^n) - b(s,X_s^{n-1}))ds$. Hence by the Burkholder-Davis-Gundy inequality with $p = 2$,

$$\begin{aligned}
E[\sup_{0\le s\le t}|X_s^{n+1} - X_s^n|^2] &\le 2E[\sup_{0\le s\le t}|\int_0^s(\sigma(u,X_u^n) - \sigma(u,X_u^{n-1}))dB_u|^2 \\
&\quad + \sup_{0\le s\le t}|\int_0^s(b(u,X_u^n) - b(u,X_u^{n-1}))du|^2] \\
&\le 2(C_{(2)}E[\int_0^t(\sigma(u,X_u^n) - \sigma(u,X_u^{n-1}))^2du] \\
&\quad + TE[\int_0^t(b(u,X_u^n) - b(u,X_u^{n-1}))^2du]) \\
&\le 2(C_{(2)} + T)K^2E[\int_0^t|X_u^n - X_u^{n-1}|^2du] \\
&\le C_TE[\int_0^t\sup_{0\le r\le u}|X_r^n - X_r^{n-1}|^2du]
\end{aligned}$$

where $C_T = 2(C_{(2)} + T)K^2$. Thus $\forall n \ge 1$, $g_{n+1}(t) \le C_T\int_0^t g_n(u)du$. Since $g_1(t) \le C'_T$, $\forall n \ge 1$, $\forall t \in [0,T]$, $g_n(t) \le C'_T(C_T)^{n-1}\frac{t^{n-1}}{(n-1)!}$. In particular, $\sum_{n=1}^\infty g_n(T)^{1/2} < \infty$
$\to \sum_{n=0}^\infty \sup_{0\le t\le T}|X_t^{n+1} - X_t^n| < \infty$ a.s. Thus $(X_t^n, 0 \le t \le T)$ converges uniformly on $[0,T]$ a.s. to a limiting process $(X_t, 0 \le t \le T)$ which has continuous sample paths. Similarly $\forall n$, X^n is adapted and therefore the same holds for X. Since σ, b are Lipshcitz in x, we get $\forall t \in [0,T]$,
$\begin{cases} \lim_{n\to\infty}(\int_0^t\sigma(s,X_s)dB_s - \int_0^t\sigma(s,X_s^n)dB_s) = 0 \\ \lim(\int_0^t b(s,X_s)ds - \int_0^t b(s,X_s^n)ds) = 0 \end{cases}$ in probability. By passing to the limit in the induction equation defining X^n, X solves $E_x(\sigma,b)$ on $[0,T]$. Then by the uniqueness part, it is a strong solution on $\mathbb{R}_{\ge 0}$ that will coincide with the solution $[0,T]$ upto time T.

$W(dw)$ stands for the Wiener measure on the canonical space $C(\mathbb{R}_{\ge 0},\mathbb{R}^m)$ of all continuous functions from $\mathbb{R}_{\ge 0}$ to $\mathbb{R}^m$.

Theorem 7.2.2. *Under the assumptions of the preceding theorem, there exists, $\forall x \in \mathbb{R}$, a mapping $F_x : C(\mathbb{R}_{\ge 0},\mathbb{R}^m) \to C(\mathbb{R}_{\ge 0},\mathbb{R}^d)$ which is measurable when $C(\mathbb{R}_{\ge 0},\mathbb{R}^m)$ is equipped with the Borel σ-field completed by the W-negligible sets and $C(\mathbb{R}_{\ge 0},\mathbb{R}^d)$ is equipped with the Borel σ-field s.t. the following properties hold :*

1. *$\forall t \ge 0$, $F_x(w)_t$ coincides $W(dw)$ a.s. with a measurable function of $(w(r), 0 \le r \le t)$*

2. *$\forall w \in C(\mathbb{R}_{\geq 0}, \mathbb{R}^m)$, the mapping $x \mapsto F_x(w)$ is continuous*

3. *$\forall x \in \mathbb{R}^d$, for every choice of the complete filtered probability space $(\Omega, \mathscr{F}, (\mathscr{F}_t), P)$ and of the m-dimensional $(\mathscr{F}_t)$-Brownian motion B with $B_0 = 0$, the process X_t defined $X_t = F_x(B)_t$ is the unique solution of $E_x(\sigma, b)$; furthermore if U is an $\mathscr{F}_0$-measurable real random variable, the process $F_U(B)_t$ is the unique solution with $X_0 = U$.*

pf. Assume wlog $d = m = 1$. Let $\mathscr{N}$ be the class of all W-negligible sets in $C(\mathbb{R}_{\geq 0}, \mathbb{R})$ and $\forall t \in [0, \infty]$, $\mathscr{G}_t = \sigma(w(s), 0 \leq s \leq t) \vee \mathscr{N}$. $\forall x \in \mathbb{R}$, we write X^x for the solution of $E_x(\sigma, b)$ corresponding to the filtered probability space $(C(\mathbb{R}_{\geq 0}, \mathbb{R}), \mathscr{G}_\infty, (\mathscr{G}_t), W)$ and the canonical Brownian motion $B_t(w) = w(t)$. This solution exists and unique by the previous theorem.

- ii

  let $x, y \in \mathbb{R}$ and let T_n be the stopping time defined by $T_n = \inf\{t \geq 0 : |X_t^x| \geq n$ or $|X_t^y| \geq n\}$. Let $p \geq 2, T \geq 1$. By the Burkholder-Davis-Gundy inequalities, $\forall t \in [0, T]$,

$$
\begin{aligned}
& E[\sup_{s \leq t} |X^x_{s \wedge T_n} - X^y_{x \wedge T_n}|^p] \\
& \leq C_p(|x-y|^p + E[\sup_{s \leq t} |\int_0^{s \wedge T_n} (\sigma(r, X_r^x) - \sigma(r, X_r^y)) dB_r|^p] \\
& \quad + E[\sup_{s \leq t} |\int_0^{s \wedge T_n} (b(r, X_r^x) - b(r, X_r^y)) dr|^p)] \\
& \leq C_p(|x-y|^p + C'_p E[(\int_0^{t \wedge T_n} (\sigma(r, X_r^x) - \sigma(r, X_r^y))^2 dr)^{p/2}] \\
& \quad + E[(\int_0^{t \wedge T_n} |b(r, X_r^x) - b(r, X_r^y)| dr)^p]) \\
& \leq C_p(|x-y|^p + C'_p t^{p/2-1} E[\int_0^t |\sigma(r \wedge T_n, X^x_{r \wedge T_n}) - \sigma(r \wedge T_n, X^y_{r \wedge T_n})|^p dr \\
& \quad + t^{p-1} E[\int_0^t |b(r \wedge T_n, X^x_{\wedge T_n}) - b(r \wedge T_n, X^y_{r \wedge T_n})|^p dr]) \\
& \leq C''_p(|x-y|^p + T^p \int_0^t E[|X^x_{r \wedge T_n} - X^y_{r \wedge T_n}|^p [dr)
\end{aligned}
$$

  where $C_p, C'_p, C'''_p < \infty$ depend on p but not on n, x, y, T. As the function $t \mapsto E[\sup_{s \leq t} |X^x_{s \wedge T_n} - X^y_{s \wedge T_n}|^p]$ is bounded, by the previous lemma, $\forall t \in [0, T]$, $E[\sup_{s \leq t} |X^x_{s \wedge T_n} - X^y_{s \wedge T_n}|^p] \leq C''_p |x-y|^p \exp(C''_p T^p t) \Leftrightarrow E[\sup_{s \leq t} |X^x_s - X^y_s|^p] \leq C''_p |x-y| \exp(C''_p T^p t)$ by letting $n \to \infty$. The topology on the space $C(\mathbb{R}_{\geq 0}, \mathbb{R})$

is defined by the distance $\mathbf{d}(w,w') = \sum_{k=1}^{\infty} \alpha_k (\sup_{s\le k} |w(s)-w'(s)| \wedge 1)$ provided $\sum \alpha_k$ converges. We may choose the coefficients α_k so that $\sum_{k=1}^{\infty} \alpha_k \exp(C''_p k^{p+1}) < \infty$. $\forall x \in \mathbb{R}$, consider X^x as a r.v. with values in $C(\mathbb{R}_{\ge 0},\mathbb{R})$. By the Jensen's inequality, $E[\mathbf{d}(X^x,X^y)^p] \le (\sum \alpha_k)^{p-1} \sum \alpha_k E[\sup_{s\le k} |X^x_s - X^y_s|^p] \le \bar{C}_p |x-y|^p$ with a constant $\bar{C}_p$ independent of x,y. By Kolmogorov's lemma, $(X^x, x \in \mathbb{R})$ has a modification with continuous sample paths, which we denote by $(\tilde{X}^x, x \in \mathbb{R})$. We set $F_x(w) = \tilde{X}^x(w) = (\tilde{X}_t(w))_{t\ge 0}$.

- i

  the mapping $w \mapsto F_x(w)$ is measurable form $C(\mathbb{R}_{\ge 0},\mathbb{R})$ equipped with the σ-field $\mathscr{G}_\infty$ into $C(\mathbb{R}_{\ge 0},\mathbb{R})$ equipped with the Borel σ-field $\mathscr{C} = \sigma(w(s), s \ge 0)$. Moreover $\forall t \ge 0$, $F_x(w)_t = \tilde{X}^x_t(w) = X^x_t(w)$ a.s. is $\mathscr{G}_t$-measurable, hence coincides $W(dw)$ a.s. with a measurable function of $(w(s), 0 \le s \le t)$.

- iii

  fix the filtered probability space $(\Omega, \mathscr{F}, (\mathscr{F}_t), P)$ and the $(\mathscr{F}_t)$-Brownian motion B. We have the process $(F_x(B)_t)_{t\ge 0}$ which has continuous sample paths and is also adapted since $F_x(B)_t$ coincide a.s. with a measurable function of $(B_r, 0 \le r \le t)$ by i and since the filtration $(\mathscr{F}_t)$ is complete. On the other hand, by the construction of F_x, $\forall t \ge 0$, $W(dw)$ a.s.

$$F_x(w)_t = x + \int_0^t \sigma(s, F_x(w)_s) dw(s) + \int_0^t b(s, F_x(w)_s) ds$$

  where the stochastic integral $\int_0^t \sigma(s, F_x(w)_s) dw(s)$
  $= \lim_{k\to\infty} \sum_{i=0}^{2^n k-1} \sigma(\frac{it}{2^{n_k}}, F_x(w)_{it/2^n k})(w(\frac{(i+1)t}{2^{n_k}}) - w(\frac{it}{2^{n_k}}))$, $W(dw)$ a.s. Here (n_k) is a suitable subsequence. Then we can replace w by B and get a.s. $F_x(B)_t$
  $= x + \lim_{k\to\infty} \sum_{i=0}^{2^{n_k}-1} \sigma)\frac{it}{2^{n_k}}, F_x(B)_{it/2^n k})(B_{(i+1)t/2^{n_k}} - B_{it/2^{n_k}}) + \int_0^t b(s, F_x(B)_s) ds$
  $= x + \int_0^t \sigma(s, F_x(B)_s) dB_s + \int_0^t b(s, F_x(B)_s) ds$. Thus we obtain $F_x(B)$ is the desired solution.

  Let U be an $\mathscr{F}_0$-measurable r.v. If, in the stochastic integral equation satisfied by $F_x(B)$, we formally substitute U for x, we obtain $F_U(B)$ solves $E(\sigma, b)$ with initial value U.
  The mapping $(w,\omega) \mapsto F_x(B)_t$ is continuous w.r.t. the variable x and $\mathscr{F}_t$-measurable w.r.t. ω. Thus this mapping is measurable for the σ-field $\mathscr{B}(\mathbb{R}) \otimes \mathscr{F}_t$. Since U is $\mathscr{F}_0$-measurable, $F_U(B)_t$ is $\mathscr{F}_t$-measurable. Hence the process $F_U(B)_t$ is adapted. $\forall x \in \mathbb{R}$, $\forall w \in C(\mathbb{R}_{\ge 0},\mathbb{R})$, we define $G(x,w) \in C(\mathbb{R}_{\ge 0},\mathbb{R})$ by $G(x,w)_t = \int_0^t b(s, F_x(w)_s) ds$ and set $H(w,x) = F_x(w) - x - G(x,w)$. Then $\forall x \in \mathbb{R}$, $W(dw)$ a.s. $H(x,w)_t = \int_0^t \sigma(s, F_x(w)_s) dw(s)$. Hence if $H_n(x,w)_t =$

$\sum_{i=0}^{2^n-1}\sigma(\frac{it}{2^n},F_x(w)_{it/2^n})(w(\frac{(i+1)t}{2^n})-w(\frac{it}{2^n}))$, $H(x,w)_t=\lim H_n(x,w)_t$ in probability under $W(dw)$, $\forall x\in\mathbb{R}$. Since given U $\mathscr{F}_t$-measurable, U,B are independent, we infer from the latter convergence that $H(U,B)_t=\lim_{n\to}H_n(U,B)_t$ in probability. The limit must be the stochastic integral $\int_0^t\sigma(s,F_U(B)_s)dB_s\to\int_0^t\sigma(s,F_U(B)_s)dB_s=H(U,B)_t=F_U(B)_t-U-\int_0^t b(s,F_U(B)_s)ds$ and therefore $F_U(B)$ solves $E(\sigma,b)$ with initial value U.

$\to\forall\epsilon\in(0,1)$, $\forall A>0$ a constant, $T>0$, $C_{\epsilon,A,T}(\omega)$ a constant s.t. if $|x|,|y|\le A$,

$$\sup_{t\le T}|X_t^x(\omega)-X_t^y(\omega)|\le C_{\epsilon,A,T}(\omega)|x-y|^{1-\epsilon}$$

7.3 Homogeneous Case$\left(\begin{cases} \sigma(t,y)=\sigma(y) \\ b(t,y)=b(y) \end{cases}\right)$

Assume σ,b are Lipschitz. Let $x\in\mathbb{R}^d$ and let X^x be a solution of $E_x(\sigma,b)$. Since weak uniqueness holds, $\forall t\ge 0$, the law of X_t^x does not depend on the choice of the solution and the law is the image of Wiener measure on $C(\mathbb{R}_{\ge 0},\mathbb{R}^d)$ under the mapping $w\mapsto F_x(w)_t$, where the mapping F_x were introduced in the previous theorem. By the theorem 6.2.2., the law of X_t^x depends continuously on the pair (x,t).

Theorem 7.3.1. *Assume $(X_t)_{t\ge 0}$ is a solution of $E(\sigma,b)$ on a complete filtered probability space $(\Omega,\mathscr{F},(\mathscr{F}_t),P)$. Then $(X_t)_{t\ge 0}$ is a Markov process w.r.t. the filtration $(\mathscr{F}_t)$ with semigroup $(Q_t)_{t\ge 0}$ defined by $Q_tf(x)=E[f(X_t^x)]$ where X^x s an arbitrary solution of $E_x(\sigma,b)$.*

pf. By the theorem 2.2.2., $Q_tf(X)=\int f(F_x(w)_t)W(dw)$. Fix $s\ge 0$ and $\forall t\ge 0$, $X_{s+t}=X_s+\int_s^{s+t}\sigma(X_r)dB_r+\int_s^{s+t}b(X_r)dr$ where B is an $(\mathscr{F}_t)$-Brownian motion starting from 0. Then set $\forall t\ge 0$, $X_t'=X_{s+t}$, $\mathscr{F}_t'=\mathscr{F}_{s+t}$, $B_t'=B_{s+t}-B_s$. The filtration $(\mathscr{F}_t')$ is complete and the process X' is adapted to $(\mathscr{F}_t')$ and B' is an m-dimensional $(\mathscr{F}_t')$-Brownian motion. Furthermore, by the approximation results for the stochastic integral of adapted processes with continuous sample paths, a.s. $\forall t\ge 0$, $\int_s^{s+t}\sigma(X_r)dB_r=\int_0^t\sigma(X_u')dB_u'$ where the RHS is computed in the filtration $(\mathscr{F}_t')$. By the above equation, $X_t'=X_s+\int_0^t\sigma(X_u')dB_u'+\int_0^t b(X_u')du$. Hence X' solves $E(\sigma,b)$ on the space $(\Omega,\mathscr{F},(\mathscr{F}_t'),P)$ and with the Brownian motion B' with initial value $X_0'=X_s$. By the theorem 2.2.2.-iii, $X'=F_{X_s}(B')$ a.s. Consequently, $\forall t\ge 0$

$$E[f(X_{s+t})|\mathscr{F}_s]=E[f(X_t')|\mathscr{F}_s]=E[f(F_{X_s}(B')_t|\mathscr{F}_s]=\int f(F_{X_s}(w)_t)W(dw)=Q_rf(X_s)$$

by the definition of Q_tf and the independence of B' and $\mathscr{F}_s$.
The properties of the transition semigroup i,iii are trivially satisfied. For the

Chapman-Kolmogorov relation, $\forall s,t \geq 0$, $Q_{t+s}f(x) = E[f(X^x_{s+t})] = E[E[f(X^x_{s+t})|\mathscr{F}_s]] = E[Q_t f(X^x_s)] = \int Q_s(x,dy)Q_t f(y)$. Thus Q_t is a transition semigroup.

Theorem 7.3.2. *The semigroup $(Q_t)_{t\geq 0}$ is Feller. Furthermore, its generator L is s.t.*

$$C^2_c(\mathbb{R}^d) \subset D(L)$$

and $\forall f \in C^2_c(\mathbb{R}^d)$,

$$Lf(x) = \frac{1}{2}\sum_{i,j=1}^{d}(\sigma\sigma^*)_{ij}(x)\frac{\partial^2 f}{\partial x_i \partial x_j}(x) + \sum_{i=1}^{d} b_i(x)\frac{\partial f}{\partial x_i}(x)$$

where σ^* denotes the transpose of the matrix σ.

pf.

- Feller
  Assume wlog σ, b are bounded. Fix $f \in C_0(\mathbb{R}^d)$. Since the mappings $x \mapsto F_x(w)$ are continuous, $Q_t f(x) = \int f(F_x(w)_t)W(dw)$ is continuous by the dominated convergence. Then since $X^x_t = x + \int_0^t \sigma(X^x_s)dB_s + \int_0^t b(X^x_s)ds$, and σ, b are bounded, $\exists C$, a constant, which does not depend on t,x s.t. $E[(X^x_t - x)^2] \leq C(t+t^2)$. By the Markov's inequality, $\forall t \geq 0 \sup_{x\in\mathbb{R}^d} P(|X^x_t - x| > A) \to 0$ as $A \to \infty$. Writing $|Q_t f(x)| = |E[f(X^x_t)]| \leq |E[f(X^x_t)\mathbb{1}_{\{|X^x_t - x|\leq A\}}]| + ||f||P(|X^x_t - x| > A)$ allow us to have $\limsup |Q_t f(x)| \leq ||f|| \sup_{x\in\mathbb{R}^d} P(|X^x_t - x| > A)$, given $f \in C_0(\mathbb{R}^d)$. Since A was arbitrary, $\lim Q_t f(x) = 0$ and therefore $Q_t f \in C_0(\mathbb{R}^d)$. $\forall \epsilon > 0$, $\sup_{x\in\mathbb{R}^d} |E[f(X^x_t)] - f(x)| \leq \sup_{x,y\in\mathbb{R}^d, |x-y|\leq\epsilon} |f(x) - f(y)| + 2||f|| \sup_{x\in\mathbb{R}^d} P(|X^x_t - x| > \epsilon)$. Since $E[(X^x_t - x)^2] \leq C(t+t^2)$, by the Markov's inequality, $\sup_{x\in\mathbb{R}^d} P(|X^x_t - x| > \epsilon) \to 0$ as $t \to 0$, hence $\limsup ||Q_t f - f|| = \limsup(\sup_{x\in\mathbb{R}^d} |E[f(X^x_t)] - f(x)|) \leq \sup_{x,y\in\mathbb{R}^d, |x-y|\leq\epsilon} |f(x) - f(y)|$ which can be made arbitrarily close to 0 by taking ϵ small. Thus $Q_t f \to f$ when $t \to 0$ and Q_t is a Feller semigroup.

- $\forall f \in C^c_c(\mathbb{R}^d)$, $Lf(x) = \frac{1}{2}\sum_{i,j=1}^d (\sigma\sigma^*)_{ij}(x)\frac{\partial^2 f}{\partial x_i \partial x_j}(x) + \sum_{i=1}^d b_i(x)\frac{\partial f}{\partial x_i}(x)$
  Let $f \in C^2_c(\mathbb{R}^d)$. By Ito's formula to $f(X^x_t)$, $X^{x,i}_t = x_i + \sum_{j=1}^m \sigma_{ij}(X^x_s)dB^j_s + \int_0^t b_i(X^x_s)ds$, $\forall i \in \{1,\dots,d\}$, given $X^x_t = (X^{x,1}_t, \dots, X^{x,d}_t)$. Then we get

  $$f(X^x_t) = f(x) + M_t + \sum_{i=1}^d \int_0^t b_i(X^x_s)\frac{\partial f}{\partial x_i}(X^x_s)ds + \frac{1}{s}\sum_{i,i'=1}^d \int_0^t \frac{\partial^2 f}{\partial x_i \partial x_{i'}}(X^x_s)d < X^{x,i}, X^{x,i'} >_s$$

  where M is a continuous local martingale. Moreover if $i, i' \in \{1,\dots,d\}$, $d < X^{x,i}, X^{x,i'} >_s = \sum_{j=1}^m \sigma_{ij}(X^x_s)\sigma_{i'j}(X^x_s)ds = (\sigma\sigma^*)_{ii'}(X^x_s)ds$. Thus if $g(x) = \frac{1}{s}\sum_{i,i'=1}^d (\sigma\sigma^*)_{ii'}(x)\frac{\partial^2 f}{\partial x_i \partial x_{i'}}(x) + \sum_{i=1}^d b_i(x)\frac{\partial f}{\partial x_i}(x)$, the process $M_t = f(X^x_t) - f(x) - \int_0^t g(X^x_s)ds$ is a continuous local martingale. Since f, g are bounded, M is a martingale. Then $f \in D(L)$ and $Lf = g$.

Corollary 7.3.1. *Suppose $(X_t)_{t\geq 0}$ solves $E(\sigma,b)$ on a filtered probability space $(\Omega,\mathscr{F},(\mathscr{F}_t),P)$. Then $(X_t)_{t\geq 0}$ satisfies the strong Markov property: if T is a stopping time and if Φ is a Borel measurable function from $C(\mathbb{R}_{\geq 0},\mathbb{R}^d)$ into $\mathbb{R}_{\geq 0}$,*

$$E[\mathbb{1}_{\{T<\infty\}}\Phi(X_{T+t},t\geq 0)|\mathscr{F}_T]=\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[\Phi]$$

where $\forall x\in\mathbb{R}^d$, $\mathbb{P}_x$ denotes the law on $C(\mathbb{R}_{\geq 0},\mathbb{R}^d)$ of an arbitrary solution of $E_x(\sigma,b)$.

e.g.

- **The Ornstein-Uhlenbeck Process**
  Let $\lambda>0$. The Ornstein-Uhlenbeck process is the solution of the stochastic differential equation $dX_t=dB_t-\lambda X_t dt$.
  $\rightarrow$ by Ito's formula to $e^{\lambda t}X_t$, $X_t=X_0e^{-\lambda t}+\int_0^t e^{-\lambda(t-s)}dB_s$. If $X_0=x\in\mathbb{R}$, X is a Gaussian process whose mean function is $m(t)=E[X_t]=xe^{-\lambda t}$ and whose covariance function is $K(s,t)=cov(X_s,X_t)=\frac{e^{-\lambda|t-s|}-e^{-\lambda(t+s)}}{2\lambda}$.
  If $X_0\sim\mathscr{N}(0,\frac{1}{2\lambda}$, X is a centered Gaussian process with covariance function $\frac{1}{2\lambda}e^{-\lambda|t-s|}$, which is stationary.

- **Geometric Brownian Motion**
  Let $\sigma>0$, $r\in\mathbb{R}$. The geometric Brownian motion with parameters σ,r is the solution of the stochastic differential equation $dX_t=\sigma X_t dB_r+rX_t dt$.
  $\rightarrow$ By Ito's formula to $\log X_t$, $X_t=X_0\exp(\sigma B_t+(r-\frac{\sigma^2}{2})t)$. If the initial value X_0 is positive, the solution remains so at every time $t\geq 0$.

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